



High-Speed III-V-on-SOI transmitters for high-density data communication

Amin Souleiman Dabar

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High-Speed III-V-on-SOI Transmitters for High-Density Data Communication

Thèse de doctorat de l'Institut Polytechnique de Paris
préparée à Télécom SudParis

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High-Speed III-V-on-SOI Transmitters for High-Density Data Communication

AMIN SOULEIMAN DABAR

December 10, 2024

“Dugsiga iyo tacliinta ha daynin, qeyrka wixii diley ha doonin, waa iga dardaarane, inuu da' macu kuu geyn”

Mohamad Aden Dugsiiye

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List of Acronyms

A

AI	Artificial Intelligence.
AM	Amplitude Modulation.
AOM	Acousto-Optic Modulator.
AR	Anti-Reflective.
AWG	Arrayed Waveguide Gratings.

B

BCB	Benzocyclobutene.
BER	Bit Error Rate.
B2B	Back-to-Back.

C

CC	Coherence Collapse.
CMOS	Complementary Metal-Oxide-Semiconductor.
CMT	Coupled Mode Theory.
CMP	Chemical Mechanical Planarisation.
CW	Continuous-Wave.

D

DAC	Digital-to-Analog Converter.
DC	Direct Current.
DER	Dynamic Extinction Ratio.
DFB	Distributed Feedback.
DML	Directly Modulated Laser.
DPSK	Differential Phase Shift Keying.
DR	Distributed Reflectivity.
DSP	Digital Signal Processing.
DUT	Device Under Test.
DWDM	Dense Wavelength Division Multiplexing

E

EAM Electro Absorption Modulator.

EDFA Erbium Doped Fiber Amplifier.

EML External Modulated Laser.

EOBW Electro-Optical Bandwidth.

ER Extinction Ratio.

F

FGC Fiber Grating Coupler.

FP Fabry-Perot.

FSR Free Spectral Range.

I

IL Insertion Loss.

IM-DD Intensity Modulation/ Direct Detection.

InP Indium Phosphide.

ISI Inter-Symbol Interference.

L

LSHB Longitudinal Spatial Hole Burning.

M

MMI multimode interference.

MQW Multi Quantum Well.

MZM Mach-Zehnder modulator.

N

NRZ Non Return to Zero.

NZ-DSF Non Zero-Dispersion Shifted Fiber.

O

OMA Optical Modulation Amplitude.

OOK On Off Keying.

OSA Optical Spectrum Analyser.

P

PAM	Pulse Amplitude Modulation.
PICs	Photonic Integrated Circuits.
PL	Photoluminescence.
PON	Passive Optical Networks.
PPR	Photon Photon Resonance.
PRBS	Pseudo Random Binary Sequence.
Q	
QAM	Quadrature Amplitude Modulation.
QCSE	Quantum-Confined Stark Effect.
QD	Quantum Dot.
QW	Quantum Well.
S	
SBG	Sampled Bragg Grating.
SCH	Separate Confinement Heterostructure.
SiC	Silicon Carbide.
SHB	Spectral Hole Burning.
SMSR	Side Mode Suppression Ratio.
SOA	Semiconductor Optical Amplifier.
SOI	Silicon on Insulator.
SSMF	Standard Single-Mode Fiber.
T	
TBPF	Tunable Bandpass Filter.
TEC	Thermo-Electric Cooler.
TE	Transverse-Electric.
TMM	Transfer Matrix Method.
TDM	Time Division Multiplexing.
TRL	Technological Readiness Level.
TUL	Tunable Laser.
TW	Traveling Wave.

W

WDM Wavelength Division Multiplexing.

WPE Wall-Plug Efficiency.

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1. Introduction

1.1 Research Background

Optical communication relies on the use of light to transmit information through optical fibers. Compared to electrical transmission methods, optical communication offers several significant advantages: substantially higher bandwidth capacity, much greater transmission distances, and complete immunity to electromagnetic interference. These benefits have enabled high-speed optical networks to play a central role in the infrastructure of modern data networks. Today, high-speed Internet connections are accessible almost everywhere in the world, largely due to advancements in microelectronics and the integration of photonic technologies, where signals travel at a fraction of the speed of light with minimal loss in optical fibers.

With the emergence of new technologies such as 6G, Light Detection and Ranging (LiDAR) systems, or artificial intelligence (AI), the demand for bandwidth continues to rise. These applications require high-capacity data transmissions with low latency, resulting in exponential growth in data traffic. Over the past decade, the volume of data created, captured, consumed, and exchanged globally has skyrocketed, increasing from 12.5 ZB in 2014 to an estimated 147 ZB in 2024 (Figure 1.1 [1]).

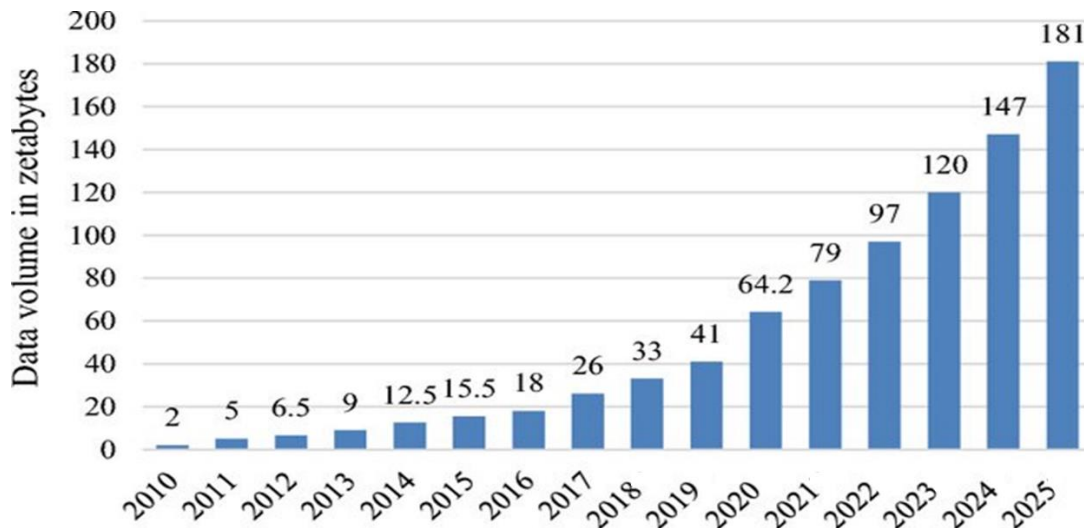


Figure 1.1: Volume of data created, captured, copied, and consumed globally from 2010 to 2024, and forecasts 2025 (adapted from [1]).

In a typical optical communication system, a transmitter converts electrical signals into optical signals, which are then modulated and transmitted through an optical fiber. The optical signal can be amplified to compensate for losses and is subsequently converted back into an electrical signal by a receiver at the other end, where it is processed and decoded. Key

components of optical communication systems include lasers, optical modulators, amplifiers, multiplexers/demultiplexers, and photodetectors.

Optical networks interconnect the world through links of various scales:

- **Long-Haul Networks:** These networks connect continents via submarine cables, such as transpacific cables (approximately 11,000 km) and transatlantic cables (approximately 6,000 km).
- **Regional and Metropolitan Networks:** These networks cover distances of several hundred kilometers, linking countries and cities.
- **Access Networks:** These networks, with ranges of 20 to 40 km, connect end-users to the core network.

Currently, long-haul links rely on coherent communication systems capable of transporting massive amounts of information across oceans, with capacities in the hundreds of terabits per second (Tbps). For instance, the Amitié submarine cable, completed in July 2023, spans 6,783 km across the Atlantic and offers a total capacity of 400 Tbps, connecting Boston in the United States to Europe, with landing points in England and France. Another even more ambitious project, Meta's Anjana cable, aims to achieve a capacity of 480 Tbps, addressing the growing global demand for data transmission [2, 3]. With the increasing volumes of data generated and transmitted worldwide, processing this data presents a significant challenge. Here, access networks play a crucial role in connecting end-users to the global infrastructure. These networks, serving urban, residential, and industrial areas, are central to the distribution of high-speed data and the reduction of latency. They ensure an optimal user experience while meeting the rising demands for bandwidth.

In the following section, we will discuss in detail the different types of optical access networks, focusing on their architecture and crucial role in modern communications.

1.2 Access network

Access networks are the segment of telecommunications networks that connect end users directly to the core communication infrastructures. They are designed to transport data from subscribers (residential users, businesses, etc.) to transport networks, thus playing a crucial role in providing access to Internet services. Among the most widespread optical access solutions are passive optical networks (PON). These networks are characterized by the use of passive optical components, such as couplers, multiplexers, and splitters, to distribute signals over a shared optical fiber without requiring active elements between the service provider and the end users. This simplified architecture significantly reduces infrastructure and maintenance costs, as no devices requiring electrical power are needed along the optical path, except at the terminal equipment.

To meet diverse bandwidth and service requirements, several PON variants have emerged, ranging from less than 1 Gbps (1G) to high-speed PON (HSP-PON) networks operating at 50G, with standards currently being developed for 100G PON networks [4]. The main PON technologies include :

- **GPON (Gigabit PON):** This is the most common PON technology, offering transmission speeds of up to 2.5 Gbps downstream and 1.25 Gbps upstream.

- XG-PON (10 Gigabit PON): A successor to GPON, this technology enables significantly higher throughput, with 10 Gbps downstream and 2.5 Gbps upstream.
- TWDM-PON (Time and Wavelength Division Multiplexed PON): This technology combines wavelength division multiplexing (WDM) and time division multiplexing (TDM), allowing for increased TDM rates based on the number of wavelengths.
- NG-PON2 (Next Generation PON 2): The latest generation of PON, NG-PON2, combines multiple wavelengths to achieve speeds of up to 40 Gbps, making it particularly suitable for the future needs of 5G networks and converged networks.
- 100G PON: The development of 100G PON networks aims to address the growing demand for bandwidth.

However, transmitting high data rates and increasing link capacities present technical challenges. The most efficient approach to tackle these challenges is through coherent detection. Nevertheless, this method requires complex and costly components. For this reason, PONs generally rely on more economical transceivers based on intensity modulation and direct detection (IM-DD) [5-6]. In order to maximize the link capacity, each wavelength can be modulated separately, corresponding to an optical channel following the ITU-T grid [7]. Modulation can be performed directly or externally, using a separate modulator. In either case, single-mode, energy-efficient, and integrable signal sources are necessary. Distributed feedback (DFB) lasers are promising candidates for this application, consequently, the vast majority of lasers used in PONs are single-wavelength DFB lasers. They can easily provide output powers on the order of several tens of mW while being energy-efficient and cost-effective. A brief introduction to this type of component will be presented in the next section.

1.3 DFB Lasers

The multimode operation of Fabry-Perot (FP) lasers presents a significant drawback for their use in optical communications due to the chromatic dispersion of optical fibers. For this reason, DFB lasers are widely used in telecommunications, particularly in PON, due to their high performance and cost-effectiveness. A DFB laser consists of a gain medium combined with a periodic structure that provides optical feedback, ensuring single-wavelength operation. The Bragg region, which acts as a wavelength-dependent reflector, is integrated into this structure with the active section. The mode that will dominate is the one with the lowest losses. The period of the Bragg grating determines the lasing wavelength and must be designed to position the wavelength within the gain spectrum of the active material. For these lasers, the appropriate active materials are typically based on InP ternary and quaternary epitaxial structures, such as InGaAsP and InGaAlAs.

DFB lasers can be directly modulated at high frequencies, allowing for rapid data transmission. Their ability to produce a narrow linewidth is crucial for minimizing signal distortion, particularly in dense wavelength-division multiplexing (DWDM) systems. As a result, they are preferred for high-speed applications where precise wavelength control is necessary. However, the direct modulation of DFB lasers can lead to frequency chirp, which may degrade signal integrity during high-speed transmission. To mitigate this issue, external modulators are often integrated with DFB lasers, providing a solution that maintains high data rates.

1.4 Heterogeneous III-V on Si platform

In response to the new bandwidth demands and growing awareness of sustainability challenges, there has been a push to design devices and storage systems with lower energy consumption to reduce the environmental impact of the industry as a whole. This has led to the exploration of new technologies aimed at achieving greater density and energy efficiency. Among these technologies, silicon photonics (Si photonics) has emerged as the leading candidate for photonic integrated circuits (PICs) due to its ability to integrate both photonic and electronic components, its potential for low cost, and its compatibility with the CMOS technology. Silicon-on-Insulator (SOI) is a platform that allows the creation of highly compact photonic circuits due to its high refractive index contrast. Numerous ultra-compact passive photonic devices have been demonstrated on this platform, ranging from simple waveguides to complex filters (ring resonators, arrayed waveguide gratings (AWG)), multimode interference (MMI) couplers, and grating couplers [8-10]. Active photonic components such as photodiodes [11, 12] and various types of modulators [13] can also be fabricated using silicon photonics technology. Furthermore, this technology spans a wide range of applications, from short-distance data communications to long-haul optical transmissions, as well as neuromorphic computing and artificial intelligence [14, 15]. Due to silicon's indirect bandgap, producing a laser source on this platform presents significant challenges. This has driven considerable research efforts towards integrating direct bandgap semiconductors, such as InP or GaAs—members of the III-V materials family—onto silicon to achieve optical gain. Additionally, materials like lithium niobate, chalcogenides, and 2D materials are being integrated to enhance the modulation and detection capabilities of silicon photonic devices. The combination of Silicon and III-V semiconductors leverages the best of both worlds: the mature, low-cost silicon technology and the superior optical properties of III-V materials. PICs on a heterogeneous III-V-on-Si platform stand out for their high integration density of lasers, modulators and detectors while maintaining competitive production costs and energy efficiency. Furthermore, the use of heterogeneous integration enables on-chip optical gain through the use of III-V materials, while leveraging the high density and maturity of CMOS foundries [16]. There are several approaches to achieving III-V on Si integration. The most established approach is flip-chip bonding, where a III-V die containing active devices (e.g., a laser, modulator, or photodetector) is directly bonded onto a photonic circuit. The coupling between the two platforms is typically achieved using edge/grating couplers. The key advantage of this method lies in its ability to process devices and circuits on their native substrates while also allowing the integration of pre-selected, known good dies onto the PIC. However, a major drawback of flip-chip bonding is that it is not a scalable technology, leading to higher costs per chip compared to other alternatives [17, 18]. Micro-transfer printing offers a promising alternative, combining the benefits of flip-chip bonding with scalability in manufacturing. Instead of bonding a fully processed chiplet, bare III-V membranes are integrated onto PICs that still require metallization steps. This technique reduces material waste, as the substrate can be reused, and allows pre-testing (at least electrically) of the chiplet membranes. Despite significant progress in the field, this technology still has a way to go before achieving a high technological readiness level (TRL), as reliability remains a concern

for large-scale adoption by major photonic manufacturers [19-21]. Finally, the direct bonding of III-V epitaxial layers onto PICs has emerged as a mature and reliable solution for developing heterogeneously integrated devices. This method involves bonding unprocessed III-V epitaxial layers onto a PIC, followed by InP substrate removal and backend processing to pattern and connect devices electrically. The University of California, Santa Barbara (UCSB) and Intel Labs pioneered the development of III-V/SOI lasers using this approach [22]. This technique has proven to be efficient and sufficiently mature to support the commercialization of Si-based transmitters [23]. It is scalable, as full wafers or multiple dies can be bonded, and it benefits from the compactness and low-loss performance of silicon-based platforms. Figure 1.2 illustrates the three methods mentioned earlier.

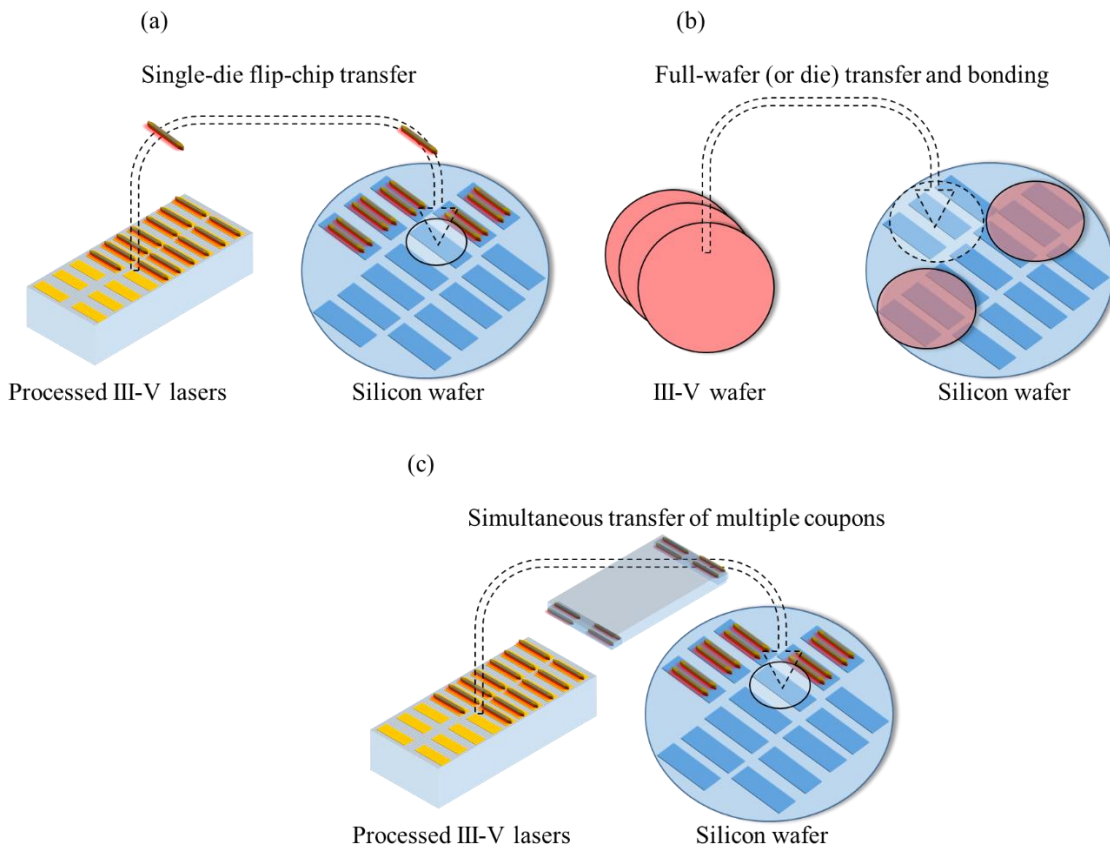


Figure 1.2: Schematic illustration of the three techniques for integrating photonic components: (a) flip-chip bonding, (b) direct bonding, and (c) μ -transfer printing.

In this thesis, we used the direct bonding technique, but with an approach different from that developed at UCSB. In their process, the silicon and III-V wafers are only oxidized prior to bonding, the bonding process used here does not involve any oxide deposits. Furthermore, the silicon waveguide is not encapsulated, there are air trenches on either side of it. They use “non-etched” III-V claddings bonded on patterned Si waveguides. In this case, the current injection area (central area) is defined by ion implantation instead of by etching. Below, we compare two cases: On the left side we can see the UCSB’s approach, with the non-etched III-V lateral claddings, and on the right we are showing the approach we use at the III-V Lab, with a central III-V P-doped mesa flanked by two BCB claddings.

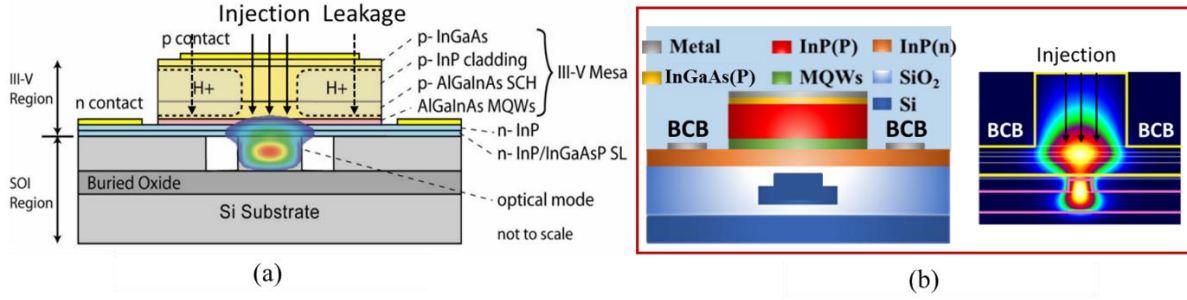


Figure 1.3: Comparisons of two cross-sections: on the left is the UCSB approach, and on the right is the approach used in the III-V lab.

Despite the controllability of ion implantation to create high-resistance III-V claddings, it remains difficult to fully eliminate current leakage from these implanted regions, particularly under high current injection. In our platform silicon is 500 nm thick and buried silicon dioxide (BOX) is usually 2 μm thick.

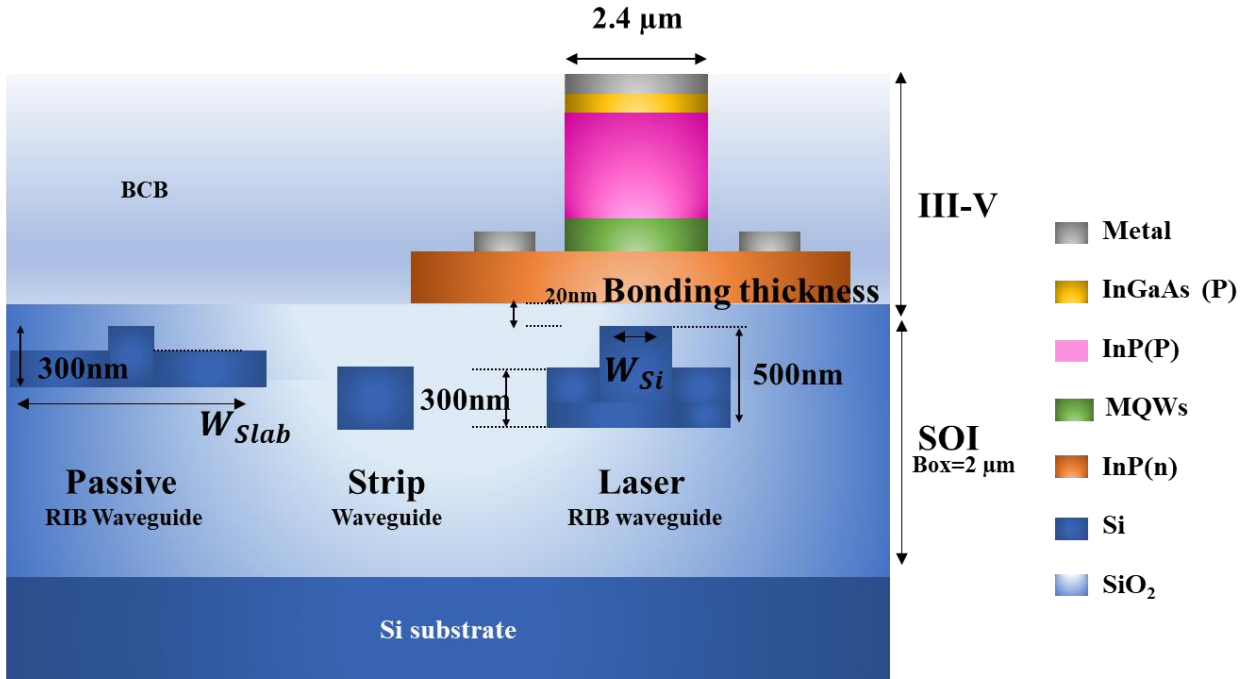


Figure 1.4: Cross-section of a passive, strip waveguide and III-V laser heterogeneously integrated on a silicon photonic circuit.

Depending on the intended application, several common types of silicon waveguides can be patterned. Specifically, three types of silicon waveguides will be used in this thesis: RIB waveguides for lasers, STRIP waveguides, and passive RIB waveguides, as illustrated in Figure 1.4. At III-V Lab, the most frequently used waveguide geometries include RIB waveguides with dimensions of 500/300 nm and 300/150 nm, as well as 300 nm thick STRIP waveguides. The 500/300 nm RIB waveguides are primarily employed for heterogeneous III-V-on-silicon devices, as they enable an efficient mode matching with the III-V waveguide, whereas the thinner waveguides are typically reserved for passive silicon functionalities. In this thesis, the standard technology used by CEA-Leti was slightly modified to enable backside processing for our components as we will see in the next chapter. The width of the silicon waveguide (W_{Si})

is defined using 193 nm deep UV lithography to ensure precise control over its size, as the waveguide's single-mode operation depends on the rib width. If W_{Si} exceeds a cut-off value, both TE^0 and TE^1 modes will be guided in the structure, as described in [24]. Meanwhile, the slab width (W_{slab}) is typically much larger than the mode size, and since it has minimal effect on the mode's properties, it is defined using the less precise 248 nm deep UV lithography [25]. After patterning, the waveguides are encapsulated in SiO_2 . The SiO_2 serves as the bonding oxide, and by adjusting its thickness, we control the coupling gap between the silicon and III-V waveguides. It is then planarized through chemical-mechanical planarization (CMP). The transition of the optical mode from a hybrid waveguide to a silicon waveguide is achieved through tapers using the principle of evanescent coupling. This mechanism is fundamental to heterogeneous III-V-on-silicon platforms, facilitating the transfer of optical mode between different materials. Because the silicon and III-V waveguides have similar refractive indices and are positioned closely, their eigenmodes couple, forming what is described as a unique supermode [26]. In evanescent coupling, the optical mode confined within the silicon waveguide extends into the overlying III-V layer through its evanescent field, allowing efficient transfer of energy and interaction between the two materials. The effectiveness of evanescent coupling depends on the overlap of the optical mode between the silicon and III-V layers, which is typically achieved through precise control of waveguide geometry and the refractive index contrast between the materials. By adjusting the width of the silicon rib waveguide (W_{Si}), it is possible to control the effective index of the silicon waveguide [27]. As the effective index changes, the distribution of the optical mode between the silicon and III-V regions also varies. Simulated field intensities for different silicon rib widths are shown in Figure 1.5, illustrating that the mode distribution is highly controllable. Heterogeneous integration platforms, where the optical mode propagates simultaneously between the silicon and III-V waveguide, differ from hybrid integration approaches, in which the optical signal is confined either within the III-V or silicon waveguide at any given time. This simultaneous mode confinement in both waveguides is crucial for efficiently transferring signal between the III-V gain region and the silicon waveguide without loss (adiabatic).

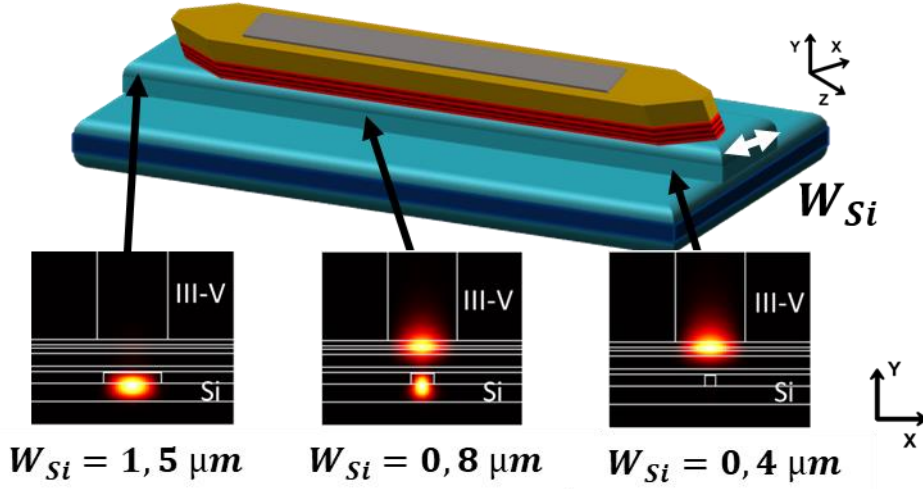


Figure 1.5: III-V on silicon waveguide 3D view. The width W_{Si} of the silicon waveguide controls the confinement in the III-V region. The inset below shows several 2D FDM cross-section simulations of the TE optical mode for different W_{Si} .

The heterogeneous integration combines two photonic platforms to create a third one that harnesses the best of both worlds. This allows us to develop high-performance transmitters, where the active components, such as lasers and modulators, benefit from the superior gain of InP, while the passive components, such as waveguides, leverage the low-loss and scalable nature of silicon photonics. As a result, we achieve efficient and compact transmitters that ambition to meet the demands of high-speed optical communication systems

1.5 III-V/Si component : topologies

There are two main modulation methods for optical transmitters: direct modulation and external modulation, each with its own advantages and disadvantages. Direct modulation, although benefiting from less complex fabrication, energy-efficient, and cost-effective, is susceptible to an undesirable phenomenon known as induced chirp (unwanted frequency modulation). In a directly modulated laser (DML), adiabatic chirp is caused by a frequency variation as a function of power of bias current, while transient chirp results from the laser's relaxation oscillations. To minimize the impact of frequency deviations and support higher data rates, the use of an external modulator, such as an electro-absorption modulator (EAM) or a Mach-Zehnder modulator (MZM), is recommended. However, in access and metropolitan networks, MZMs are associated with higher costs, larger footprints, and increased power consumption. In contrast, EAMs, being more compact, can be directly integrated with lasers, helping to reduce both footprint and insertion optical losses.

1.5.1 Direct modulation

DMLs are key components for short-distance optical interconnect applications [28]. Their simplicity offers a cost-effective alternative to external modulated lasers (EMLs). Intensity Modulation with Direct Detection (IMDD) systems using DMLs as transmitters are the most economical solutions. However, in addition to the previously mentioned limitations, direct

modulation systems are also constrained by a lower extinction ratio compared to externally modulated lasers [29-31]. The primary limitation of DMLs in terms of modulation speed, which makes them slower than external modulators, is their -3 dB electro-optical bandwidth (EOBW). The intrinsic 3-dB bandwidth of DMLs is limited by the relaxation oscillation frequency (f_r), and rarely exceeds 30 GHz [32-36]. Since the modulation speed is approximately given by $1.55 \times 1.3 \times f_r$ [37, 38], their typical modulation speed ranges between 40 and 56 Gbps. Recent demonstrations exceeding 50 Gbps have been achieved with DFB lasers based on the InP platform, operating in both the O-band (1300 nm) and C-band (1550 nm), capable of direct modulation up to 56 Gbps [36, 39]. However, in recent years, silicon photonics (SiPh) technology has emerged as a serious competitor by employing both electrical and optical approaches to extend bandwidth. Various techniques, such as photon-photon resonance [31, 40-42], detuned loading [43, 44], injection locking [45], and push-pull modulation [46], have been implemented to overcome these intrinsic limitations, significantly increasing the 3-dB bandwidth beyond the relaxation oscillation resonance frequency of a standard laser. In the following sections, we will explore some of these techniques.

1.5.1.1 Bandwidth extension

Increasing the modulation speed of DMLs is essential for enabling high-speed optical transmitters. To make DMLs competitive for high-performance, it is crucial to extend their EOBW. The following sections will provide a brief overview of techniques used to increase the bandwidth of DMLs.

Photon-photon resonance

One promising method for extending the electrical-optical bandwidth (EOBW) is the photon-photon resonance (PPR). This phenomenon occurs when the frequencies of photons resonate with specific energy levels in the material. When the frequency of a photon matches the excitatory state of a quantum transition within the material, allowing a collective excitation state, thereby enhancing the efficiency of stimulated emission [47]. This concept is illustrated in Figure 1.6.

In the Intensity Modulation (IM) response of a laser, the first peak is referred to as the carrier-photon resonance peak. A second peak, known as the photon-photon resonance, arises when one of the optical modulation sidebands aligns with a resonance of a compound cavity. This phenomenon can occur in all laser diodes, but for the photon-photon resonance to effectively extend the laser's modulation bandwidth, the second peak must be strong and occur at a frequency close to the first peak. In other words, the carrier-photon resonance must happen at relatively high frequencies, and the frequency separation between the main resonance and the sideband mode should be small. Achieving a high-frequency first peak typically requires short cavity designs and high bias currents.

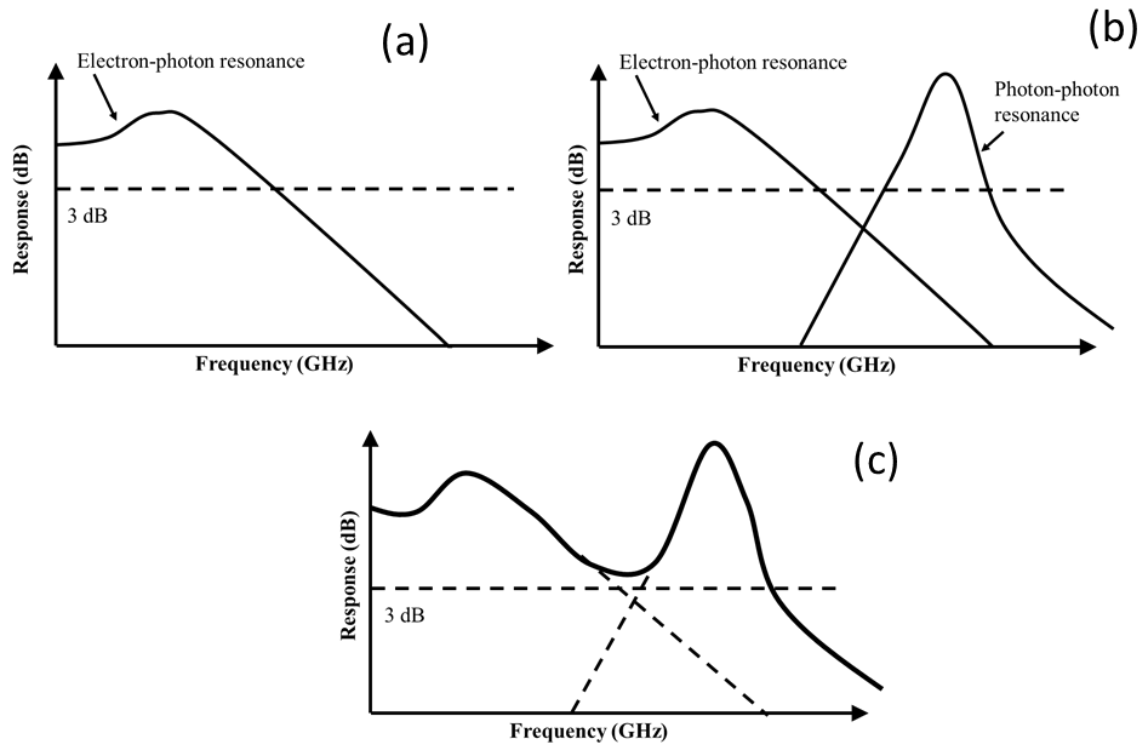


Figure 1.6: Diagrammatic representation showing the positions of (a) electron-photon and (b) photon-photon resonances within the modulation transfer function (right and left sections). (c) The lower section depicts a scenario in which the modulation bandwidth significantly expands due to the overlap of these two resonances (figure adapted from [49]).

The position of the second resonance peak (PPR) can be precisely adjusted by introducing an external cavity with an optimally chosen optical length. Based on the Fabry-Perot cavity model, a longer external cavity reduces the frequency separation between the main and side modes, bringing them closer in resonance and thereby improving the modulation bandwidth of the laser. Recently, S. Yamaoka et al. [48] demonstrated a directly modulated membrane laser with a 3-dB bandwidth exceeding 110 GHz with an active region length of just 50 μm . Additionally, measurements at high temperatures revealed a moderate bandwidth reduction, with the 3 dB bandwidth remaining at 74 GHz even at 85°C. It is crucial to note that the injection current and the cavity length significantly impacts the modulation bandwidth. High injection currents are often avoided due to the risks of thermal heating within the laser cavity, which can degrade performance. Effective thermal management, therefore, becomes a critical factor in optimizing the performance of photonic devices. This was particularly evident in [48], where a III-V DML was fabricated on a high thermal conductivity silicon carbide (SiC) substrate. While SiC offers superior thermal performance compared to silicon, it is more costly to produce.

Detuned loading

Detuned loading is another technique employed to extend the bandwidth of photonic devices [43, 50]. This approach involves intentionally offsetting the resonant frequency of the optical components to create a condition where the system can respond to a broader range of frequencies. By strategically designing the optical cavity or waveguide structures, the device can be tuned to operate at frequencies that are not directly aligned with its natural resonance. In III-V/Si lasers, detuned loading can be implemented by adjusting the geometric parameters

or the refractive index profile of the waveguide [51]. This results in an enhanced interaction between the signal and the active material, allowing for a wider operational bandwidth. The benefits of this technique include improved modulation capabilities and the ability to accommodate high-speed data rates without significant degradation of signal quality.

1.5.2 External Modulation

Although the bandwidth of DML has been improved through the previously mentioned techniques, external modulation offers key advantages. It provides higher bandwidth, reduced noise, and improved extinction ratios. Unlike DMLs, where modulation is integrated, external modulation utilizes a separate device, allowing for better control. Commonly used devices include EAM and MZM. These devices can be monolithically integrated with lasers, creating EML. Recently, A. Uchiyama et al. demonstrated a lumped EAM integrated with an InP-based DFB laser, achieving a bandwidth of 85 GHz for modulation suitable for 400 Gbps operations using multi-level modulation [52]. Additionally, an EML with a traveling wave (TW) modulator configuration demonstrated a bandwidth of 100 GHz [53, 54]. The lumped EAM is particularly advantageous for high-density circuits, allowing the integration of more components on a single chip. Another study showed lumped EML chips with a bandwidth of 93 GHz, producing clear optical eye diagrams at 400 Gbps [55].

In spite of these breakthroughs, transmitters based on the InP monolithic platform face significant challenges in cost reduction, scalability, and integration with other components or materials.

1.6 Outline

The thesis work will focus on the study of III-V/Si high-speed transmitters, aiming to enhance the intrinsic bandwidth and data rate of these devices. The ultimate goal is to demonstrate a compact III-V/Si high-speed transmitter, which could serve as a cost-effective solution for DWDM in metropolitan and access networks. To achieve this, the work is divided into two parts. The first part will involve the development of a directly modulated laser (DML), specifically a directly modulated DFB laser with a wide intrinsic bandwidth. The second part will focus on the demonstration of an external modulated laser (EML).

The thesis is organized into three chapters. The second chapter serves as an introduction, presenting various aspects of the heterogeneously integrated III-V-on-Silicon DFB laser. Initially, we will cover the theoretical foundations of DFB lasers, relying on rate equations and coupled mode theory. In the second part, we will investigate the impact of the coupling coefficient (κ) on this type of component through simulation bias. We will propose a solution to reduce it by leveraging the properties of the III-V/Si heterogeneous platform by etching the Bragg grating at the backside of the silicon waveguide. We will conclude this chapter by improving the intensity distribution within the laser cavity, proposing a sample distribution of the Bragg grating and comparing it to that of a conventional $\lambda/4$ DFB laser. These two types of DFB lasers will be studied subsequently.

The third chapter will focus on the experimental study of DFB lasers to demonstrate a high-speed directly modulated heterogeneously integrated DFB laser, utilizing the theoretical knowledge gained in the previous chapter. The chapter will begin with the static

characterization of the two types of DFB lasers. We successfully demonstrated 32 Gbps NRZ-OOK direct modulation and transmission over a 2 km standard single-mode fiber (SSMF), achieving an intrinsic modulation bandwidth of up to 21 GHz. By using advanced modulation formats, we demonstrated 4-level Pulse Amplitude Modulation (PAM-4) formats at 50 Gbps. The remainder of the chapter will explore the robustness of these components against optical feedback. We successfully demonstrated significant optical feedback tolerance of DFB lasers for high-speed direct modulation.

The fourth chapter will be dedicated to the development of an EML using an EAM. We will first introduce the operational principles of this modulator. Next, we will design a lumped EAM and we will initially integrate it with the previously studied DFB lasers. To address the insertion losses induced by the EAM, we will subsequently integrate a semiconductor optical amplifier (SOA). To facilitate fabrication, the co-integrated components will share the same III-V epitaxial stacks, albeit with a different InP mesa for each component. Once this transmitter is validated, we will focus on the tunable laser. We will begin by studying the design and evaluate static performance through experimentation. Subsequently, we will demonstrate a compact III-V/Si high-speed transmitter for DWDM, consisting of a tunable laser, an EAM, and an SOA sharing the same mesa. We successfully demonstrated 56 Gbps NRZ-OOK transmission over 10 km through SSMF.

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2. Distributed Feedback Lasers

2.1 Introduction to DFB Lasers

Since Maiman's demonstration of the first laser in the early 1960s [1], semiconductor lasers have undergone remarkable advancements. The progress in material purity and epitaxial growth techniques has led to a variety of semiconductor lasers covering a wide range of wavelengths. Semiconductor laser diodes are optoelectronic devices that emit coherent signal when electrons and holes recombine in a semiconductor PIN junction. To achieve a laser effect, it is essential that lasers incorporate a gain medium (active layer) capable of generating and amplifying signal, along with an optical resonator providing the necessary feedback to sustain laser oscillation. A fundamental condition for achieving laser action is the interaction of the amplifying medium with radiation, where gain is created through stimulated emission. This involves the recombination of an electron in the conduction band with a hole in the valence band [2, 3]. During this process, energy and momentum must be conserved. The energy difference between these two states determines the energy of the emitted photon, in accordance with energy conservation. Momentum conservation, on the other hand, requires the use of direct bandgap materials. In these direct-gap semiconductors, the energy bands exhibit an extremum for the same wave vector $\vec{k}$, vertically aligning the minimum of the conduction band with the maximum of the valence band in reciprocal space. Direct bandgap materials are typically III-V semiconductor compounds such as AlGaAs, InGaAs, and InGaAsP. The bandgap of these materials can be adjusted by varying the composition of ternary and quaternary compounds, thereby allowing the modification of the operating wavelength and covering a large portion of the visible, near-infrared, and mid-infrared spectrum. A direct transition between the conduction and valence bands allows the emission of a photon with energy:

$$\hbar\omega = E_c - E_v = E_g \quad (2 - 1)$$

Where $\hbar$ is the reduced Planck's constant and ω the angular frequency defined as $2\pi\nu$. E_v and E_c are the energy levels of electrons in the valence and conduction bands, respectively, and E_g represents the bandgap width.

Optical amplification is achieved only when population inversion is produced by injecting excited carriers, favoring stimulated emission over absorption. For the material to amplify photons, they must satisfy the Bernard-Durrafourg condition [4]. Due to their compactness, efficiency, and ability to emit signal over a wide range of wavelengths, semiconductor laser diodes are essential in numerous fields, from telecommunications to medicine, as well as scientific instrumentation and information technology. Their small footprint also makes them ideal for integration into high-speed, low-power photonic circuits.

However, the multimodal operation of Fabry-Perot (FP) lasers presents a major challenge for optical communication applications due to the chromatic dispersion of optical fibers. The reflectivity on the facets of the FP cavity is identical for all longitudinal modes, with spectral discrimination only arising from the gain curve's width and mode competition, drawing from the same carrier reservoir. The wavelengths λ of these longitudinal modes satisfy the phase condition

$$m\lambda = 2n_e(\lambda)L \quad (2 - 2)$$

Where L is the cavity length and $n_e(\lambda)$ the effective refractive index. The gain material's spectral width typically spans several longitudinal modes, resulting in a laser spectrum with multiple peaks, indicating multimode operation. A schematic representation of a simple Fabry-Perot laser with two cleaved facets and its lasing spectrum is illustrated in Figure 2.1-a. To achieve mode selection, wavelength-dependent feedback is necessary, which involves different intracavity losses for various longitudinal modes. This concept, proposed by Kogelnic and Shank in 1971 [5, 6], has been implemented in distributed feedback (DFB) lasers. These lasers are unique in that signal feedback is no longer provided by terminal facets but by a periodic variation in the thickness or width of a waveguide layer, introducing a periodic change in the effective refractive index along the entire length of the laser, similar to a Bragg grating (Figure 2.1-b). The successive reflections on these layers create coupling between forward and backward propagating waves, similar to the reflections in a Fabry-Perot structure. The emission wavelength can be tuned by slightly adjusting the grating period or the effective index of the waveguide. The mode with the lowest losses will dominate. Since their initial demonstration, DFB lasers have been continuously optimized in terms of power, bandwidth, and spectral stability, becoming key components in modern optical communication systems [7-10]. The fundamental working principle of DFB lasers revolves around the Bragg condition, given by:

$$\lambda_B = \frac{2n_{eff}\Lambda}{m} \quad (2 - 3)$$

where λ_B is the Bragg wavelength, n_{eff} is the effective refractive index of the mode, m is the bragg order and Λ is the grating period. This condition ensures that signal with a wavelength matching λ_B experiences constructive interference, resulting in strong feedback and efficient lasing at this wavelength.

Distributed reflectivity (DR) induces coupling between counterpropagating waves within the laser cavity. This coupling is characterized by a factor denoted κ , expressed in cm^{-1} , and primarily determined by the refractive index variation within the medium. The Bragg grating coupling factor, typically expressed as the product κL where L is the cavity length, is a crucial parameter for characterizing DFB lasers [5]:

$$\kappa L = \frac{\pi\Delta n}{\lambda} L \quad (2 - 4)$$

Where Δn represents the maximum refractive index difference of the grating relative to the average refractive index n_{eff} , and λ is the laser emission wavelength. This factor will be discussed in more detail later in the chapter.

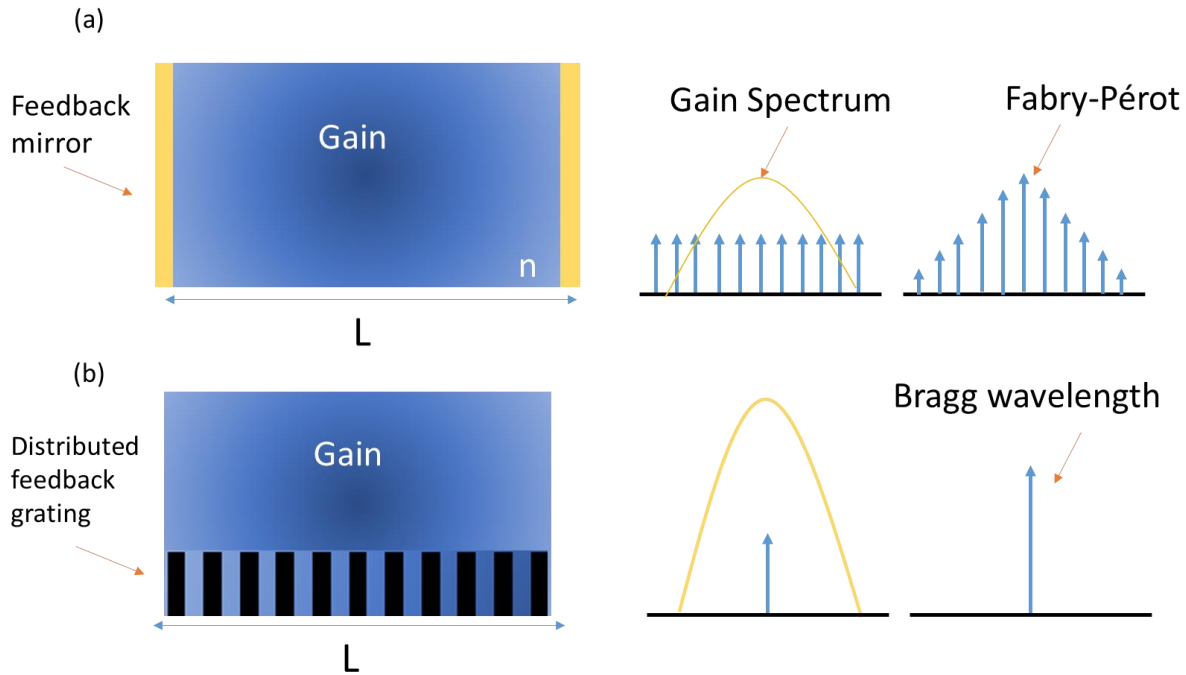


Figure 2.1 : Schematic representation of a Fabry-Perot laser cavity with its lasing spectrum (a) and for DFB laser cavity with its lasing spectrum (b).

The distributed nature of the feedback allows for stable single-mode operation, which is essential for many applications. In optical communication systems, their narrow linewidth and stable single-mode operation minimize chromatic dispersion and crosstalk in optical fibers, ensuring signal integrity over long distances, which is critical for maintaining high-speed data transmission [11]. In DWDM systems DFB lasers are the primary signal sources for each channel, allowing multiple wavelengths to be multiplexed onto a single optical fiber, thereby significantly enhancing the data throughput [12]. Their precise wavelength control supports advanced modulation formats like Quadrature Amplitude Modulation (QAM) and Differential Phase Shift Keying (DPSK).

This chapter delves into the realm of silicon-integrated DFB lasers, exploring the challenges associated with optimizing their performance. To gain a comprehensive understanding of DFB laser performance, we will delve into two key models: rate equation theory and coupled-mode theory. These will serve as our tools to analyze the fundamental characteristics of DFB lasers.

2.2 Rate equation theory

Rate equations are a fundamental tool for understanding and modeling the behavior of semiconductor lasers. These equations describe the dynamic relationship between photon density and electron density within the laser cavity. The primary aim is to maintain a balance between photon generation and electron-hole recombination, ensuring particle and energy conservation. There are two principal approaches to modeling semiconductor laser operation: the photon/electron particle exchange method and the electromagnetic field interaction method. The former focuses on the rates of photon absorption and emission, and electron-hole recombination, while ensuring detailed balance at every stage. These rate equation

approaches, although robust, are well documented [10, 13-15] which are crucial for analyzing both static and dynamic properties of the device.

The rate of change of photon density S in a laser cavity is given by the equation:

$$\frac{dS}{dt} = \Gamma G(N)S - \frac{S}{\tau_p} + \beta_{sp} \frac{N}{\tau_{sp}} \quad (2-5)$$

Where, Γ is the confinement factor, representing the fraction of the optical mode that overlaps with the active region, $G(N)$ is the optical power gain coefficient, which is a function of the electron density N and τ_p is the photon lifetime, representing the average time a photon stays in the cavity before escaping. β_{sp} is the spontaneous emission factor, indicating the fraction of spontaneous emission that contributes to the lasing mode and τ_{sp} is the spontaneous emission lifetime.

The first term on the right-hand side represents the stimulated emission of photons, the second term accounts for photon losses from the cavity, and the third term represents the contribution of spontaneous emission to the lasing mode. The losses occurring within the cavity are determined by the photon lifetime τ_p and can be separated into two components :

$$\frac{1}{\tau_p} = v_g(\alpha_{int} + \alpha_{mirror}) \quad (2-6)$$

Where v_g the group velocity, α_{int} the internal cavity losses and α_{mirror} is the mirror loss. The carrier density N in the active region of a DFB laser evolves according to several mechanisms, represented by the rate equation:

$$\frac{dN}{dt} = \frac{I}{qV} - \frac{N}{\tau_r} - G(N)S \quad (2-7)$$

Here, I is the injection current, τ_r the carrier recombination lifetime, q the electronic charge, V the active region volume.

The first term on the right-hand side represents the rate of electron injection into the conduction band. The second term represents the spontaneous recombination rate, which includes radiative and non-radiative recombination processes. The third term accounts for the stimulated recombination due to lasing.

In semiconductor lasers, maintaining a detailed balance between photon generation and electron-hole recombination is crucial. The recombination rate can be expressed as:

$$\frac{N}{\tau_r} = AN + BN^2 + CN^3 \quad (2-8)$$

The AN term represents non-radiative recombination, typically due to defects. The BN^2 term represents bimolecular (radiative) recombination, where B is the bimolecular recombination coefficient. And the CN^3 term represents Auger recombination, a non-radiative process involving three carriers.

In practical applications, several factors can introduce modifications to the rate equations governing semiconductor lasers. One significant factor is gain compression, which occurs at high photon densities. This phenomenon is often approximated using the gain compression factor, ε , where the gain function $G(N) = (\frac{G_0}{1+\varepsilon S})$, reflects reduced gain with increasing photon density. Another consideration is carrier transport delay, influencing the modulation speed particularly in quantum well lasers, as carriers take time to reach the active region. Additionally, the assumption of charge neutrality ($N \approx P$, where P is the hole density) simplifies analysis by ensuring electrical neutrality within the device. These factors collectively contribute to the complexity and accuracy of modeling laser behavior in real-world conditions. For DFB lasers, where the refractive index or permittivity changes periodically, coupled-mode equations are essential. These equations describe how a fraction of the forward-traveling field is coupled into the reverse-traveling field in the presence of a periodic structure, such as a Bragg grating. These equations will be further explored in the next section.

2.3 Coupled mode theory

The interaction of coupled modes is described by the Coupled Mode Theory (CMT), which is derived from perturbation theory. This theory is extensively detailed in numerous books [13, 16, 17] and publications [18, 19]. CMT is essential for analyzing the interaction between forward and backward propagating waves in the periodic structure of a Distributed Feedback (DFB) laser. It enables the management of a complex feedback system, such as a DFB laser, where the carrier density and photon density depend on the longitudinal coordinate. This method relies on Maxwell's equations, coupled with the carrier rate equations, to examine the static and dynamic characteristics of the laser along its length.

The field inside the laser diode can be written as the sum of a propagating wave $A(z)$ and contra-propagating wave $B(z)$:

$$E(z) = A(z) \exp(j\beta z) + B(z) \exp(-j\beta z) \quad (2 - 9)$$

With β the wave propagation constant. The grating introduces coupling between the forward and backward propagating waves. The reflected waves constructively interfere at the Bragg wavelength. Typically, this coupling is described by the coupling coefficient κ , which is proportional to the variation in the effective index Δn . The spatial evolution of the forward and backward propagating waves can be described by the coupled mode equations as [21, 22]:

$$\frac{dA(z)}{dz} = (\gamma - j\Delta\beta)A(z) - j\kappa B(z) \quad (2 - 10)$$

$$\frac{dB(z)}{dz} = -(\gamma - j\Delta\beta)B(z) - j\kappa A(z) \quad (2 - 11)$$

where $A(z)$ and $B(z)$ are the amplitudes of the forward and backward propagating waves, κ is the coupling coefficient, z is the position along the waveguide, γ the gain for the field in a DFB laser with non-reflecting facets and $\Delta\beta = \frac{2\pi n_{eff}}{\lambda} - \frac{\pi}{\Lambda}$ the Bragg deviation, with Λ being the grating period.

These equations constitute a set of two coupled, linear, first-order, ordinary differential equations for the functions $A(z)$ and $B(z)$. They can be solved if we know A and B at a given point, for example $z = z_0$. Within the Transfer Matrix Method (TMM) formulation of coupled mode theory [20], the propagating and contra-propagating waves amplitudes $A(z)$ and $B(z)$ at the end of the structure are the product of the device matrix $\bar{\bar{M}}$ times the initial amplitudes $A(z_0)$ and $B(z_0)$:

$$\begin{pmatrix} A(z) \\ B(z) \end{pmatrix} = \bar{\bar{M}} \begin{pmatrix} A(z_0) \\ B(z_0) \end{pmatrix} \quad (2 - 12)$$

For a given κL product, solutions of those equations are the allowed wavelengths and the corresponding threshold gain α_{th} . The general relation between the fields at a point z and the fields at a point z_0 can be written as:

$$\bar{\bar{M}} = \begin{pmatrix} \cosh(\sigma z) + \frac{\gamma - j\Delta\beta}{\sigma} \sinh(\sigma z) & -\frac{j\kappa}{\sigma} \sinh(\sigma z) \\ \frac{j\kappa}{\sigma} \sinh(\sigma z) & \cosh(\sigma z) - \frac{\gamma - j\Delta\beta}{\sigma} \sinh(\sigma z) \end{pmatrix} \quad (2 - 13)$$

With the complex propagation constant σ equal to :

$$\sigma = \sqrt{(\gamma - j\Delta\beta)^2 + \kappa^2} \quad (2 - 14)$$

In the simplest scenarios, we can solve these equations by knowing the boundary conditions. Assuming both facets are anti-reflective (AR) coated, making the reflectivity of each mirror zero, and setting $z = L$ and $z_0 = 0$, we apply the boundary condition and the oscillation condition is given by [20]

$$\bar{\bar{M}}_{22} = 0 \quad (2 - 15)$$

This equation can be solved numerically to determine the oscillating modes in a gain versus detuning curve. To solve this, we have plotted in Figure 2.2 the normalized threshold gain versus the Bragg deviation $\Delta\beta L$ for different optical modes. Such a configuration leads to a dual-frequency laser because the modes on either side of the stopband have the same threshold gain.

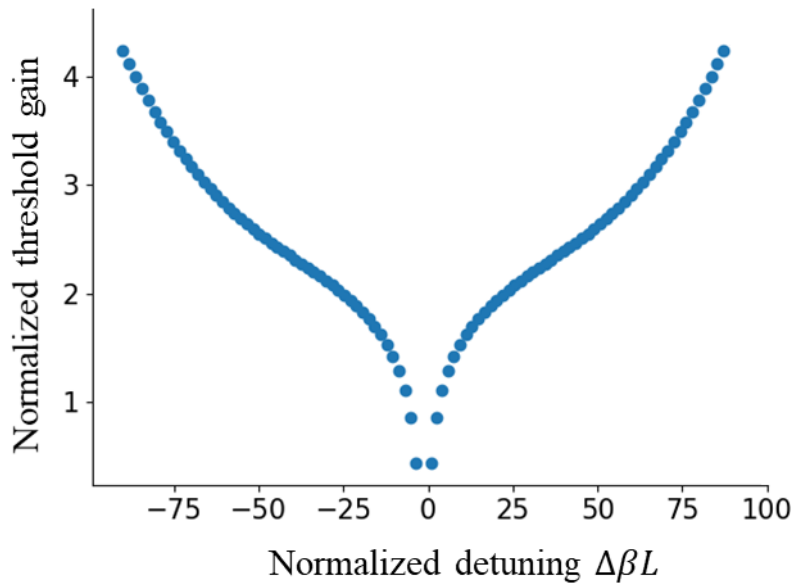


Figure 2.2: Normalized threshold gain vs. normalized detuning for uniform DFB laser.

The stopband is an important parameter because it provides insight into the laser's coupling coefficient, which influences the mode selection and stability of the DFB laser. It's indicated by the spacing between the two main modes. The stopband is a range of wavelengths or frequencies within which signal cannot propagate through the periodic structure of a DFB laser [22]. This phenomenon is a direct result of the periodic modulation of the refractive index within the grating. When signal encounters this periodic structure, certain wavelengths are reflected constructively, overall performance of the laser. The width of the stopband is directly related to the coupling coefficient κ , which represents the strength of the interaction between forward and backward propagating waves within the grating. A larger coupling coefficient generally results in a wider stopband, indicating stronger reflection and more robust feedback for lasing action. The coupling coefficient κ will be discussed in more detail in Section 2.5.1. For the Bragg grating without loss or gain, the relation between coupling coefficient and stopband width $\Delta\lambda$ can be expressed as:

$$\Delta\lambda = \frac{\lambda_B^2}{\pi n_{eff} L} \sqrt{(\kappa L)^2 + \pi^2} \quad (2 - 16)$$

There are several methods to achieve a single-mode DFB laser. Mode degeneracy can be eliminated by gain coupling [5] or by adding facet reflectivity [22]. Another effective technique involves introducing a phase shift in the middle of the grating to facilitate lasing at the Bragg frequency [23, 24], which is the simplest and most systematic approach to achieving single-mode operation. In this context, the most fundamental method is to introduce a $\lambda/4$ phase shift at the center of the DFB cavity [25], while ensuring there are no reflections in the integrated circuit.

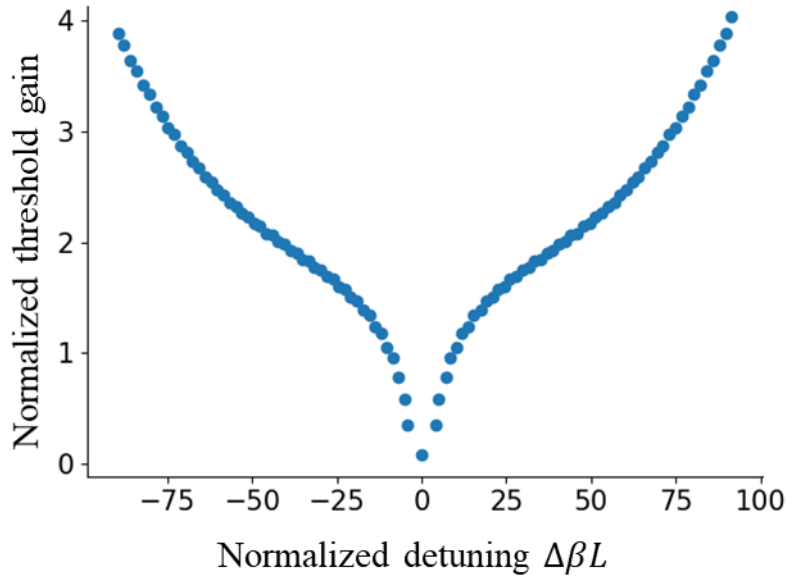


Figure 2.3: Normalized threshold gain vs. normalized detuning for phase shifted DFB laser.

The structure with a $\lambda/4$ phase shift exhibits a unique fundamental mode at the Bragg wavelength, with lower losses compared to other modes, as sketched in figure 2.3, thus favoring its predominance at low polarization. For our DFB lasers fabricated on the III-V on silicon platform, the coupling grating is etched onto the silicon waveguide, where the mastery of CMOS technology enables high-precision etching, allowing the use of lasers with a $\lambda/4$ phase shift. These phase-shifted lasers offer a high threshold gain margin because the phase shift enhances mode discrimination, reduces mirror losses for the fundamental mode, and suppresses side modes more effectively. This results in an SMSR greater than 50 dB. However, far above the threshold, multimode operation can sometimes occur due to the phenomenon of longitudinal spatial hole burning [25].

Coupled Mode Theory also helps in understanding the effect of grating chirp and apodization on the performance of DFB lasers. Chirped gratings, where the grating period varies along the length of the laser, can broaden the reflection bandwidth, improving the laser's tolerance to temperature and fabrication variations. Apodization, where the grating strength is gradually varied, can reduce side-lobe reflections and improve the spectral purity of the laser output. In this section, I have introduced the Coupled Mode Theory and the transfer matrix method. In the next section, I will present the design of heterogeneously integrated DFB lasers on the III-V on silicon platform.

2.4 Heterogeneous Integration

Although Si and its oxide (SiO_2) can be effectively used to create compact passive optical components, the fabrication of lasers on this platform presents a more complex challenge. This challenge primarily arises from the indirect nature of silicon's electronic band structure. As a result, only a small fraction of electrons in the conduction band generate radiative

transitions when returning to a less excited state, making it an unsuitable candidate for signal generation.

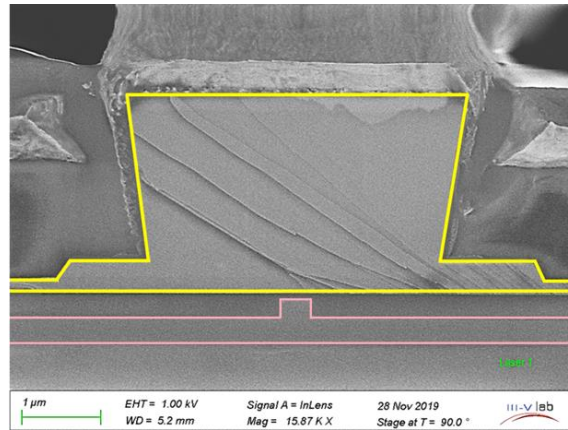


Figure 2.4: Cross-sectional scanning electron microscope (SEM) image of a III-V heterostructure integrated on silicon. The active region (highlighted in yellow) is deposited on top of the silicon waveguide (outlined in pink).

Faced with this obstacle, several strategies have been developed to efficiently couple signal with silicon photonic circuits. One of these strategies involves integrating III-V materials family, directly onto silicon. To design III-V chips on silicon, the direct bonding heterogeneous integration method [27] offers several advantages, including a wide variety of design possibilities, allowing for the integration of a significant number of components on a single chip. The heterogeneous integration via wafer/die bonding method involves bonding an untreated III-V epilayer stack onto a pre-processed silicon wafer containing the passive photonic circuits. A thin bonding layer (5-100 nm) is used [28]. This technique efficiently couples optical mode emitted by the III-V components to silicon waveguides. The fabrication of III-V components is carried out after the bonding process. This means there is no additional alignment issues between the two parts other than the intrinsic alignment precision of the lithographic equipment, as the second part is manufactured directly above the first, as shown in the SEM image of a DFB in Figure 2.4. The final chip consists of two distinct parts: a silicon part containing the passive components and a III-V part containing the active medium. A colored cross-section is illustrated in Figure 2.4, showing two stacked waveguides. The silicon waveguide is outlined in magenta, while the III-V waveguide containing the QW responsible for the optical gain is in yellow. Bonding oxide separates the two. Evanescent coupling occurs between the two waveguides, and this coupling is adiabatic in nature, meaning there is no energy loss. Adiabatic coupling is achieved by adjusting the dimensions of each waveguide to achieve a mode effective index crossover for a specific combination of waveguide widths. The details of this process will be discussed in the next section.

2.4.1 DFB structure

When carrier injection is induced in the III-V waveguide, an optical mode is generated and partially confined in both the III-V and silicon waveguides, forming a hybrid mode. The III-V stack ensures optical gain, while the mode is also present in the silicon waveguide, which contains the Bragg grating. To extract the optical mode, it is coupled into the silicon waveguide, where it can propagate with minimal loss, as illustrated in figure 2.5. For efficient coupling of the optical mode between waveguides, we use adiabatic tapers. By gradually

adjusting the width of the silicon waveguide adiabatically, the optical mode can be efficiently transferred between the two waveguides. This adiabatic coupling method between III-V materials and silicon was proposed by A. Yariv in 2007 [29, 30]. This type of coupling offers a considerably broader spectral bandwidth compared to other coupling devices such as directional couplers. Figure 2.6 illustrates the optical simulation of the taper transition, demonstrating both the longitudinal propagation of the optical mode and the cross-section (insets) for different levels of optical confinement in the quantum wells (Γ_{QWS}). We represent the case of a mode that is tightly confined in the III-V region initially and coupled into the silicon waveguide across three figures, showing the transition of the optical mode from the III-V waveguide to silicon for three different Γ_{QWS} . Figure 2.6-a shows an optical simulation of the propagation of the optical mode coupling for $\Gamma_{QWS} = 15\%$. Figure 2.6-b presents a scenario with a reduced optical confinement factor of $\Gamma_{QWS} = 10\%$. Here, the optical mode is less confined in the III-V region and begins to spread more into the silicon waveguide. Figure 2.6-c depicts an even lower optical confinement of $\Gamma_{QWS} = 5\%$, where the optical mode is predominantly confined in the silicon waveguide, resulting in only weak coupling to the III-V region. This highlights the critical role of adiabatic tapers in controlling optical mode confinement and ensuring efficient mode transfer between the III-V and silicon waveguides. Additionally, locating the mode tapers outside the laser cavity contributes to reducing the optical intra-cavity losses in the laser (notice that this is not the case for III-V-on-SOI extended-cavity lasers, where the tapers are defined inside the laser cavity, hence contributing to the internal loss of the device).

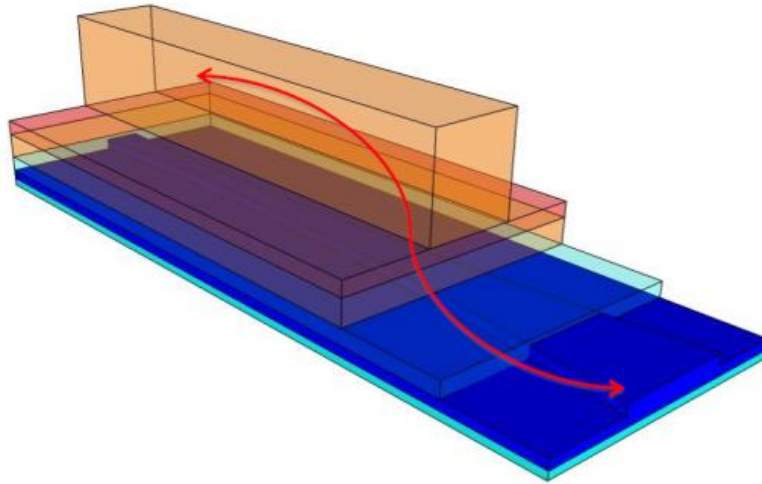


Figure 2.5: Schematic 3D representation of the heterogeneously integrated DFB laser.

A typical heterogeneously integrated DFB laser on the Si-on-insulator (SOI) platform is shown schematically in Figure 2.5. In the case of traditional III-V DFB lasers, the formation of the Bragg grating involves a series of successive etching and regrowth steps during epitaxy, a complex choice due to various technological constraints. However, in the context of III-V-on-silicon heterogeneous lasers, the Bragg grating formation occurs within the silicon region, leveraging the maturity of SOI processes and the high index contrast between Si and SiO_2 , thus simplifying the fabrication process of the III-V devices as no regrowth steps are required.

Moreover, the hybrid architecture offers the additional advantage of structuring the silicon waveguide beneath the III-V waveguide, improving mode coupling between the two materials.

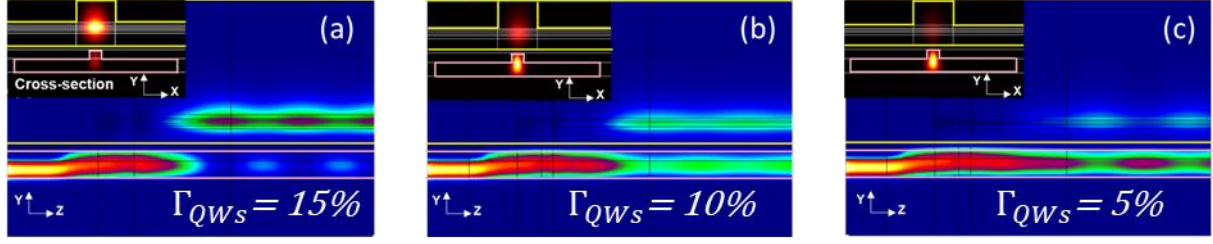


Figure 2.6: Optical simulations of the TE mode propagation for different quantum well confinement factors. (a) $\Gamma_{QWs} = 15\%$, (b) $\Gamma_{QWs} = 10\%$, and (c) $\Gamma_{QWs} = 5\%$. The main panels show the field distribution along the y-z plane, while the insets illustrate the corresponding 2D FDM cross-section simulations in the y-x plane.

An earlier batch of III-V/Si DFB lasers was produced at III-V Lab with a Bragg grating etched in the upper part of the laser silicon waveguide [28]. However, with this approach, we observe a strong coupling of the optical mode in the silicon waveguide, leading to increased confinement in the bragg grating, which in turn increases the index difference and causes an increase in the κ .

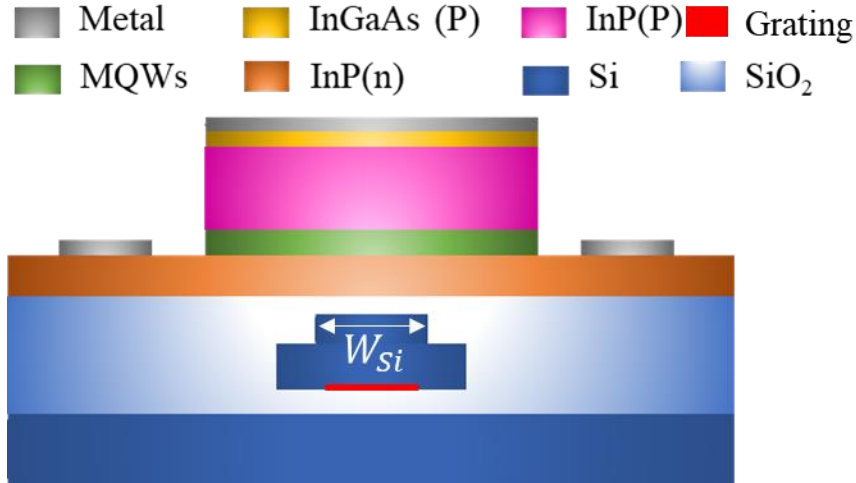


Figure 2.7: III-V on silicon DFB structure: Transverse cross section.

As we will see in section 2.5, in this thesis work the Bragg gratings are designed at the backside of the Si waveguide, achieving a low κ . Thanks to this low grating strength, we can increase the length of the laser cavities without increasing the κL value. Transverse cross section is shown in figure 2.7. With the Bragg grating located in the lower part of the silicon waveguide, the optical mode is attracted to it. This is especially true when the silicon waveguide width (W_{Si}) is too wide. W_{Si} defined in Figure 2.7 controls the mode properties. As a result, we end up with an optical mode fully confined within the silicon waveguide, with no overlap with the quantum wells. To avoid this, we have implemented optical mode engineering, which will be discussed in the next section, to achieve an optimal balance between the confinement of the optical mode in the quantum wells and maintaining a low grating strength (κ).

2.4.2 Optical design

The optical mode confinement in QWs is a critical parameter for achieving high-speed direct modulation lasers. It directly affects the relaxation oscillation frequency formula and influences parameters such as the threshold carrier density and the differential gain (section 3.3.1). Variation in W_{Si} and the bonding layer thickness during fabrication provides flexibility to adjust the confinement factor in both QWs and Si waveguides. The evolution of Γ_{QWs} as a function of silicon waveguide width for different bonding layer thickness is illustrated in Figure 2.8.

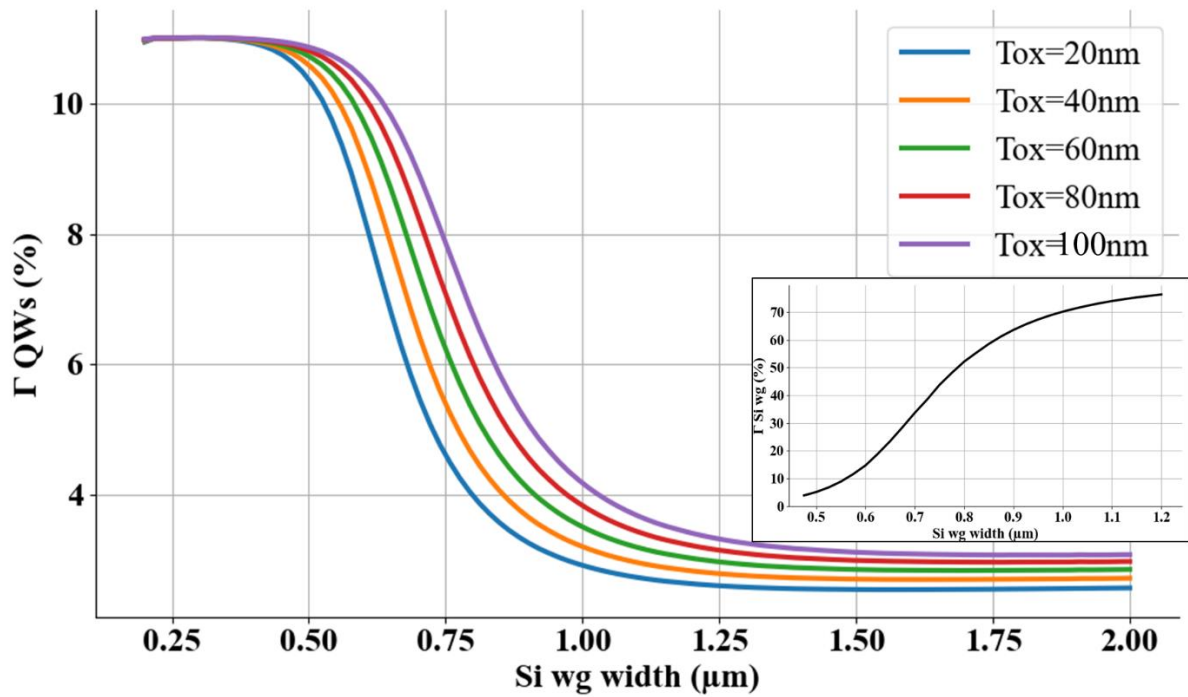


Figure 2.8: The simulated confinement factor in QW as a function of silicon waveguide width (W_{Si}) for different bonding layer thicknesses. The inset shows the silicon waveguide confinement factor versus silicon waveguide width for a bonding layer of 20 nm.

As the bonding layer thickness increases, the mode shifts towards the active waveguide, thereby increasing confinement in the active region. For example, with a bonding layer thickness of 20 nm, the confinement factor in quantum wells reaches 10.5% for or a waveguide width W_{Si} of 0.5 μm (1.75% per QW). In this scenario, the mode predominantly resides in the III-V active waveguide, while the confinement factor in the Si waveguide decreases to approximately 6% (see inset). Due to the p-doped InP layer introducing significant internal optical losses (Inter-Valence Band Absorption, IVBA), it is imperative to minimize these losses by reducing optical mode overlap in this layer. This necessitates decreasing the bonding layer thickness to enhance confinement in Si waveguides, which exhibit significantly lower propagation losses compared to III-V waveguides (more than an order of magnitude difference). While ensuring optical mode overlap with the gain section, we selected a bonding layer thickness of approximately 20-25 nm for our DFB laser.

Additionally, the inset in Figure 2.8 depicts the evolution of Γ_{Si} as a function of Si waveguide width, demonstrating an inverse trend. Narrow Silicon waveguide results in strong confinement in III-V waveguide, whereas wider silicon waveguides confine the mode within silicon. To better understand the choice of W_{Si} , we compared two different Si waveguide widths ($W_{Si} = 0.7 \mu m$ and $W_{Si} = 0.82 \mu m$). The wider Si waveguide nearly doubles Γ_{Si} (55% vs. 33%) and provides a Γ_{QWS} of 4.5% (6% for the design with $W_{Si} = 0.7 \mu m$). Consequently, these parameters yield a $\Gamma_{QWS} / \Gamma_{Si}$ ratio 3 times lower for the laser design with W_{Si} of 0.825 μm compared to that with W_{Si} of 0.7 μm . Hence, we opted for a W_{Si} of 0.825 μm for our DFB laser.

2.4.3 Process flow

The silicon manufacturing process is conducted at CEA-Leti, while the III-V wafers are epitaxially grown at III-V Lab in Palaiseau, as well as the the III-V back-end-of-line (BEOL). The III-V-on-Si DFB lasers were fabricated using a heterogeneous platform consisting of a reverse III-V multilayer PIN structure (total thickness $\approx 3 \mu m$) with a PIN junction composed of a P-doped InP layer for current injection. The intrinsic region includes quaternary AlGaInAs quantum wells (MQWs) for gain, sandwiched between two separate confinement heterostructures (SCH), and a N-doped InP layer for current collection from the lateral electrodes. Such N-doped layer, if moderately doped, does not introduce massive propagation losses in the structure. These InP-based layers are grown using metal organic chemical vapor phase epitaxy (MOVPE). A 2 μm -thick Silicon On Insulator (SOI) wafer containing passive photonic circuits (waveguides, adiabatic tapers and grating couplers) is planarized and bonded to the III-V wafer using a direct bonding process. The integration of these circuits with the III-V multilayer can be summarized in the following six distinct steps (see Figure 2.9):

- i. SOI preparation & III-V preparation: As this involves a backside process, we begin with etching the Bragg grating (BG) for our DFB lasers to a depth of 30 nm, along with the passive Si waveguides (300 nm-thick). The 30 nm deep etched DFB grating is defined by e-beam lithography and etched with reactive ion etching. The total thickness of the Si rib waveguide will be 500 nm, including the etched section containing the Bragg grating. Simultaneously, a III-V structure composed of p-doped InGaAs along with an inverted vertical p-i-n junction, is epitaxially grown on an InP substrate. In order to protect the epitaxial layers, two sacrificial layers are grown : InGaAs and InP layers.
- ii. Bonding of a Second SOI Wafer: Once the grating and passive components are etched, SiO₂ cladding is deposited, followed by planarization. Subsequently, a bulk Si wafer is bonded atop the processed wafer (Si carrier). This additional Si layer allows us to flip the structure before starting the backside process and contributes to maintaining the structural integrity of the wafer stack.
- iii. Flipping the Wafers for Backside Processing: After the bonding process, the wafers are flipped to expose the Si substrate. This orientation is crucial for accessing the Si waveguides from the backside, facilitating subsequent processing steps. The flipping process must be carried out with precision to avoid misalignment or damage to the delicate structures on the front side of the wafer.

- iv. Removal of the sacrificial layers: With the wafers flipped, the next step involves the removal of the silicon substrate and the silicon dioxide layer to enable the patterning of the Si waveguides. Two distinct Si rib waveguides were defined: a 500 nm-thick core rib waveguide for the laser section, and a passive rib waveguide with a 300 nm-thick core designed for optical mode transmission and defining grating couplers.
- v. Planarization and Growth of a Silicon Dioxide Layer: Finally, a new layer of silicon dioxide is grown on the exposed silicon surface. This silicon dioxide layer is then planarized to a thickness of approximately 20-25 nm. The precise control of the oxide thickness is crucial as it affects the optical properties and the bonding quality of the subsequent III-V layer. The two top sacrificial layers of the III-V structure are then removed by wet etching before being bonded to the SOI wafer.
- vi. Front-side back-end-of-line : A benzocyclobutene (BCB)-passivated shallow-rib III-V waveguide with a width of 2.4 μm was defined to better confine the optical mode and guarantee a low current leakage under high current injection. Finally, a front-side BEOL was carried out to establish electrical contacts to the devices [27].

Thick Si rib waveguides were used under the gain region and at the III-V-to-Si transition to ensure a tight control of the Γ_{QWS} and an efficient optical mode transfer between the III-V multi-stack and the Si rib waveguide. As detailed in the initial step of the backside processing, the grating was etched on the backside of the Si rib waveguide within the slab.

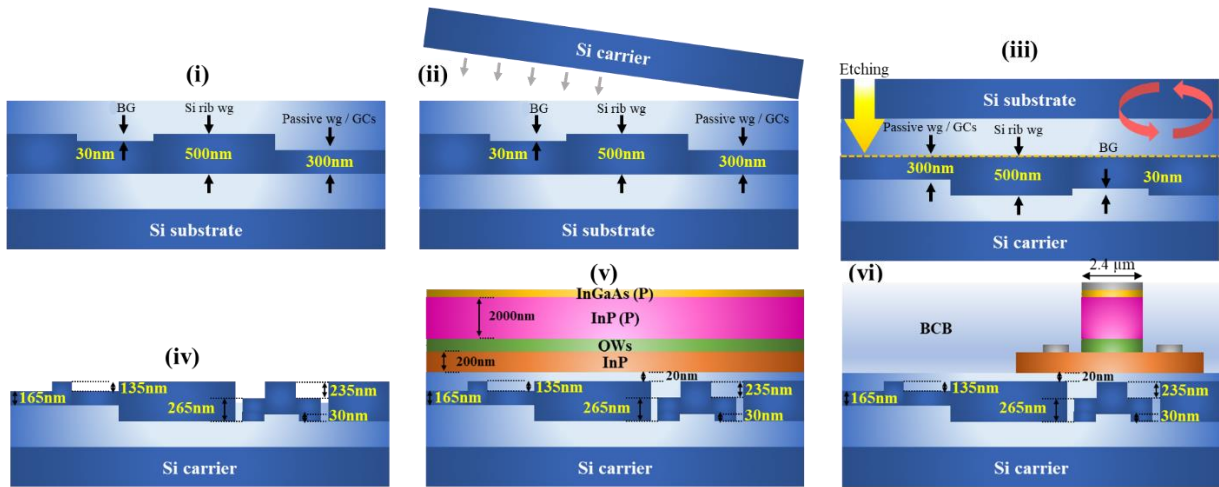


Figure 2.9: Backside wafer processing (i-iv), heterogeneous molecular bonding (v) and front-side back-end-of-line (vi).

Thiessen et al. [31] previously demonstrated back-side-on-BOX III-V-on-Si DFB lasers in the O-band, utilizing a relatively thin Si rib waveguide with a thickness of 300 nm. This design resulted in a strongly confined optical mode within the III-V waveguide. To ensure effective transmission of the optical mode between the III-V waveguide and the Si rib waveguide, they needed to add a local deposition of 200 nm of amorphous Si in the transition region. Our approach, however, employs a thicker Si rib waveguide, measuring 500 nm. This increased thickness allows us to tightly control Γ_{QWS} and κ while enabling an adiabatic mode transition by simply adjusting the width of the Si rib waveguide (W_{Si}). This flexibility eliminates the need for an additional amorphous Si layer deposition.

2.5 Low- κ design

2.5.1 Coupling factor

The coupling between counter-propagating waves is caused by the distributed reflectivity present in the laser cavity. This coupling is characterized by a factor denoted as κ , expressed in cm^{-1} , and it primarily depends on the refractive index variation within the medium. In general, for a heterogeneously integrated DFB laser, the coupling coefficient (κ) also strongly depends on W_{Si} as it is proportional to the grating confinement. As we will see in Figure 2.10. Fundamentally, this coefficient allows us to estimate the strength of the interaction between optical fields. The Bragg grating coupling factor, κ , is related to the position of the grating with respect to the active region, the grating profile, as well as the thickness and composition of the different layers of the waveguide. Furthermore, the coupling coefficient, typically expressed as the product κL , where L is the length of the cavity, is a highly significant parameter for DFB lasers and can be expressed, from equation (2 – 16) as [5]:

$$\kappa = \frac{\pi(n_{Si}^2 - n_{SiO_2}^2)A\Gamma}{\lambda n_{eff}} \quad (2 - 17)$$

Where n_{Si} and n_{SiO_2} represent the refractive indices of silicon and its oxide, silica, respectively, λ is the wavelength of the laser emission, and A is a coefficient expressed as follows: $A = \frac{\sin(\pi\Lambda_{Si}/\Lambda)}{\pi}$, where Λ is the pitch and Λ_{Si} is the length of the unetched region of the lattice. $A = \frac{1}{\pi}$ when the lattice pitch is twice the length of the unetched region [18, 32]

It has been demonstrated that lasers capable of achieving both narrow linewidth and high power can be realized either through high coupling coefficients or by employing longer cavities [34]. However, both theoretically and experimentally, it has been observed that lasers with high coupling coefficients or longer cavities, and thus a high κL product, tend to exhibit multimode behavior even at relatively low to moderate power levels [34, 35]. This effect can be attributed to the non-uniform longitudinal distribution of carrier density. Known as "Longitudinal Spatial Hole Burning" (LSHB), this distribution is particularly pronounced in $\lambda/4$ phase-shifted DFB lasers, where signal intensity is concentrated near this phase-shift position. To reduce the effects of LSHB in DFB lasers, several methods have been proposed and studied in the literature. LSHB can be mitigated by introducing longitudinal variations in the coupling coefficient and/or the net gain [36]. In his thesis, S. Zhao describes various DFB laser structures designed to improve performance, including DFB laser structures with asymmetric sections to attenuate the effects of LSHB. The integration of combed grating structures into DFB lasers can help to homogenize the gain profile along the cavity. These structures allow for a more uniform distribution of carriers, thereby reducing the effects of LSHB [37]. The use of sampled bragg gratings in the cavity of DFB lasers has also been proposed in the literature to reduce the effect of LSHB. This approach leads to a more uniform axial distribution of mode intensity within the laser cavity compared to a conventional single-phase-shift DFB laser [38-41]. Consequently, spatial hole burning is less effective at destabilizing single-mode operation, and the performance of these modified DFB lasers is less dependent on operating power. Maintaining a sufficiently low κL product is necessary to prevent the onset of lateral modes due to LSHB [42].

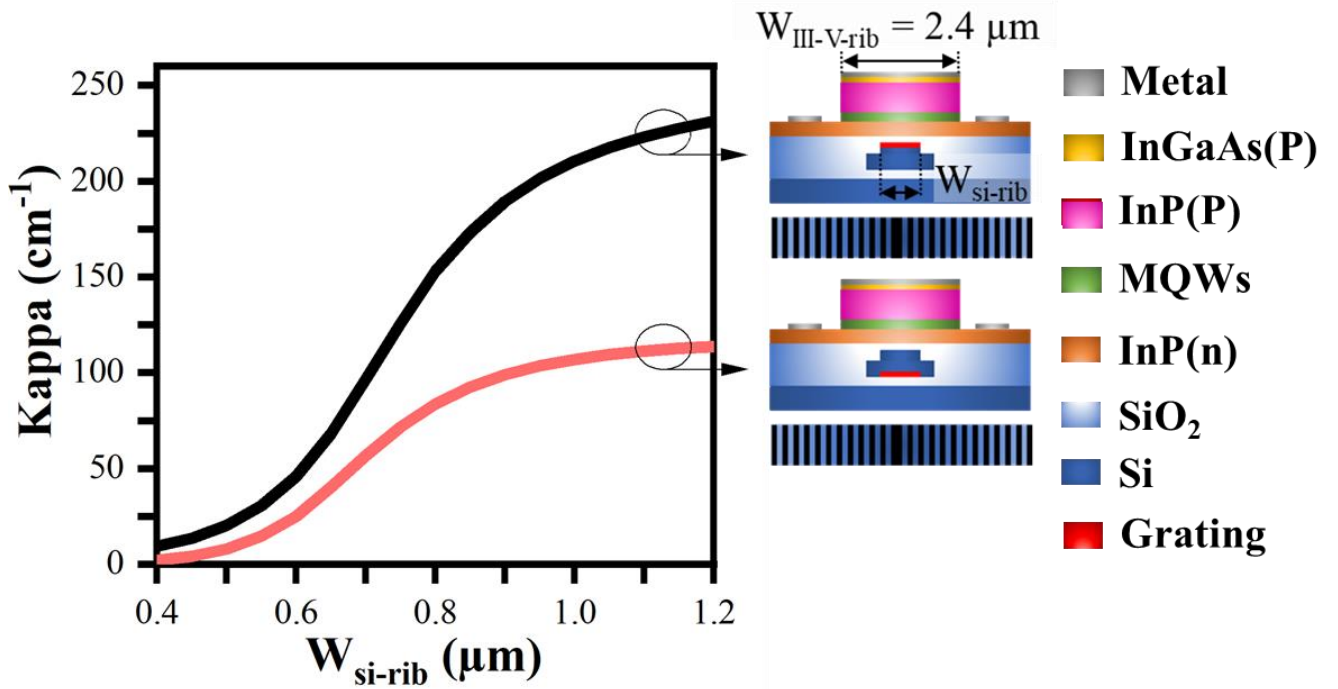


Figure 2.10: Simulated κ as a function of W_{Si} for the classic design (black curve), the backside grating design (magenta curve).

However, a significant reduction in cavity length typically results in decreased laser output power, as it also reduces the total active volume of the gain section. This means that we need to reduce κ to mitigate the effects of LSHB while maintaining the same performance. In this context, we compared in Figure 2.10 the evolution of κ as a function of silicon waveguide widths, W_{Si} , for two DFB laser designs. The first design (top) corresponds to a typical III-V-on-Si DFB design at $\lambda/4$ with a Bragg grating etched above the silicon waveguide [43]. The other design shows the same $\lambda/4$ DFB design but with the Bragg grating etched at the bottom of the silicon waveguide, in the slab. As illustrated, the configuration with the laser grating above the silicon waveguide (black line) exhibits the highest κ value, regardless of the silicon waveguide width (W_{Si}). As a consequence, laser designs with narrow W_{Si} values and a relatively short laser length need to be adopted to maintain the κL product sufficiently low and ensure a high threshold gain margin between the fundamental lasing mode and the side modes. Moreover, the κ - W_{Si} curve exhibits a steep slope around these narrow W_{Si} values, making the design highly sensitive to minor variations in the thickness of the oxide bonding layer. To address this issue, a backside grating design can be considered, as depicted in the second cross-sectional view (dark gray curve). In this configuration, the κ - W_{Si} curve shows considerably lower values with saturation around 110 cm^{-1} . Despite this reduction, an alternative method to mitigate the LSHB will be discussed in the next section.

2.5.2 Sampled Bragg grating

As discussed previously, one proposed method to mitigate the LSHB effect is to reduce the κ value of the DFB laser. This can be achieved by using an SBG-DFB design. In an SBG-DFB structure, in addition to having a phase shift in the middle of the cavity to enable single-mode operation, the grating is periodically interrupted along the cavity. Unlike conventional DFB

laser (see cross section in figure 2.12), which have uniform periodicity throughout the cavity, this structure introduces a more uniform axial distribution of the mode intensity inside the laser cavity, thereby helping to alleviate LSHB effects. To illustrate this, Figure 2.11 shows the longitudinal distribution of normalized optical intensity along the cavity for both a $\lambda/4$ phase-shifted DFB and an SBG-DFB. In both cases, the grating pitch (Λ) is 240 nm. The SBG has a pitch of $\Lambda_{\text{SBG-DFB}} = 2.4 \mu\text{m}$ and a duty cycle of $\text{DC}_{\text{SBG-DFB}} = 50\%$. The black curve represents the intensity distribution in a $\lambda/4$ DFB laser, where the optical field is highly concentrated at the cavity center. In contrast, the red dashed curve shows the distribution for the SBG-DFB, which displays a broader, more evenly spread intensity profile. This highlights key differences in how photon density varies with the laser structure. In a classic $\lambda/4$ phase-shifted DFB laser, the concentration of the optical field at the center ensures that a larger number of photons experience effective index changes in the central phase shift. This, however, leads to non-uniform carrier distribution along the cavity, as the carrier density is lower at the cavity center due to increased stimulated emission. As the bias current increases, this non-uniformity in carrier density becomes more pronounced, particularly in the $\lambda/4$ DFB laser. The refractive index within the cavity varies with carrier density, impacting the laser's operational mode. Since the laser emission wavelength (Bragg wavelength) is closely tied to the refractive index, fluctuations in carrier density can affect the stability and performance of the laser. By contrast, the SBG-DFB design, with its broader intensity profile, provides a more uniform carrier distribution along the cavity, potentially enhancing performance by reducing mode instability and mitigating LSHB effects.

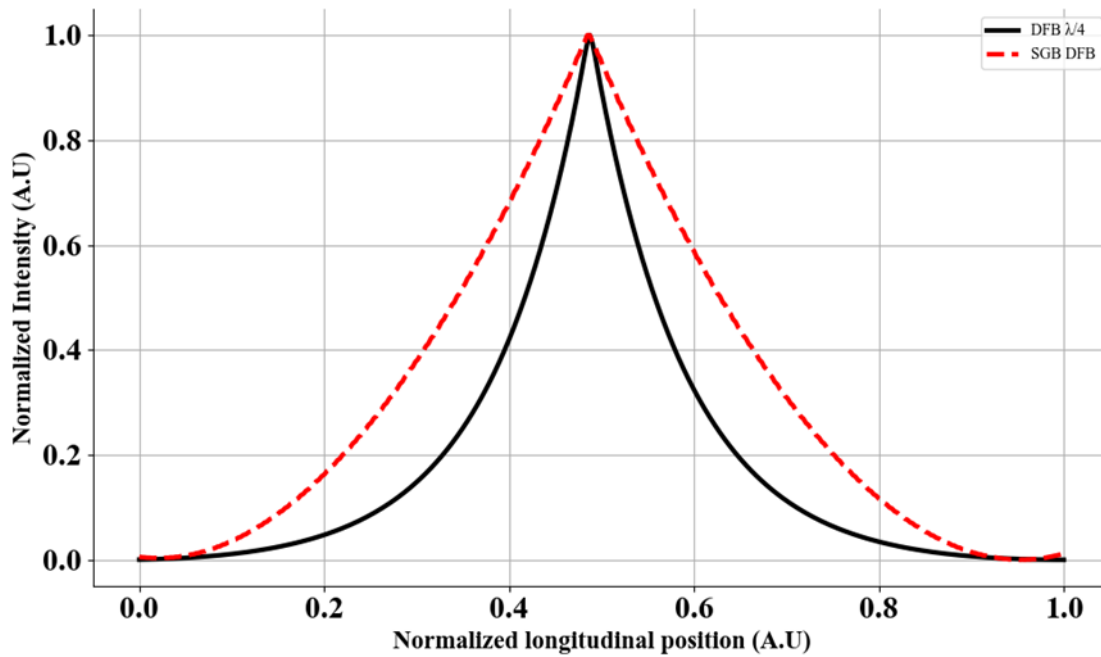


Figure 2.11: Simulated longitudinal distribution of normalized optical intensity in the cavity for two DFB laser designs: a $\lambda/4$ phase-shifted DFB (black curve) and a sampled Bragg grating DFB (SBG-DFB) (dashed red curve).

To better understand this reduction, we revisited Figure 2.10 and incorporated the κ - W_{Si} values for the new design, as illustrated in Figure 2.12. The SBG-DFB design achieves a low κ factor even at high W_{Si} values ($\kappa \leq 50 \text{ cm}^{-1}$), while maintaining an optimal balance between

optical gain and internal losses by adjusting W_{Si} , which in turn modifies Γ_{Si} and Γ_{QWS} , as needed. To visualize this, I simulated the 2D optical mode distribution for our two laser designs with approximately the same κ but at different W_{Si} values. In the classic $\lambda/4$ phase-shifted DFB laser (simulation outlined in magenta), we observe that the optical mode strongly overlaps with the InP-P, which induces internal losses. In contrast, the SBG-DFB design (outlined in light gray) displays a mode distribution that is more confined to the silicon waveguide, thereby reducing losses.

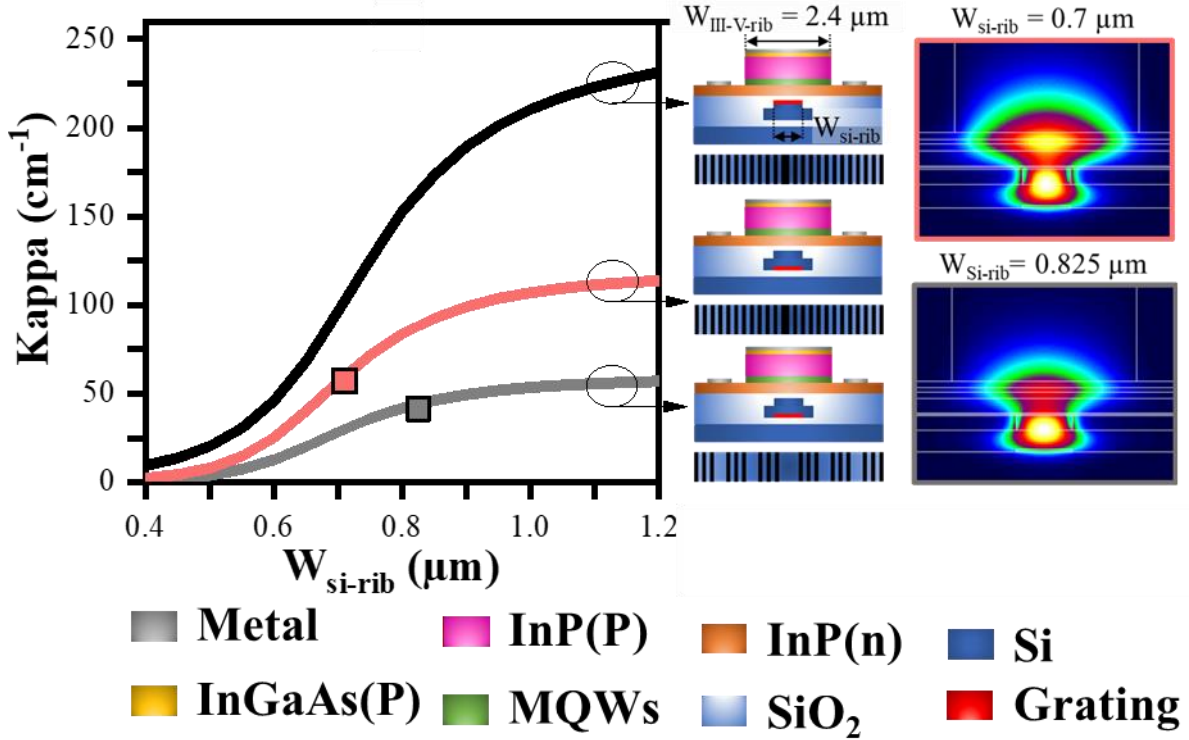


Figure 2.12: Simulated κ as a function of W_{Si} for the classic design (black curve), the backside grating design (magenta curve) and the backside sampled Bragg grating design (light gray curve). A simulation of the 2D optical mode distribution of our two laser designs with different W_{Si} are shown on the right.

This approach facilitates the extension of laser cavity lengths without increasing the κL value due to the reduced κ , resulting in enhanced optical power and improved coherence, which positively impacts narrow linewidth laser designs, as we will see in the following chapters.

2.6 Chapter Conclusion

This chapter explores heterogeneously integrated III-V-on-Silicon DFB lasers, with a particular focus on their design, fabrication and optimization within this platform. We first established a fundamental understanding of DFB lasers through a brief introduction. Then, we delved into the theoretical framework of rate equations and coupled mode theory, which are essential tools for analyzing and predicting laser behavior.

Our attention then shifted to DFB lasers integrated on a 500 nm-thick Si platform, examining in detail their design, structure, and fabrication process. Coupled mode theory allowed us to calculate the conditions for single-mode operation. Traditionally, a $\lambda/4$ phase shift at the

center of the DFB cavity has been considered optimal. However, we recognized that the non-uniform distribution of longitudinal carrier density can introduce limitations. This phenomenon is particularly pronounced in $\lambda/4$ phase-shifted DFB lasers due to the concentrated signal intensity near the phase shift region. To address LSHB, we explored optimization strategies for the III-V-on-Si DFB lasers. Critical parameters such as the coupling coefficient and confinement factor were carefully analyzed to improve laser performance. We used a sampled grating to improve the longitudinal intensity distribution, and to reduce the coupling coefficient, thus mitigating the effects of LSHB.

The insights gained here provide a solid foundation for the upcoming discussion on high-speed III-V/SOI directly modulated lasers. The next chapter will focus on the characterization of these devices, emphasizing their high-speed performance with or without external optical feedback perturbation.

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3. High-speed III-V/SOI Directly modulated lasers

In this chapter, we will focus on the study of high-speed III-V/SOI directly modulated lasers, which is a natural progression following our exploration in Chapter 2 of the design of heterogeneously integrated DFB lasers on silicon. Building on the theoretical foundation and design concepts previously studied, we will examine how these lasers operate under different modulation conditions. The main objective will be to analyze their performance in terms of speed and stability, while also considering potential disturbances caused by external optical feedback. Static measurements will be presented for each of the designs seen in the previous chapter. This will be followed by the study of the dynamic performance through the measurement of the small-signal scattering parameters to assess the electro-optical bandwidth, followed by large-signal measurements. Eye diagrams and bit error rate (BER) curves at different baud-rates will be shown to evaluate the characteristics of a communication link. These features will be analyzed both in a back-to-back configuration and after transmission over an optical fiber. Finally, the last section of this chapter will provide a brief discussion and conclusion. The results presented in this chapter have been published in [1], [2], [3], [4] and [5].

3.1 Introduction

The silicon photonics platform allows for the design of III-V-on-SOI directly modulated lasers (DMLs), particularly well-suited for low-cost high-speed transmission systems requiring broad direct modulation bandwidth, high optical power, and stable single-mode operation. Such characteristics are of interest for optical communications, specifically targeting applications in high-capacity optical data center systems [6]. Since, ultra-high-capacity Ethernet standards, such as Ethernet 800Gbps (800GbE), are increasingly being considered [7], [8]. Therefore, it is imperative to increase the modulation speed of DMLs for high-speed optical transmitters. However, achieving higher modulation speeds relies on enhancing the electro-optical bandwidth (EOBW). The intrinsic -3 dB EOBW of DMLs limits their utility as 50 GBaud-class optical transmitters, [9], [10], and relegates them to a secondary position compared to external modulators and lasers with integrated EAM (which we will see in the next chapter) for applications beyond 400GbE. It then becomes imperative to increase the modulation bandwidth while maintaining low power consumption. The relaxation oscillation frequency (f_r) represents a fundamental limit on the intrinsic modulation bandwidth of directly modulated lasers. The DML bandwidth is intrinsically characterized by : the relaxation oscillation frequency f_r , the damping factor, and the RC time constant. Among these, f_r constitutes the main bottleneck. This quantity is proportional to the square root of the differential gain, the optical confinement factor Γ , and the current density above the threshold current density as described in [29], and is more precisely expressed in Equation 3 – 5 :

$$f_r \propto \sqrt{\frac{\Gamma dg}{dN}} \frac{1}{LWN_w t_w} (I_p - I_{th}) \quad (3-1)$$

Where Γ is the optical confinement factor in the active region, dg/dN is the differential gain, N_w is the number of wells, W is the active region width, L is the active region length, t_w is the wells' thickness, I_{th} is the threshold current, and I_p is the bias current. It becomes evident that to increase f_r , it is essential to enhance the optical confinement, the differential gain, and the operating current of the laser. To improve the differential gain, it is crucial to use strain-compensated multi-quantum wells (MQW) structures. In our devices, we use AlGaInAs alloys for the MQWs, which are known to have a higher conduction band offset compared to the InGaAsP MQWs, providing lower electron leakage and thus higher differential gain. Another way to increase differential gain is to design the laser's Bragg wavelength with a certain detuning (blue-shift) with respect to the optical gain's central wavelength [11] (This technique has been employed to enhance the bandwidth of our component). On the other hand, according to Eq. 3-1, increasing f_r requires a reduction in the dimensions of the active region, the length, width, thickness, and number of quantum wells. However, decreasing these parameters will be mainly compensated by the reduction in optical confinement. Alternatively, we can design lasers that can be operated under high current injection as seen in the previous chapter.

3.2 Static characterization

The lasers were fabricated with a $\lambda/4$ phase shift in the middle of the cavity. The active region consists of 6 MQWs and the photoluminescence (PL) peak is at 1520. More details on the epitaxial layer structure were provided in Chapter 2. Figure 3.1 shows the top view of the device, showing the ground-source-ground electrical pads. The signal from the DFB lasers was coupled from the silicon waveguide to a single-mode fiber using a grating coupler. The propagation and coupling losses of this grating were initially assessed using the cut-back method, with optical waveguides of varying lengths integrated onto the same chip as the tested lasers. We used passive waveguides of three different lengths (total length of passive waveguides per cell): 0.78 mm, 0.945 mm, and 5.5 mm. The setup, illustrated in Figure 3.2-a, consists of coupling a widely tunable laser source, covering part of the S-band to the entire L-band, through a polarization controller and then into the chip via a vertical grating coupler. The signal then passes through the passive component under study and is coupled back into the fiber using another grating coupler. We analyze the transmission spectrum of the component by sweeping the tunable laser source, synchronized with the optical spectrum analyzer. The transmission spectrum of a single grating coupler is represented in Figure 3.2-b. These measurements were obtained by optimizing the coupling for the operating wavelength, at $\lambda = 1545$ nm. It is observed that for this grating coupler, the wavelength at which transmission is maximized does not coincide with the operating wavelength of our components. This represents an opportunity for improvement in future runs by recentering the maximum transmission on the transmitter's operating wavelength. The insertion losses per grating coupler were determined by dividing the total insertion losses by two, resulting in

a range of 4 to 5 dB per grating coupler. Additionally, the propagation losses, on the order of 2 dB/cm, are considered negligible.

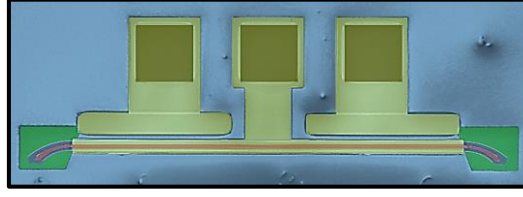


Figure 3.1: Top view of the fabricated DFB lasers.

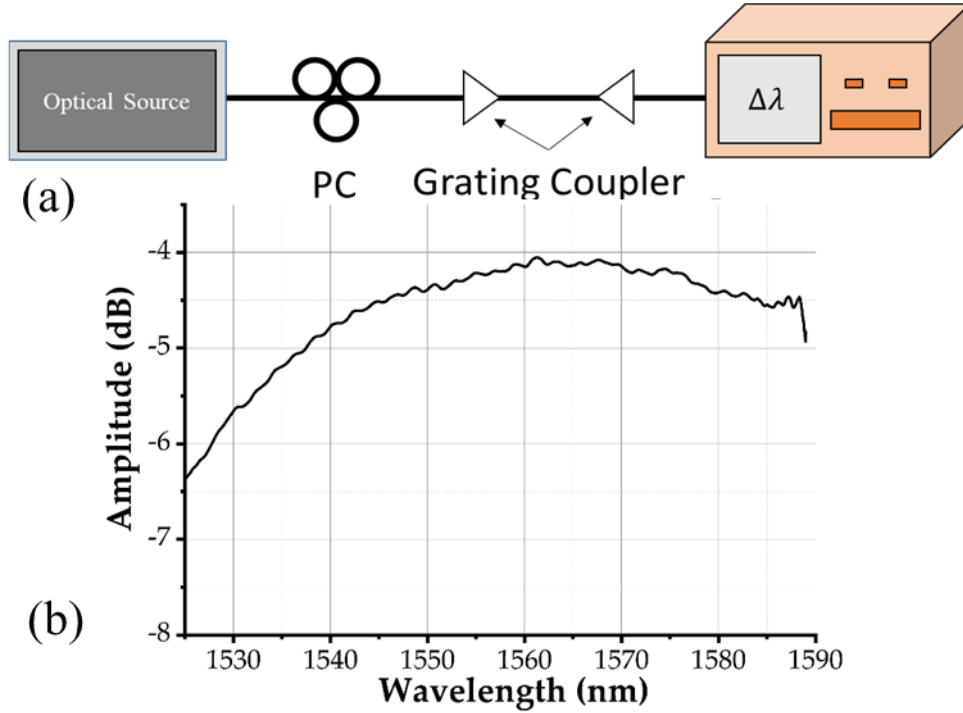


Figure 3.2: (a) Experimental setup for the measurement of insertion losses. (b) Measured transmission spectrum of a single fiber grating coupler.

3.2.1 $\lambda/4$ Phase-Shifted Short-Cavity Device

We fabricated lasers with three different cavity lengths: 400 μm , 600 μm , and 880 μm . The first-order DFB grating period is 245 nm. However, the longer lasers (880 μm) exhibited multimode behavior at low currents ($\approx 2I_{\text{th}}$), which is related to their high κL value. The laser is mounted on a stage equipped with a thermo-electric cooler (TEC) for temperature stabilization. The static characteristics of the lasers were measured using a standard Keithley 2400 current source. All measurements were performed at a controlled temperature of 20°C.

The L-I-V curve and optical spectra of a typical 400 μm laser are shown in Figure 3.3, allowing us to extract a I_{th} of 26 mA. The coupled optical power in the waveguide exceeds 13 mW for an injected current of $I = 140$ mA ($\approx 5I_{\text{th}}$), as indicated on the same graph. The coupling coefficient was extracted from the Bragg stopband observed in the laser spectrum. The measured stopband width ($\Delta\lambda$) allows us to calculate the grating coupling coefficient κ using Eq. (2-16). From this relation, we estimate $\kappa \approx 54$ cm^{-1} , which is consistent with simulated

values and previously reported results in the previous chapter. Combined with the cavity length of 400 μm , this yields a κL product close to 2. As expected, the laser maintains a single-mode operation even at high current levels. However, a kink is observed in the L-I curve for injection currents around 90 mA. This behavior is attributed to mode hopping to a mode at a shorter wavelength than the DFB's lasing wavelength, a phenomenon occurring only at an injection current of around 90 mA, which disappears thereafter, as confirmed by monitoring the laser spectrum for different injection current values. The optical spectrum shown in Figure 3.3-b, measured at 20°C, reveals a slight reduction in the Side Mode Suppression Ratio (SMSR) from over 50 dB (for $I < 90$ mA) to 37 dB for $I = 90$ mA. Nevertheless, the laser remains in a single-mode regime, and SMSR values greater than 50 dB are achieved again for $I > 90$ mA. Furthermore, it should be noted that the lasing wavelength ($\lambda_{\text{DFB}} = 1544$ nm) of the DFB laser is shifted by 7 nm (at $T = 20^\circ\text{C}$) towards shorter wavelengths compared to the central wavelength of the optical gain spectrum to increase the differential gain, as explained before. Additionally, from the V-I curve, we obtain a differential resistance of $R = 6.5 \Omega$. These values are consistent with results published in the literature for III-V/Si-based DFB lasers [12, 18]. The maximum wall-plug efficiency (WPE) of the device is 7%. Recently, J. Rahimi et al demonstrated more than 12% WPE for a 200 μm DFB laser [23].

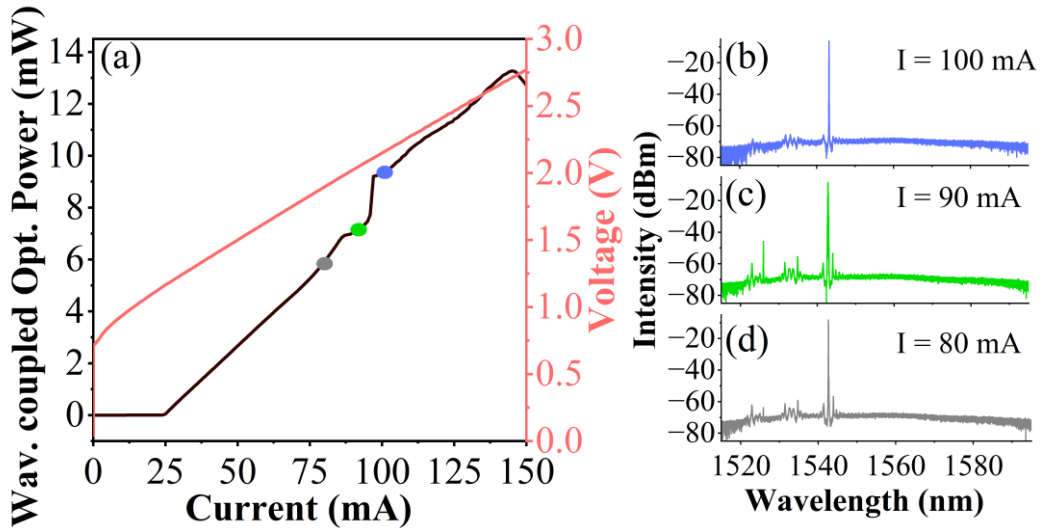


Figure 3.3: a) Light-current-voltage (L-I-V) characteristics of the 400 μm cavity DML laser. The colored markers correspond to the bias points where the emission spectra were recorded. (b-c-d) Optical spectra measured at injection currents of 100 mA (blue), 90 mA (green), and 80 mA (gray), respectively.

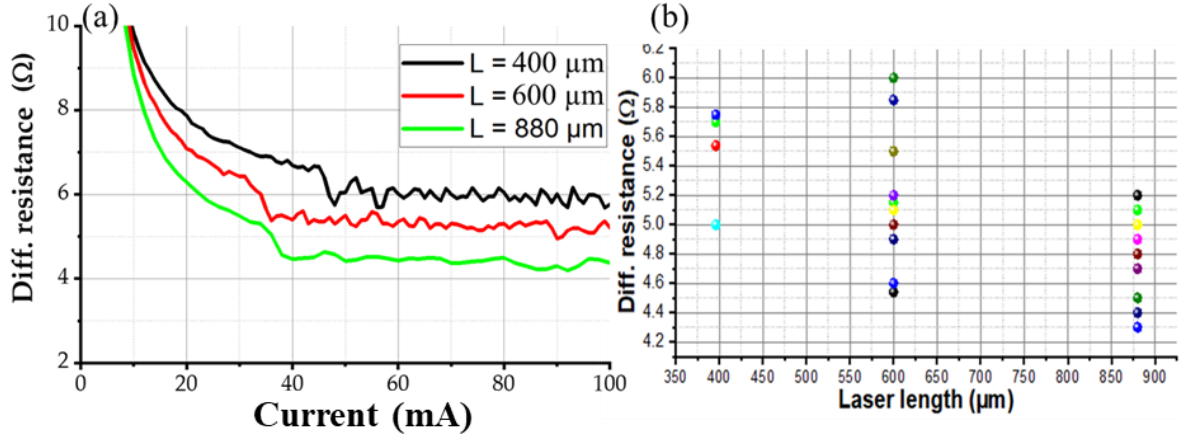


Figure 3.4: (a) Differential resistance as a function of bias current for different laser cavity lengths, (b) Differential resistance as a function of laser cavity length.

Based on the V-I curve, we were able to deduce the differential resistance, which represents the modulation efficiency and the electrical response of the laser to changes in current. This measurement was carried out on three different laser cavity lengths. Figure 3.4 presents the results, showing that the differential resistance decreases as the cavity length increases. The resistance given by : $R = \rho \frac{l}{S} = \rho \frac{l}{L.W}$ where ρ is the resistivity, l is the electronic pathway, and S is the cross-sectional area. R decreases as the cross-sectional area increases, making it inversely proportional to the cavity length L . We observe that the appearance of the plateau coincides with the current range where lasing occurs. This indicates a constant carrier density in the conduction band, which is one of the key consequences of the lasing effect. Notably, we can predict the current threshold without directly measuring the optical power, as it corresponds with the onset of the plateau. It can be inferred that measuring differential resistance is a key indicator for predicting optical changes in the laser. This also highlights the importance of monitoring these fluctuations to optimize the stability and performance of semiconductor lasers. Figure 3.4-b illustrates the differential resistance for several laser samples with varying cavity lengths for the same current. The general trend indicates that shorter laser cavities exhibit higher differential resistance, while longer cavities provide lower resistance. However, there are some outliers observed for the 600 μm samples. It is important to note that although we have not measured a sufficient number of lasers to fully assess the wafer yield, the results presented here are repeatable and consistent across the different laser designs along the fabricated wafer.

After these measurements, the operating temperature of the stage on which the sample was placed was adjusted to determine the characteristic temperature and thermal impedance. The experimental setup for these measurements was the same as for the previous tests. The thermal study was carried out in the 600 μm long components. Figure 3.5-a shows the continuous-wave L-I characteristics for various stage temperatures, ranging from $T = 25^\circ\text{C}$ to $T = 50^\circ\text{C}$ in 5°C increments.

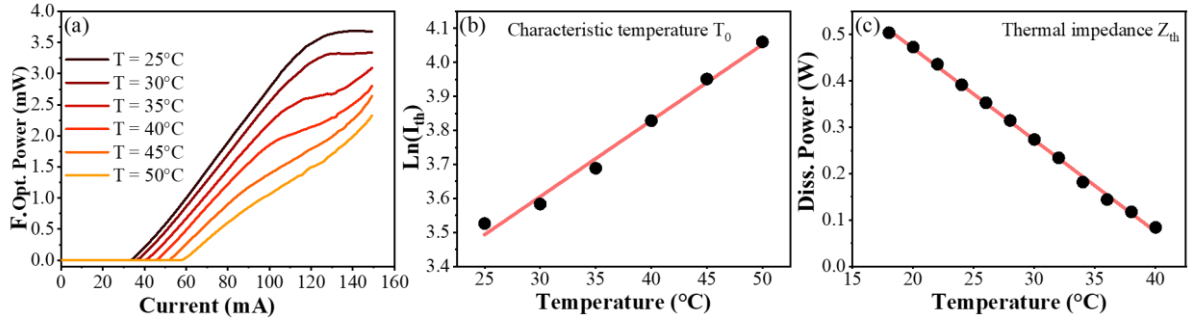


Figure 3.5: (a) L-I characteristic measured with increasing temperature, (b) Evolution of the natural logarithm of I_{th} as a function of the temperature at the TEC stage. (c) Dissipated power as a function of the TEC stage temperature for a lasing wavelength of the 600μm cavity DML laser.

The optical power increases sub-linearly as the bias current is raised due to non-linear phenomena, such as gain compression, and also due to the thermal roll-off. Noticeably, there is no mode hopping at any temperature. In addition, the increase in temperature leads to a redistribution of carriers (electrons and holes) in the conduction and valence bands, making the radiative recombination process less efficient [16]. Furthermore, the observed reduction in optical power at higher temperatures might also be linked to an increase in leakage currents within the quantum wells. At elevated temperatures, carriers (particularly electrons) gain more kinetic energy, which allows them to escape more easily from the quantum wells and overcome the potential barriers. Nonlinear phenomena, such as carrier-induced refractive index changes or saturation effects, tend to be more pronounced at higher optical powers. Therefore, if the optical power decreases, these nonlinear effects could diminish as well. It is important to note that this is a hypothesis, as we currently lack direct evidence to fully confirm the link between carrier leakage and the suppression of nonlinear effects. Further experimental validation and theoretical modeling would be needed. Exponentially spaced lasing thresholds are observed, which is linked to the relationship between temperature and threshold current. The characteristic temperature, T_0 , can be determined using the following equation [13]:

$$I_{th} = I_0 \exp\left(\frac{T}{T_0}\right) \quad (3-2)$$

Using the parameter I_0 , we successfully obtained a T_0 value of 44.7°C for this component by fitting the natural logarithm of I_{th} as a function of temperature, which aligns with previous findings (see Figure 3.5-b), in agreement with previously reported work [14]. Furthermore, we measured the thermal impedance, Z_{th} , using a method based on the athermalization condition described in reference [15]. Such method enables us to estimate the Z_{th} using a DC current source uniquely, thus avoiding the modification of the setup to integrate a pulsed current source, as is the case for the “standard” measurement [15]. In our case, we have measured the dissipated power required to obtain a stable lasing mode at $\lambda = 1544$ nm for $T = 20^\circ\text{C}$. Then, we have compensated the drift towards the longer-wavelength side induced in the lasing wavelength when the temperature is increased by a decrease in the injected current, hence maintaining the same lasing wavelength all over the temperature sweep. The results of this measurement are presented in Figure 3.5-c and demonstrate an excellent fit,

yielding a Z_{th} value of 32°C/W. These values are well in-line with those previously reported in similar cavity-length DFB lasers [15].

3.2.2 Sampled Grating Long-Cavity Device

In Section 2.5 of the previous chapter, we theoretically demonstrated that using a SBG-DFB design significantly reduces the κ coefficient of the lasers, thereby mitigating the LSHB effect. This approach results in a more uniform axial distribution of the mode intensity within the laser cavity compared to a conventional $\lambda/4$ phase-shifted DFB laser. Consequently, spatial hole burning is less effective in destabilizing single-mode operation, making the performance of these modified DFB lasers less dependent on the operating power [16]. The SBG length is 880 μm , with a coupling coefficient derived from the measured stopband in the optical spectrum, $\kappa \approx 37 \text{ cm}^{-1}$, which closely matches the simulation results, chapter 2 section 2.5.2, and providing $\kappa L \approx 3.5$. Longer laser cavities with a high mode overlap with the distributed feedback grating are desirable for reducing laser linewidth, as this decreases mirror losses (α_m). However, they are also prone to exhibit multi-longitudinal mode operation induced by LSHB under high bias currents [17]. A quarter-wavelength shift was introduced at the center of the grating to create a resonant mode in the middle of the photonic stopband. The nominal etching depth is 30 nm, the grating width is 2.4 μm , and the grating pitch (Λ_{DFB}) is 240 nm with a duty cycle (DC_{DFB}) of 50%. The SBG has a pitch of $\Lambda_{\text{SBG-DFB}} = 2.4 \mu\text{m}$ and a duty cycle of $\text{DC}_{\text{SBG-DFB}} = 50\%$.

The L-I-V characteristics under continuous-wave operation were measured for different stage temperatures ranging from $T = 20^\circ\text{C}$ to $T = 45^\circ\text{C}$ in 5°C increments. The L-I curve at 20°C is shown in Figure 3.6-a. The optical power follows a sub-linear trend with increasing current bias (due to gain compression) without mode jumps. A lasing threshold of $I_{th} = 35 \text{ mA}$ (1.65 kA/cm^2) was observed, with a maximum waveguide-coupled optical power of over 20 mW from each side of the SBG-DFB laser, confirming the expected symmetry of the field distribution inside the cavity. The inset provides a zoom-in of the L-I curve around I_{th} for various temperatures. The total WPE, defined as the sum of the left and right optical power, is shown in Figure 3.6-b along with the V-I curve. A maximum WPE of 17% and a differential resistance of $R = 4.5 \Omega$ were extracted. These values are consistent with those previously reported for III-V/Si DFB lasers [18]. The evolution of the optical spectrum as a function of stage temperature is shown in Figure 3.6-c, with a gain peak shifting towards longer wavelengths with increasing temperatures and a SMSR exceeding 45 dB.

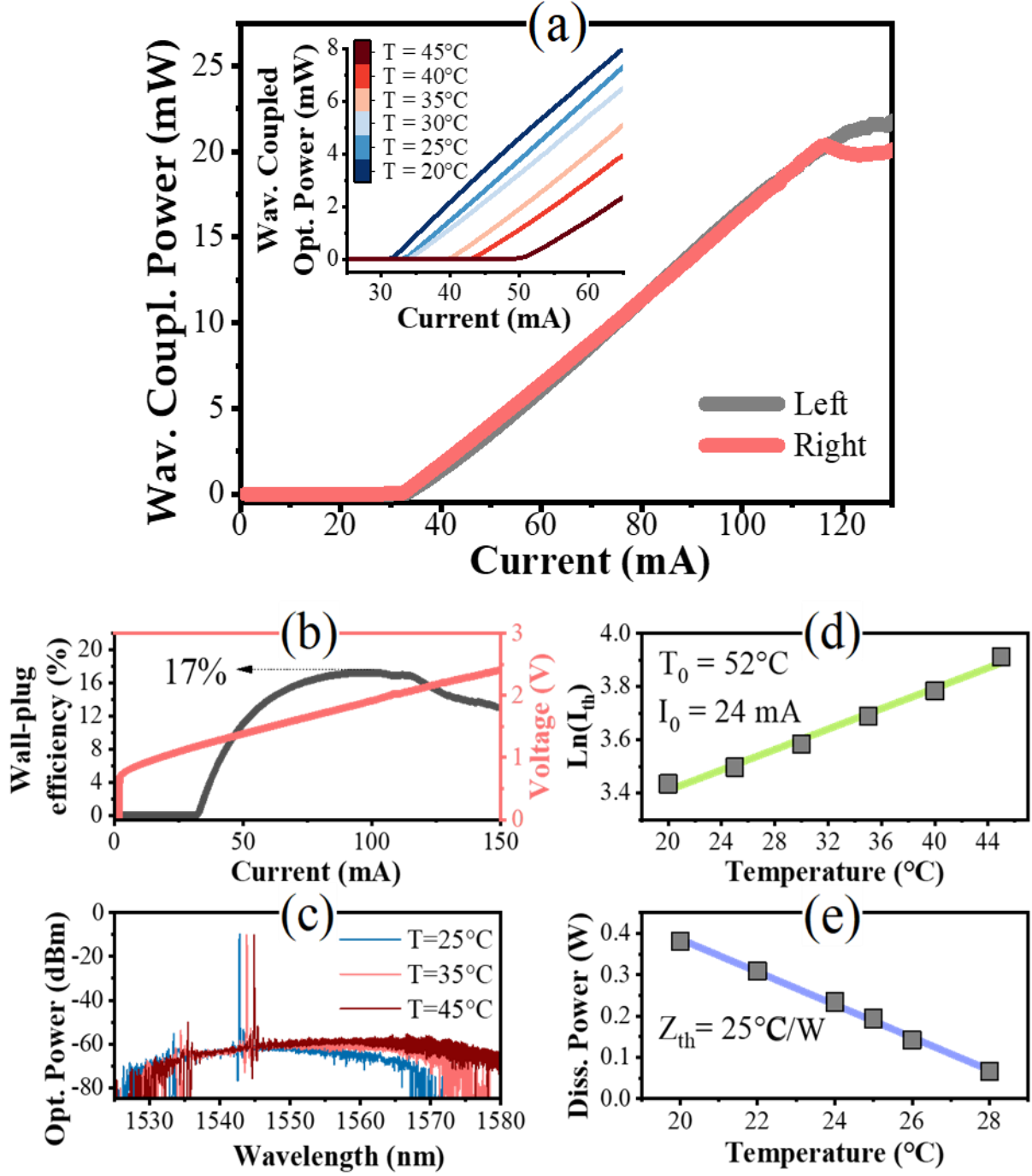


Figure 3.6: (a) L-I characteristic measured from either the left (grey curve) or the right hand side (magenta curve) of the SBG-DFB laser. The inset shows the evolution of I_{th} with increasing temperature. (b) Total wall-plug efficiency (left + right optical power) and measured voltage as a function of the injected current. (c) Optical spectra at different temperatures. (d) Evolution of the natural logarithm of I_{th} as a function of the temperature at the TEC stage. (e) Dissipated power as a function of the TEC stage temperature for a lasing wavelength of $\lambda = 1544 \text{ nm}$.

It is worth noting that the wavelength shift of the laser as a function of dissipated electrical power was also examined, yielding a value of 4 nm/W . The characteristic temperature, T_0 , can be derived from the L-I curves at different temperatures (inset of Figure 3.6-a) using equation (2). We obtain a value of $T_0 = 52^\circ\text{C}$ (see Figure 3.6-d), which is in agreement with previously

reported results [14]. The thermal impedance, Z_{th} , was also measured using a method based on athermalization, and the results are presented in Figure 6.6-e, showing $Z_{th} = 25^\circ\text{C/W}$.

Subsequently, we performed a laser linewidth measurement using the delayed self-heterodyne technique. Figure 3.7 shows the setup. The device is driven by a low-noise electrical source (BE2102, iTEST), and a two-stage optical isolator is used to prevent optical feedback from the setup. The optical signal is split by a 90/10 coupler to either measure the laser linewidth using the delayed self-heterodyne technique or to measure the L-I-V characteristic and the optical spectrum. The delayed self-heterodyne setup consists of an erbium-doped fiber amplifier (EDFA) that provides 10 dB of optical gain, coupled to a JDS Fitel tunable band-pass filter (TBPF) with a 5 nm bandwidth and a fiber-based Mach-Zehnder interferometer (MZI). An acousto-optic modulator (AOM) operating at 40 MHz is placed in one arm of the MZI to shift the frequency, while the other arm contains a fiber coil (50.6 km) to induce a phase delay. Both signals are coupled to a photodiode (XPDV-1120R), and their mode beating is measured by an electrical spectrum analyzer (Agilent E4448A). The second optical line of the second output is coupled to either a power meter (Thorlabs) or an optical spectrum analyzer (Advantest Q8384).

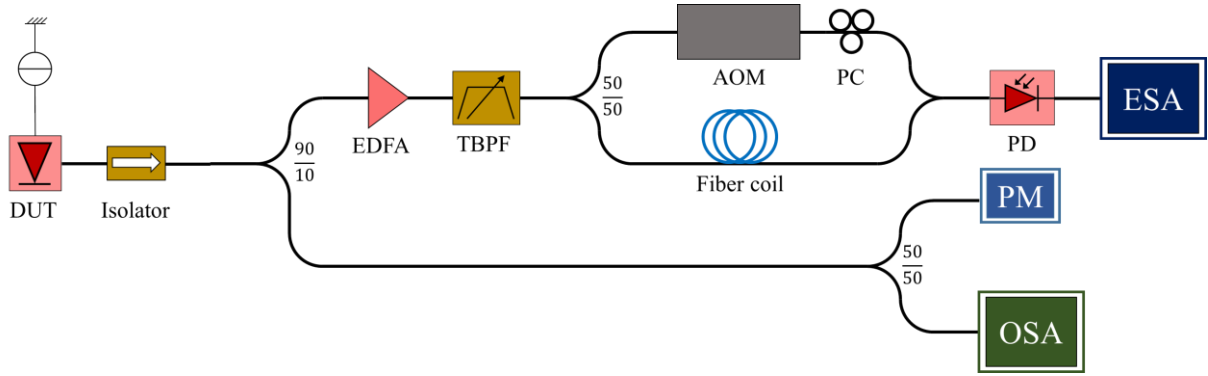


Figure 3.7: Self-heterodyne setup.

Figure 3.8-a shows the measured laser linewidth using the delayed self-heterodyne method. Figures 3.8-b, 3.8-c, and 3.8-d present the measured electrical spectra for three representative current biases: $I = 40$ mA, $I = 80$ mA, and $I = 110$ mA. A Voigt function was used to separate the intrinsic Lorentzian linewidth from the Gaussian technical noise. As seen in Figure 3.8-a, the laser linewidth decreases monotonically with increasing output power, in accordance with the modified Schawlow-Townes formula [19]:

$$\Delta\nu = \frac{v_g^2 \cdot h\nu \cdot n_{sp} (1 + \alpha_H^2) \ln\left(\frac{1}{R_m}\right)^2}{8\pi P \eta L^2} \quad (3-3)$$

Where v_g is the group velocity, R_m is the facet power reflectivity, η is the quantum efficiency, $h\nu$ is the photon energy, n_{sp} is the spontaneous emission factor, P is the output power and α_H is the linewidth enhancement factor. Moreover, Equation 3 – 3 was adapted in [51] by accounting for additional factors including the mirror loss coefficient α_m , which can be related to the coupling coefficient κ .

A minimum laser linewidth of $\Delta\nu = 118$ kHz is measured at $I = 110$ mA. For higher bias current, the laser linewidth re-broadens (see the last point of Figure 3.8-a, $I = 120$ mA), whose origin has been attributed to three main mechanisms [20]:

- i) Spontaneous emission fluctuations of non-clamped side modes in the cavity;
- ii) Local current density fluctuations in the cavity; and
- iii) Perturbations in the carrier/photon density feedback that generate LSHB and gain compression.

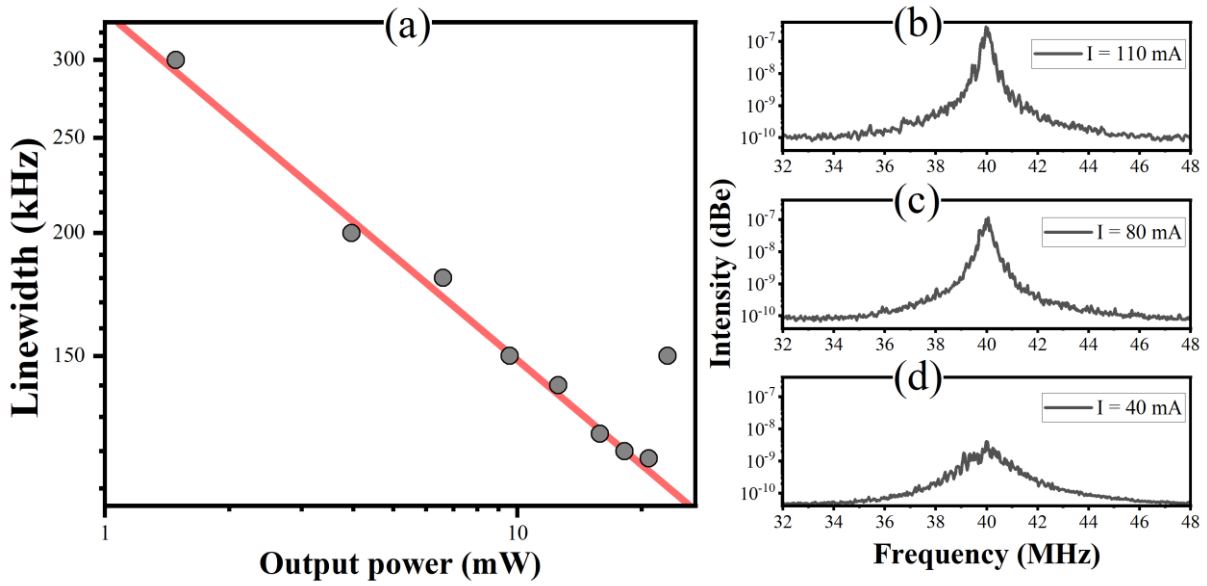


Figure 3.8: (a) Evolution of the measured laser linewidth as a function of the output power. The magenta line corresponds to the linear fitting of experimental data, excluding the last value at $I = 120$ mA. Measured electrical spectra under a bias current of (b) 110 mA, (c) 80 mA and (d) 40 mA.

3.3 Dynamical characterization & Transmission experiments

In what follows, we will focus on two specific designs: Firstly, the long-cavity SBG design, referred to as the "Sampled Grating Long-Cavity Device". These narrow linewidth lasers may be of interest for applications requiring hybrid schemes that combine intensity-modulated emitters with complex receivers capable of simultaneously recovering both intensity and phase information. The second design is a shorter device with a traditional $\lambda/4$ phase-shifted grating, referred to as the " $\lambda/4$ Phase-Shifted Short-Cavity Device". These short-cavity lasers are designed to have a broader bandwidth and are suitable for high-speed IMDD due to the reduced photon lifetime in the cavity.

3.3.1 Small signal Electro-optical bandwidth (EOBW) measurement

The dynamic characterization of the sample was performed to determine the small-signal EOBW. The bandwidth measurement evaluates the ability of the device to follow a modulated signal as the excitation frequency is progressively increased. In practice, a sinusoidal electrical signal with varying frequency is applied, and the response at the output is monitored. At low frequencies the device reproduces the signal accurately, while at higher frequencies physical limitations such as carrier transit time, resistive-capacitive effects, or device geometry lead to attenuation and phase shift. The bandwidth is defined as the frequency range over which the

response remains effective, more specifically by the -3 dB cutoff frequency at which the signal amplitude drops by half. A large bandwidth indicates a fast response and the ability to transmit data at very high rates, which is a key requirement for components in modern optical communication systems. The experimental setup is illustrated in Figure 3.9.

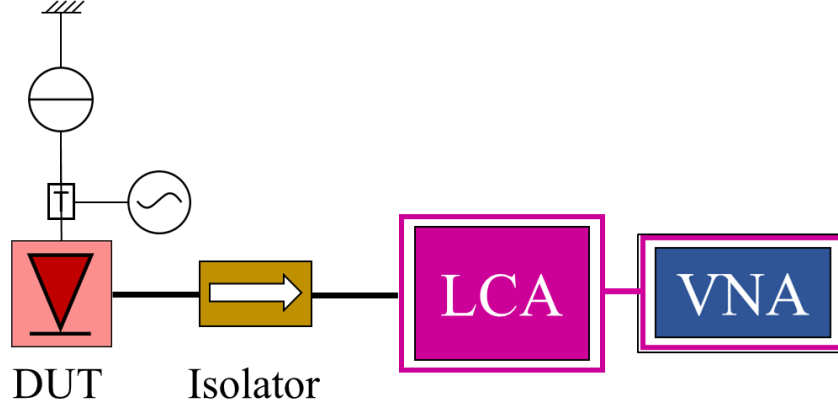


Figure 3.9: Small signal Bandwidth setup.

The small-signal modulation response (S_{21}) was measured by driving the laser with a combination of a DC current bias and a small RF signal generated by a vector network analyzer (VNA). The output power was directed to a lightwave component analyzer (LCA Agilent N5242A/N4375B) for opto-electronic conversion and analysis by the VNA to obtain the S parameters. The modulation frequency was swept from 0 to 25 GHz to characterize the laser's transfer function. Figure 3.10-a shows the small-signal measurement, revealing the characteristic peak at the relaxation oscillation frequency, f_r , followed by a frequency cut-off and a gradual decay of the amplitude modulation (AM) signal. The measured small-signal bandwidth of a directly modulated laser, S_{21} parameter, is composed by two contributions: the intrinsic response of the device and the parasitic response of its equivalent circuit, which is driven by the time constant ($\tau_c = RC$). The product of both transfer functions describes the measured frequency response:

$$|S_{21}(f)| = \frac{f_r^2}{\sqrt{\left((f^2 - f_r^2)^2 + \frac{f^2 \gamma^2}{(2\pi)^2}\right)}} \frac{1}{\sqrt{(1 + (2\pi f \tau_c)^2)}} \quad (3 - 4)$$

Where f_r is the relaxation oscillation frequency and γ is the damping factor. Similarly, f_r can be defined as [29]:

$$f_r = \frac{1}{2\pi} \sqrt{\frac{v_g}{q} \Gamma \eta_i \frac{dg}{dN} \frac{I - I_{th}}{V}} \quad (3 - 5)$$

The damping factor γ increases proportionally to the square of the resonance frequency following the relationship [29]:

$$\gamma = K f_r^2 + \gamma_0 \quad (3 - 6)$$

Where v_g is the group velocity, q is the elementary charge, η_i is the internal quantum efficiency, $\frac{dg}{dN}$ is the differential gain, γ_0 the damping factor offset, which can be expressed in a simplified way as $\gamma_0 = \frac{1}{\tau_c}$, τ_c the carrier lifetime and K a coefficient, which can be expressed in a simplified way as $K = 4\pi^2\tau_p$, τ_p the photon lifetime, and V is the active volume. Since the parasitic response is independent from the current bias, we can extract the intrinsic small-signal response of the laser just by subtracting two curves measured at different current bias [21]. For large resonance frequencies, the K -factor describes the damping of the response, and as such is an important parameter in the characterization of high-speed lasers. The damping factor offset γ_0 is important at low powers where the relaxation resonance frequency is small. In practice, K and γ_0 are used as fitting parameters to be extracted from the laser modulation response.

3.3.1.1 Sampled bragg grating Long-cavity device

Figure 3.10 compares the measured ($f_r^{meas.}$) and intrinsic ($f_r^{intrin.}$) 3 dB bandwidth with the f_r parameter, derived from the fitting of the intrinsic small-signal response, as a function of $\sqrt{I - I_{th}}$. A maximum intrinsic bandwidth of 14 GHz was extracted, with a $f_r = 9$ GHz. Remarkably, the last f_r value, obtained from a bias current of 110 mA, slightly deviates from the linear trend and undergoes saturation. This fact indicates the apparition of optical nonlinear phenomena in the laser, which induce gain suppression in the laser cavity [22]. Noticeably, such bias current coincides with the beginning of the saturation of the L-I curve and the laser linewidth in Figure 3.6-a and Figure 3.8-a, respectively. In Figure 3.10-b, we plotted the D-factor, which is defined as the slope of f_r , as a function of $\sqrt{I - I_{th}}$. For the SBG-DFB laser, D-factor was determined to be 1 GHz/mA^{1/2}. The τ_c and τ_p values were also extracted by fitting the measured γ factor as a function of the square of f_r , as seen in Figure 3.10-c, obtaining a $\tau_c \sim 123$ ps and a $\tau_p \sim 5.8$ ps. These values align well with those previously reported by other research teams, such as NTT [12, 22] and the Eindhoven University of Technology [46], in similar integrated III-V/Si DFB lasers.

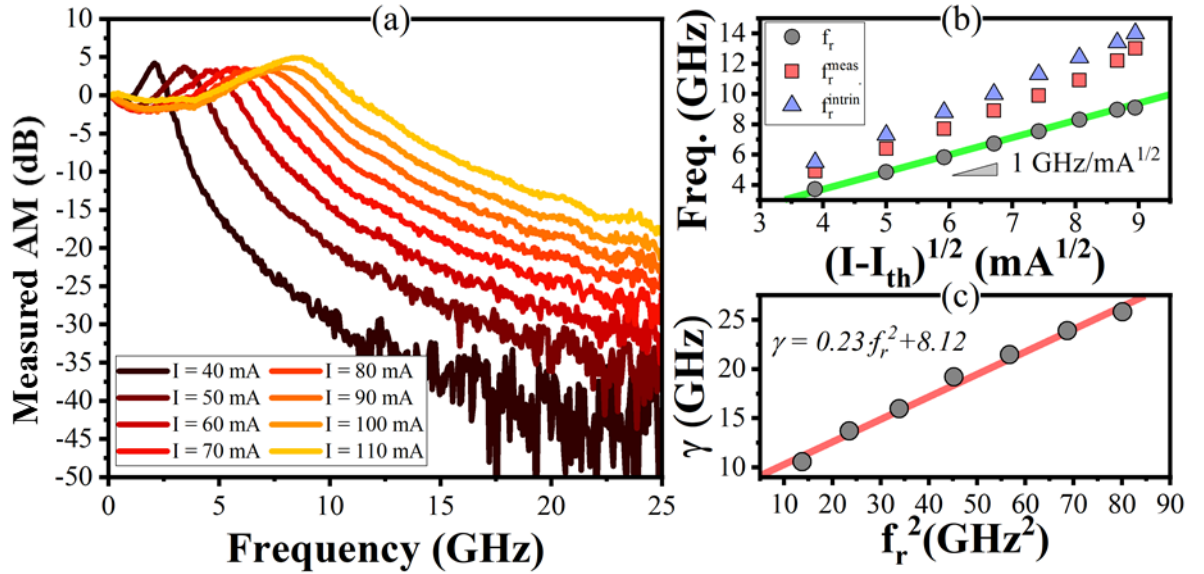


Figure 3.10: (a) Small-signal measurements for different bias current. (b) Evolution of the relaxation oscillation frequency (f_r) the measured (f_r^{meas}) and intrinsic (f_r^{intrin}) 3 dB bandwidth as a function of the square root of the bias current above the threshold current. (c) Damping factor (γ) as a function of the square of the relaxation oscillation frequency.

3.3.1.2 $\lambda/4$ phase-shifted short-cavity device

The small-signal modulation response (S21) was measured for $\lambda/4$ phase-shifted short-cavity DFB lasers. According to the equation 5, the relaxation oscillation frequency depends on the bias current of the laser. To illustrate this fact, in Figure 3.11, the modulation response of the laser is shown versus frequency for different bias currents. It can be observed that the relaxation oscillation resonance moves towards higher frequencies as the bias current increases. The highest intrinsic modulation bandwidth measured was 21 GHz, extracted for a relaxation oscillation frequency $f_r = 14$ GHz. The higher modulation bandwidth of this laser, compared to other demonstrated III-V/Si DFB lasers without external cavity [23], can be attributed to the ability to inject a very high current while maintaining a single-mode behavior, which results in a larger f_r . The presence of the Bragg grating at the bottom allows us to pull the optical mode downward (into the Si), which reduces internal losses since the III-V material typically has much higher losses. This configuration also decreases the confinement of the mode within the III-V region. Consequently, the mode is less sensitive to nonlinear effects that arise in the III-V material when subjected to high current injection, such as localized spatial hole burning and gain compression. As a result, the laser remains single-mode for currents exceeding five times the threshold current.

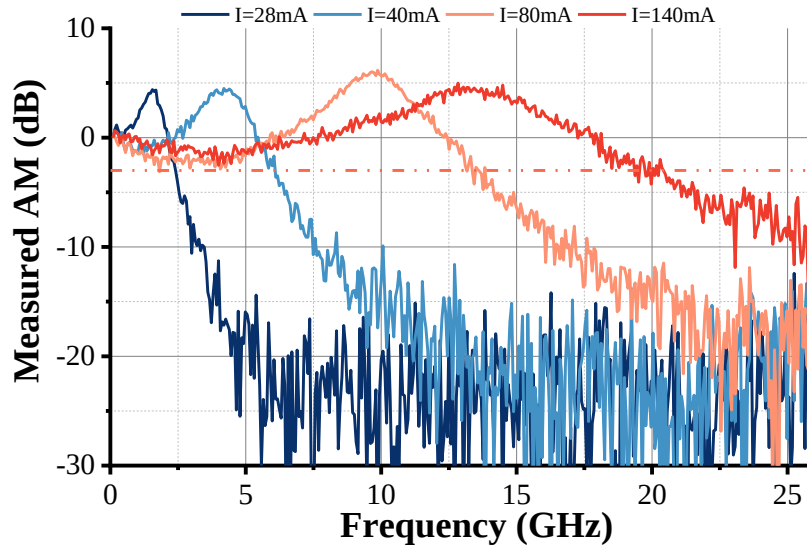


Figure 3.11: Small signal bandwidth of the DFB laser at different bias current values.

The maximum possible intrinsic modulation bandwidth f_{max} (3-dB bandwidth) is solely determined by the K-factor, a smaller K-factor yields superior performance for high-speed operation. The K-factor depends not only on the photon lifetime, but it is also significantly affected by nonlinear gain effects and carrier transport effects. As defined in equation 3-6, we can extract a photon lifetime $\tau_p = 3.7$ ps, and $1/\tau_c$ to the carrier lifetime of 0.17 ns. This short photon lifetime reduces the damping effect and maximizes the achievable f_{max} [22]. Those results are comparable with bandwidths of III-V on silicon lasers reported by other groups [12].

In this section, I have presented the small-signal EOBW properties of two different types of III-V on silicon lasers: the long-cavity and short-cavity devices. The long-cavity laser

demonstrated a photon lifetime exceeding 5 ps, contributing to a smaller modulation bandwidth. On the other hand, the short-cavity laser achieved a higher cut-off frequency of 21 GHz, making it suitable for high-speed data transmission, with the potential to support bit rates above 30 Gbps. In the following section, I will discuss the results of the large-signal modulation measurements performed on these lasers.

3.3.2 Large signal modulation

3.3.2.1 NRZ Modulation

The next step in the characterization procedure involves large signal measurements. Figure 3.12 shows a schematic representation of the setup for transmission measurements. A standard electrical source (Keithley 2400) is used to drive the laser, which can be combined with a Pseudo-Random Binary Sequence (PRBS) generator (Anritsu MU183020A) with adjustable pattern lengths for eye diagram and Bit Error Rate (BER) measurements. To prevent optical feedback, an optical isolator is employed. The signal then passes through segments of SSMF with adjustable lengths to simulate different transmission distances. The output is coupled to an Erbium-Doped Fiber Amplifier (EDFA) with 10 dB of gain, followed by a TBPf to amplify and filter the received signal to the desired level. On the receiver side, a variable optical attenuator (VOA) (Agilent 81577A) and a high-speed photodiode (XPDV-2320R) are used. The amplified photocurrent is either sampled by a high-bandwidth Keysight storage oscilloscope with a 70 GHz bandwidth (Agilent 86118A) for eye diagram measurements, ensuring that the receiver's bandwidth limitations are nearly negligible, with data stored for offline processing; or it is connected to the Anritsu MP1800A Signal Quality Analyzer, a multi-channel Bit Error Rate Tester (BERT), for BER measurements aimed at assessing the integrity of high-speed digital data signals.

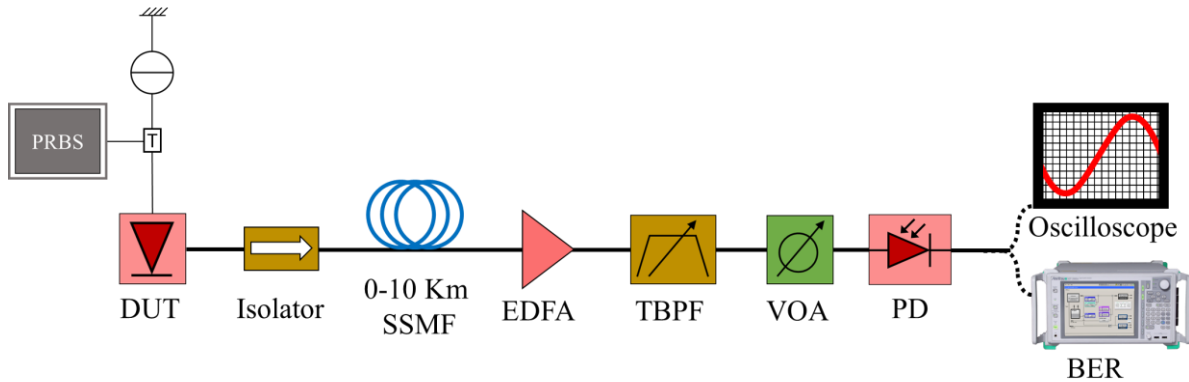


Figure 3.12: Experimental setup used for the transmission measurement of lasers.

$\lambda/4$ phase-shifted short-cavity device

Eye diagrams as function of modulation rate using $2^{31} - 1$ pattern length are shown in Figure 3.13 for the Back-to-back (B2B) case. The laser's bias current was set around 140 mA, and a swing voltage of 3.5 V peak-to-peak (assuming a 50 Ω load) was set to achieve large-signal modulation. It is worth to remark that such voltage values are nominal for a 50 Ω load. Since the laser is not terminated in 50 Ω , a lower voltage (> 1 V) swing amplitude is applied on the laser.

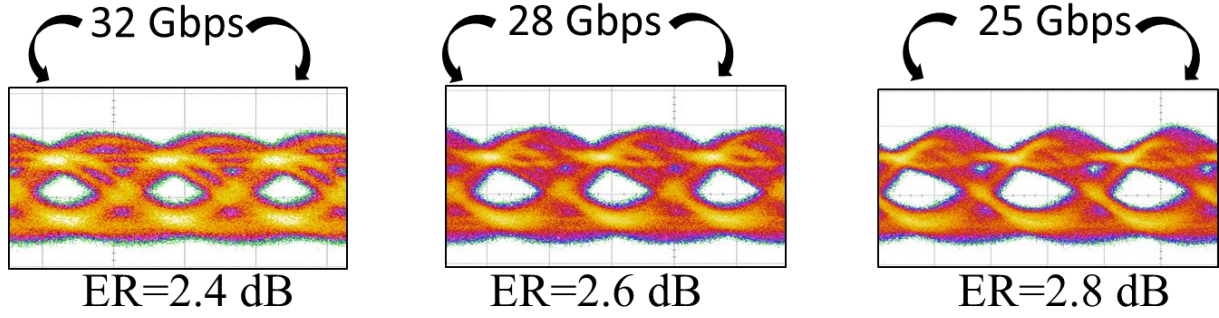


Figure 3.13: Measured optical eye diagrams for an NRZ signal with a PRBS $2^{31} - 1$ pattern at modulation speeds of 32 Gbps, 28 Gbps, and 25 Gbps in a back-to-back setup.

For BER measurements, we initially focused on testing the system's limits in terms of data rate and transmission distance. To do this, we used a shorter pattern length: $2^7 - 1$. Once the initial limits are established, we then compare the performance under various data rates and pattern lengths. Specifically, we compare the larger $2^{31} - 1$ pattern length, which is more representative of real data transmission, to observe how the signal degrades over longer distances and higher bit rates. First, BER measurements were conducted using 28 Gbps and 32 Gbps using $2^7 - 1$ NRZ PRBS pattern signals. These measurements were carried out in three distinct configurations: in a B2B setup and following a transmission over a distance of 1 km and 2 km of SSMF. The results are presented in Figure 3.14. The eyes are open at both 28 Gbps and 32 Gbps, even after propagating over 2 km of fiber. A dynamic extinction ratio (DER) ranging from 3.7 dB to 2.1 dB was measured, a degradation attributable to several factors such as increased data rate, losses, or fiber dispersion. Less than 1.5 dB power penalty at a BER of 10^{-3} was observed for 1 km transmission over SSMF at 28 Gbps. We confirmed that all bit error rates were below the hard-decision forward error correction threshold (HD-FEC), demonstrating the stable operation of the component within a relatively wide current range. For a transmission speed of 28 Gbps, error free results were successfully obtained even after 1 km of transmission. However, after 2 km of transmission, a notable degradation is observed, to the point where the B2B measurement at 32 Gbps (pink curve) is transposable to this one. Negligible power penalty was observed when comparing the 32 Gbps data rate B2B measurement (pink curve) with the 28 Gbps after transmission over 2 km (dark blue curve). However, it should be noted that we used SSMF. Additionally, we did not apply any electrical amplification at the receiver or perform signal equalization. A much longer transmission distances could be obtained by using non-zero dispersion shifted fibers (NZ-DSF) as shown elsewhere [24]. One of the key advantages of these lasers is their effective trade-off between high modulation speed and high modulated optical output power, which is typical of 3 mW. This characteristic is significant because high-speed lasers often face a limitation in modulated optical power due to their smaller size. A comparison with directly modulated lasers from NTT and IMEC highlights this advantage. NTT's lasers, while fast, typically exhibit lower coupled power of less than 1 mW [22].

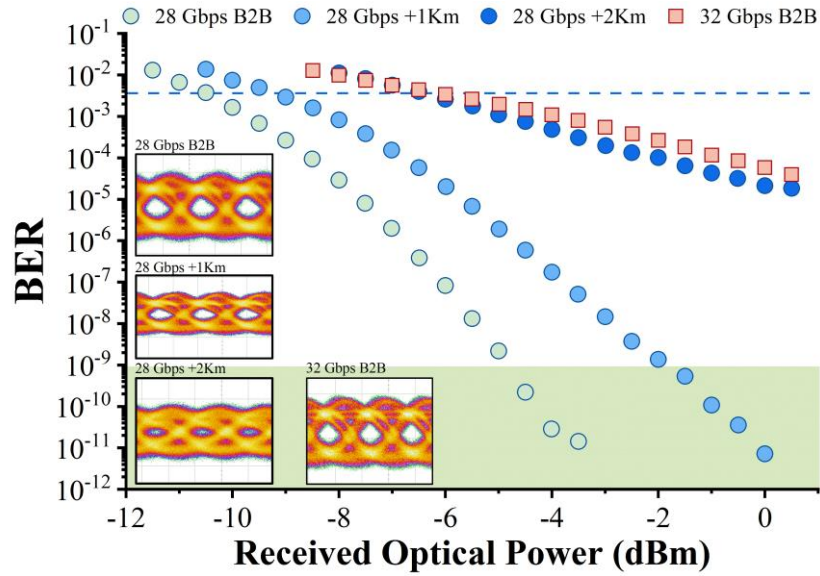


Figure 3.14: Measured BER at 28-Gbps and 32Gbps as a function of the received power. A laser current of $I_{\text{Laser}} = 140$ mA and a peak-to-peak voltage of $V_{pp} = 3.5$ V were used in all cases. The inset shows the eye diagrams at different modulation speeds and distances. Blue horizontal lines mark the HD-FEC limit, and green part represents the onset of error-free transmission.

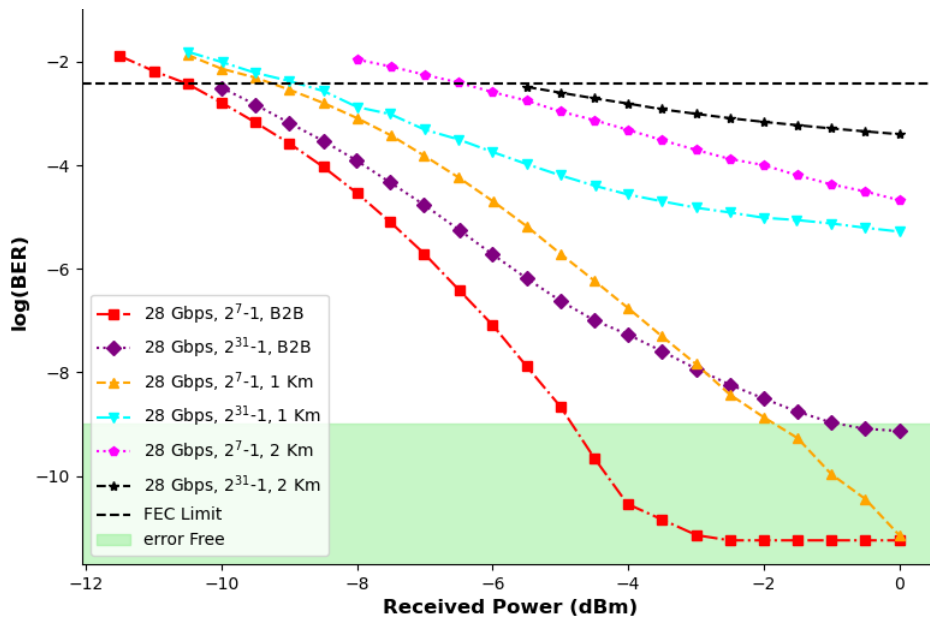


Figure 3.15: 28-Gbps BER measurement as a function of the received power for different PRBS pattern lengths, for B2B, 1 km and 2 km SSMF. Gray horizontal lines mark the HD-FEC limit, and green part represents the onset of error-free transmission.

Then, BER measurements were conducted to verify the data transmission quality at 28 Gbps using different PRBS pattern lengths (see Figure 3.15). The results indicate noticeable degradation in performance as the PRBS pattern length increases, particularly for longer word lengths. This degradation is linked to the low-frequency roll-off of the laser's small-signal response, which affects the transmission of longer sequences. The longer PRBS patterns introduce more low-frequency components, which the laser struggles to transmit efficiently due to its limited low-frequency response. This leads to increased inter-symbol interference (ISI) and consequently higher error rates.

To mitigate this issue, employing encoding techniques such as 8b/10b could be beneficial. This encoding method ensures a balanced transmission by limiting the run-length of consecutive identical bits, thus reducing the dependency on low-frequency components. It introduces DC balance and improves the spectral characteristics of the transmitted signal, potentially minimizing the pattern effect and enhancing the overall transmission reliability, especially at high data rates [41].

Sampled bragg grating Long-cavity device

Although the SBG-DFB laser is suitable for transmissions requiring a narrow optical linewidth, it is limited in high-speed applications due to its restricted electro-optic bandwidth of 14 GHz, its long cavity length of 880 μm , and the high photon lifetime within its cavity. We also evaluated the large-signal intensity modulation of the SBG-DFB laser using data transmission experiments with an NRZ signal and a $2^{31}-1$ PRBS pattern length. We chose this pattern length and limited the data rate to 25 Gbps. The DC bias current operating point was set to 100 mA. A voltage swing of 2 V peak-to-peak was used.

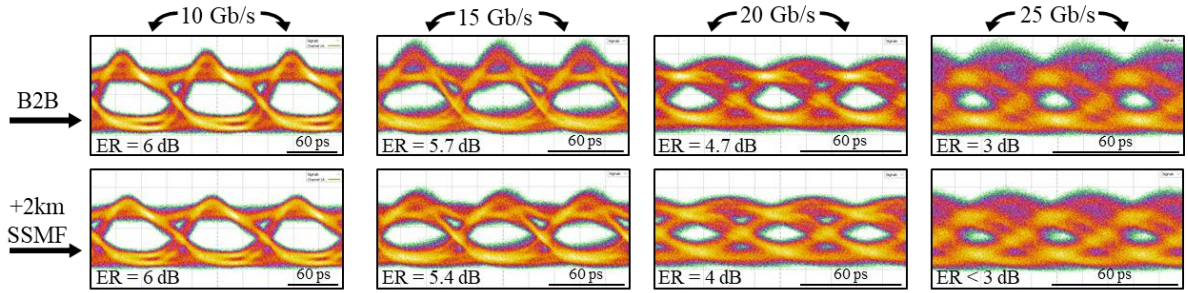


Figure 3.16: Measured optical eye diagrams obtained from an NRZ modulation signal with a PRBS pattern length of $2^{31} - 1$ for various modulation speeds (10 Gbps, 15 Gbps, 20 Gbps and 25 Gbps) and for a back-to-back configuration or after transmission over a 2 km optical fiber.

Figure 3.16 shows open eye diagrams for a B2B configuration and after transmission over a 2 km SSMF, with a gradual degradation of the dynamic extinction ratio (DER) from 6 dB at 10 Gbps down to 3 dB at 25 Gbps modulation speed. A moderate penalty of less than 1 dB is observed when comparing the B2B measurements with the transmission over the SSMF. These results highlight the limitations of SBG-DFB lasers as directly modulated sources. On the other hand, their narrow linewidth makes them highly attractive for hybrid schemes that combine DML transmission with advanced receiver-side techniques for intensity and phase recovery [25].

3.3.2.2 Pulse Amplitude Modulation (PAM-4)

The experimental setup for PAM-4 direct modulation of the III-V-on-silicon DFB laser (Device Under Test, DUT) is depicted in Figure 3.17. To drive the transmitter module, a standard electrical source is utilized in conjunction with a Digital-to-Analog Converter (DAC). The DAC is driven by two NRZ electrical signals carrying decorrelated PRBS of lengths $2^7 - 1$ bits at 25 GBd, originating from a pattern generator. We implement a delay on one arm to align these two signals before injecting them into the DAC. Subsequently, the two streams are merged and converted into a four-level electrical signal. The converter is then connected to a two-stage variable-gain RF-linear amplifier (SHF S807 C) with ultrawide bandwidth. Finally, the four-level amplified electrical signal directly drives the laser. The laser's bias current was set

around 140 mA, and a swing voltage of 3.5 V peak-to-peak (assuming a 50 Ω load) was set to achieve large-signal modulation. The optical signal generated by the laser is collected from the vertical grating coupler using a lensed fiber. To prevent optical feedback, an optical isolator is employed. The output was connected to an EDFA with 10 dB of gain and a TBPF to amplify and filter the received signal accordingly. On the receiver side, a 3dB optical attenuator is placed before a high-speed photodiode (XPDV-2320R). The amplified photocurrent is sampled by a high-bandwidth Keysight storage oscilloscope with a bandwidth of 70 GHz (Agilent 86118A). This ensures that the receiver's bandwidth limitations are nearly negligible, and the data is stored for offline processing.

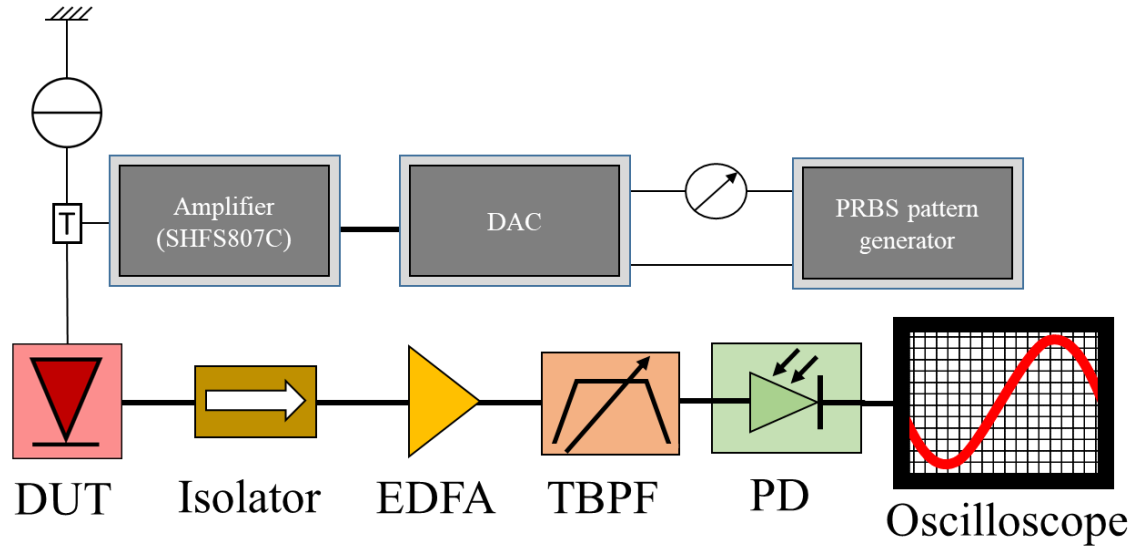


Figure 3.17: Schematic of the setup for a PAM-4 direct modulation experiment.

We subjected the III-V/Si DML to a multi-level modulation scheme to evaluate its performance under high bit-rate PAM-4 modulation. These measurements were performed on the previously discussed $\lambda/4$ phase-shifted DFB design. The modulation was tested at 25 Gbps, a commonly used data rate, and at 50 Gbps (25-Gbaud) to assess its capabilities at higher data rates. The electrical signals after the PAM-4 multiplexer were measured, as shown in Figure 3.18-a. Figure 3.18-b presents the resulting optical eye-diagrams for B2B transmission, without any equalization. The results at 25 Gbps demonstrated stable signal quality with PAM-4 modulation, confirming the device's ability to reliably support transmission at this speed. The modulation was evaluated using a PRBS length sequence pattern (2^7-1) in a B2B configuration. However, at 50 Gbps PAM-4 modulation, the limitations of the DAC and distortion introduced by the RF amplifier become apparent, as seen in the input eye diagram (Figure 3.18-a). This likely contributes to the degradation observed in the optical eye diagram. Another contributing factor is that the laser may have reached the bandwidth limit of its modulation capabilities, although it still successfully produced four distinct optical levels.

It is important to note that this experiment was conducted on a standalone DML without an external cavity and without any digital signal processing (DSP) at the receiver. With appropriate DSP and efficient equalization, the noise margin could be significantly improved.

Despite these limitations, the results suggest that the III-V/Si DML is suitable for 50 Gbps PAM-4 modulation, demonstrating its potential for high-speed optical communication systems.

In this section I presented the direct modulation properties of III-V on silicon lasers which are compatible with high speed operation. In the next section I will discuss its chirp properties.

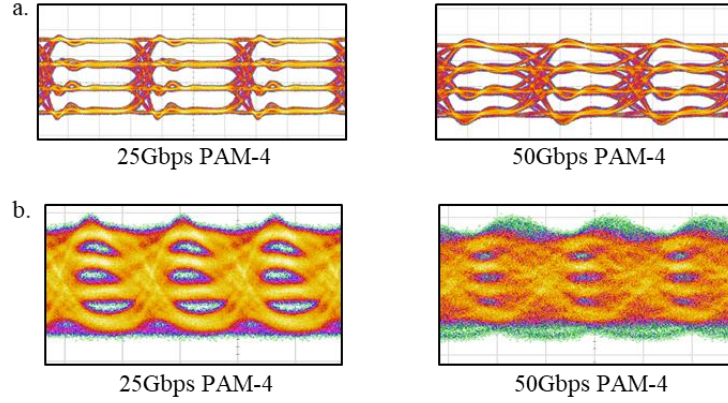


Figure 3.18: (a) Electrical eye diagrams after the MUX for 25 and 50Gbps PAM-4 signals. (b) Optical eye diagrams for 25 and 50 Gbps PAM-4 signals.

3.3.3 Chirp measurement

To investigate the chirp properties of the component, we measured both the frequency and amplitude response of the device under large-signal modulation. The measurement was performed using the experimental setup depicted in Figure 3.19.

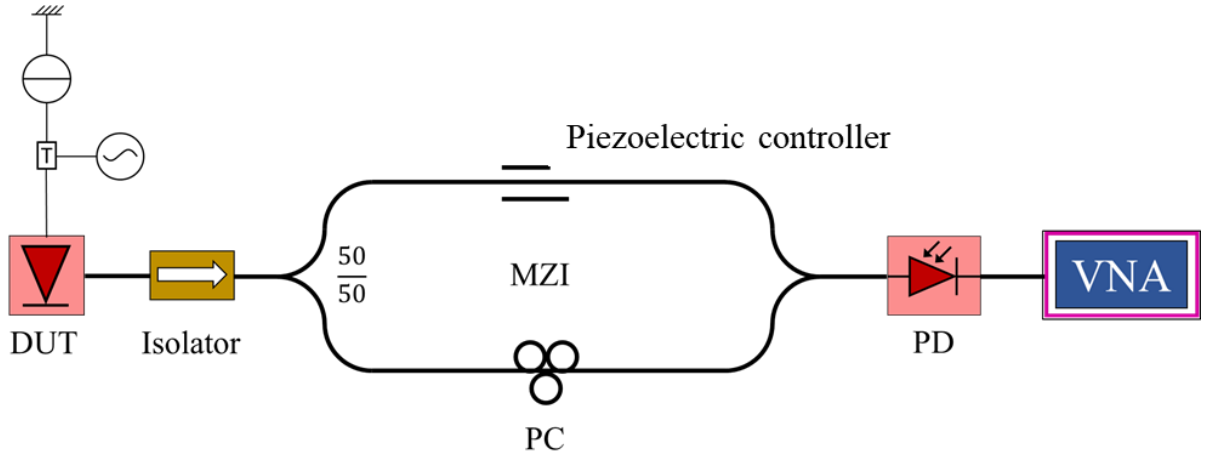


Figure 3.19: Experimental setup for measuring amplitude and frequency response.

We measure intensity and frequency modulation response with a Mach-Zehnder interferometer (MZI) locked on the laser wavelength with a feedback loop.

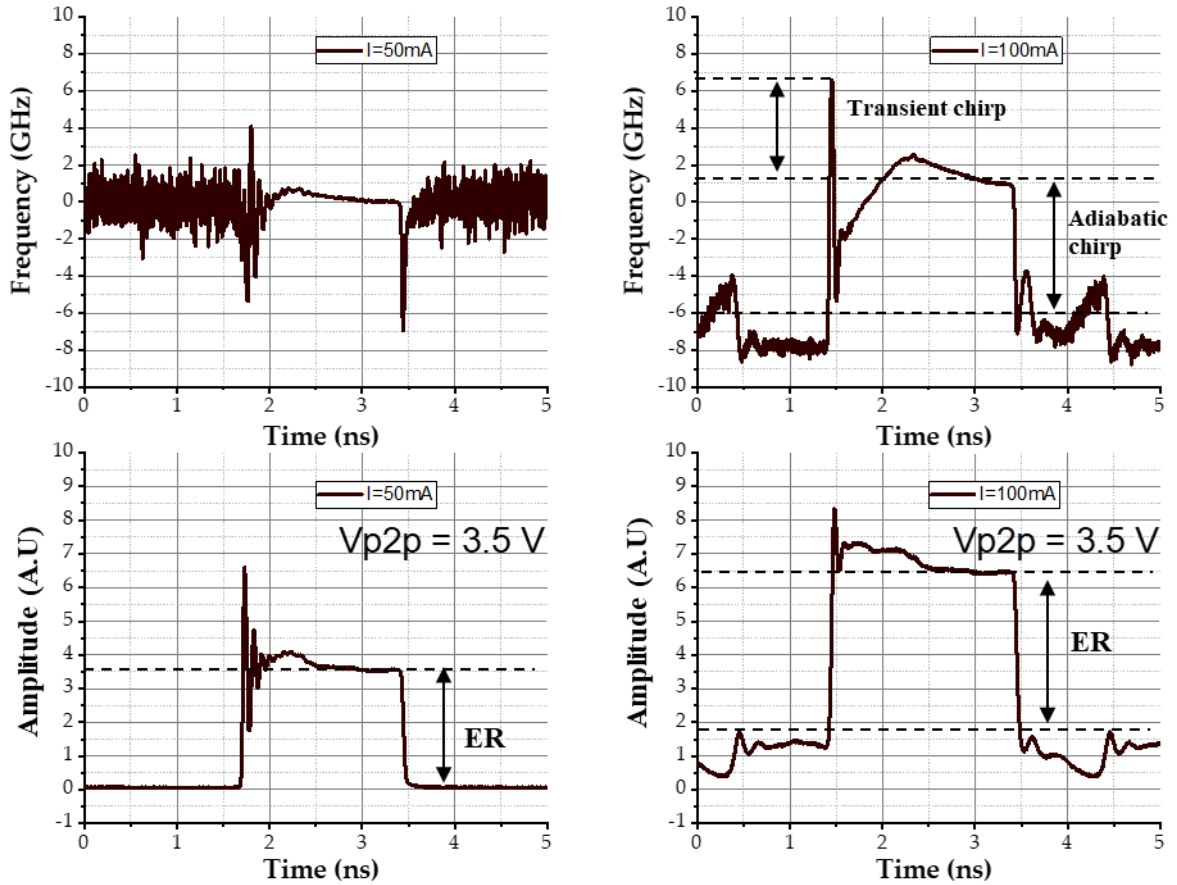


Figure 3.20: Large signal frequency (top figures) and amplitude (bottom figures) response for different bias current.

The MZI converts frequency fluctuations into intensity fluctuations, allowing us to analyze the chirp characteristics of the laser. We measured the intensity and frequency modulation response using a MZI, locked to the laser wavelength through a feedback loop. The output signal from the laser first passes through an optical isolator and then injected into the Mach-Zehnder interferometer. The free-spectral range (FSR) of the MZI is the inverse of the differential delay $|T_2 - T_1|$ between the two arms, T_1 and T_2 correspond to the propagation times in the two arms, respectively.

A polarization controller is used to maintain parallel polarization states of the two signals. This guarantees optimal interference at the output. To accurately control the optical path difference between the arms, a cylindrical piezoelectric transducer is installed in one arm of the MZI. This transducer adjusts the optical path length by altering the fiber length, and it is directly controlled by an external locking circuit. After passing through the interferometer, the signal is analyzed by a VNA. The MZI allows us to decouple the amplitude and frequency response of the device, providing a clear view of the laser's chirp characteristics. Detailed explanations and derivations of this method can be found in [42].

These measurements were performed on the previously discussed $\lambda/4$ phase-shifted DFB design. From the frequency response measurements, we extracted both the transient and adiabatic chirp, which are influenced by the bias current and swing voltage. The large-signal amplitude and frequency responses are shown in Figure 3.20 for different bias currents and a

swing voltage of 3.5 V. The large-signal amplitude response displays an extinction ratio between 3.5 dB and 5 dB. From the large-signal frequency response, it is possible to distinguish the transient and adiabatic chirp. As described in [43], adiabatic chirp is defined as the frequency shift between two different intensity states. In the case of NRZ OOK modulated signals, this frequency deviation occurs between the power levels 0 and 1. On the other hand, transient chirp refers to the frequency variations observed during the rising and falling edges of transitions between levels. It is caused by refractive index modulation induced by transient carrier variations in the laser cavity that accompany any regime change. A rapid increase in injection current is followed by a rise in carrier density, which in turn increases the photon density and the rate of stimulated emission. The device shows a positive adiabatic and transient chirp of 6 GHz. Positive adiabatic chirp is typical in lasers without reflective facets (such as anti-reflection-coated or integrated lasers), as explained in [44] and [45]. Both adiabatic and transient chirp degrade the eye diagram after transmission. However, as noted in [43], adiabatic chirp has a significantly weaker impact on transmission quality.

Now that we have performed the characterization of our components, we will compare them with the state of the art. As shown in Table 3.1, the fabricated devices are competitive with state-of-the-art III-V/Si DFB lasers demonstrated by other research institutions.

State of the art	Group	Year	λ (nm)	I_{th} (mA)	L (μm)	κ (cm^{-1})	WPE (%)	On-chip P_{out} (mW)	EOBW (GHz)	Modulation Speed (Gbps)
J. Rahimi et al. [23]	IMEC	2022	1550 C-band	10.5	200	50	17	6	15	20
A. Gallet et al [49]	III-V Lab	2018	1550 C-band	35	600	-	-	-	13	25
C. ZHANG et al [50]	UCSB	2016	1550 C-band	7.5	400	-	3	5	24 @ 6dB	40
A. Abbasi et al [51]	IMEC	2017	1550 C-band	32	340	200	3.3	5	34 (PPR extended)	43
S. Yamaoka et al [12]	NTT	2023	1310 O-band	2.9	60	600	-	0.82*	100 (PPR extended + SiC substrate)	112
S. Yamaoka et al [22]	NTT	2021	1310 O-band	2.75	50	1000	-	0.73*	108 (PPR extended + SiC substrate)	256 – PAM4
This work-SBG [2]	III-V Lab	2022	1550 C-band	35	880	37	17	20	14	25
This work- $\lambda/4$ [1]	III-V Lab	2023	1550 C-band	26	400	54	7	12	21	50 – PAM4

Table 3.1 : Representative work on heterogeneously integrated III-V/Si DMLs.

In the following section, we will examine how optical feedback affects transmission reach.

3.4 Feedback and Stability Analysis

Semiconductor lasers, particularly Index-Coupled DFB lasers, are highly sensitive to optical feedback. Even a slight reflection of signal of less than 1% within the laser cavity can significantly impact their performance. Undesired reflections have long been recognized as potential disruptive elements that can be detrimental to the stability and performance of lasers [26], [27]. External Optical Feedback can exert a considerable influence on

* The maximum optical output power coupled to the fiber.

semiconductor laser performance. Among the wide range of potential effects, whose manifestation depends on parameters such as the external cavity length and the strength of external optical feedback, Coherence Collapse (CC) stands out as the most influential and penalizing factor, affecting the stability and spectral purity of the device [28]. Traditionally, optical isolators are used to address this issue, but finding a balance between minimal insertion losses and good isolation power in commonly used isolators can be challenging. Furthermore, it is crucial to note that one of the main motivations for using III-V-on-silicon heterogeneous circuits is their compactness. However, the large size of isolators and their incompatibility with CMOS manufacturing processes have so far hindered their integration, inevitably leading to cost increases sometimes exceeding that of the laser chip itself. Therefore, designing and producing lasers that are ultra-stable and insensitive to optical feedback would constitute a necessary step for future photonic technologies. More and more researchers are considering the design and realization of feedback-insensitive lasers to do away with optical isolators.

In response to these challenges, the development of lasers insensitive to optical feedback has become a key objective in silicon photonics integration, with the intention of reducing the risk of undesired reflections. The critical feedback level, denoted as f_{crit} , for a semiconductor laser, is defined as the fractional proportion of the laser's output power that can be externally reflected and reintroduced into the cavity before coherence collapse occurs. In other words, it represents the maximum feedback ratio that can be practically tolerated to ensure stable laser operation within a communication system [29], is given as:

$$f_{crit} = \frac{\tau_p^2 \gamma^2}{16C_e^2} \left[\frac{1 + \alpha_H^2}{\alpha_H^4} \right] \quad (3 - 7)$$

Where τ_p the photon roundtrip time in the laser cavity, γ the damping factor, $C_e = \frac{1-R}{2\sqrt{R}}$ is the coupling strength of the laser cavity to the external cavity with facet reflectivity R and α_H the linewidth enhancement factor. Equation 3 – 7 can be harnessed for the design of feedback-insensitive lasers. This can be realized through: i.) minimizing the α_H -factor, ii.) enhancing the damping factor γ , or iii.) achieving a high Q-factor, signifying a low-loss laser with an extended photon lifetime. A recent study [30] highlighted two demonstrations that exhibit a near-insensitivity to reflection. The first device is a DFB laser, derived from a previous batch of lasers manufactured at III-V Lab, III-V-on-silicon, with high-quality factor [31]. The second is an epitaxial Quantum Dot (QD) Fabry-Perot laser on silicon [32]. It is noteworthy that lasers based on quantum dots typically exhibit a lower α_H value due to the three-dimensional carrier confinement, when compared to quantum well lasers. This characteristic renders QD-based lasers less susceptible to adverse effects of optical feedback [33]. However, it is important to emphasize that the fabrication of QD lasers is currently more intricate and less well-controlled, posing an additional challenge in their utilization and advancement. Other achievements include InAs/GaAs QD lasers integrated on a silicon platform operating without an optical isolator for basic I/O applications [34], as well as semiconductors equipped with an intracavity optical isolator [35], Fano nanolasers in photonic crystal [36], and parity-time symmetric Distributed Feedback (DFB) lasers [37], among others, demonstrating significant advances in the field. However, despite the notable progress thus far, a signal source completely

insensitive to feedback has yet to be reported, particularly at the scale of complete systems. Therefore, studying the resistance of lasers to feedback is an important parameter for their characterization. In this section, the previously discussed $\lambda/4$ phase-shifted designs of DFB lasers subjected to external feedback will be presented. We studied first the non-linear dynamics of the component in static regime by observing the appearance of CC, signaling the beginning of chaotic instability, at different polarization currents by comparing $\lambda/4$ phase-shifted DFBs with different coupling coefficients. This will be accomplished using Chraplyvy and Tkach's mapping [38] to analyze this phenomenon. Chaotic behavior is characterized by a reduction in coherence and spectral broadening of the laser line. We then attempted to modulate our laser under different levels of feedback at 28 Gbps, and analyzed the transmission of our component under these conditions. Its impact on the optical spectrum, eye diagram, and BER is discussed.

3.4.1 Results and discussion

Optical feedback occurs when a part of the generated optical beam is redirected back into the laser cavity. By creating an external cavity capable of redirecting signal into the laser cavity, for example, or by aligning a mirror with the laser beam's path, an optical feedback mechanism is established. The feedback strength, denoted as f_{ext} , quantifies the proportion of laser's emitted optical power that is reintroduced into the laser cavity. It is defined as the ratio of the reflected optical power to the output power of the laser facet subject to this optical feedback.

$$f_{ext} = \frac{P_{reflected}}{P_{emitted}} \quad (3 - 8)$$

The distance between the laser and the mirror, denoted as L_{ext} , is related to the time it takes for signal to complete a round trip in the external cavity, defined as:

$$\tau_{ext} = \frac{2n_{ext}L_{ext}}{c} \quad (3 - 9)$$

Where n_{ext} is the refractive index of the external cavity, and c represents the speed of light in a vacuum.

The device under study is the short-cavity $\lambda/4$ phase-shifted DFB previously studied. The device had a grating depth of 30 nm and a grating tooth width of 2.25 μm . Its L-I-V curve can be found in Figure 3.3-a.

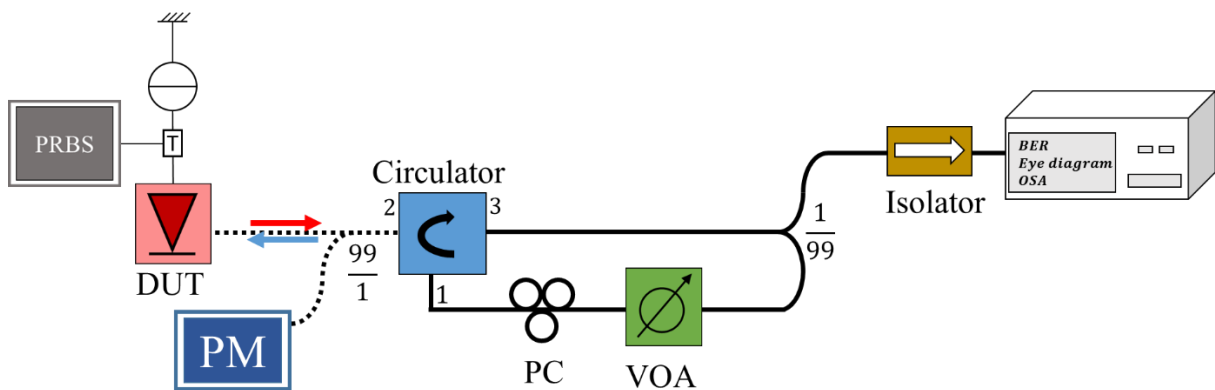


Figure 3.21: Schematic diagram of the experimental setup used for optical feedback.

The schematic diagram of the experimental setup is shown in Figure 3.21. To drive the laser, we use a standard electrical source that can be combined with a Pseudo-Random Binary Sequence (PRBS) generator (Anritsu MU183020A). The optical signal generated by the laser is collected from the vertical grating coupler using a lensed fiber with 4 dB coupling loss. The laser threshold current at room temperature (@20°C) was 26 mA. The optical signal passed through a 99/1 coupler before traversing an optical circulator. A second 99/1 splitter is used to direct 1% of the light through an isolator; this 1% is then either sent to an optical spectrum analyzer (OSA) to observe the optical spectrum or to an EDFA with 10 dB of gain and a TBPf to amplify and filter the received signal accordingly. The signal was then received by a high-speed photodiode (XPDV-2320R) after passing through a 3 dB optical attenuator. The amplified photocurrent is then either sampled by a high-bandwidth Keysight storage oscilloscope with a bandwidth of 70 GHz (Agilent 86118A) to generate the eye diagram or transmitted to the Anritsu MP1800A Signal Quality Analyzer, a multi-channel BERT for BER measurements aimed at assessing the integrity of high-speed digital data signals. The remaining 99% of the signal was controlled by a VOA and a PC and then sent to the circulator and subsequently into the 99/1 coupler. 1% of the reinjected signal was directed to a power meter to measure the power reinjected into the laser, with the remainder being sent back into the laser. The PC was adjusted to obtain the maximum feedback effect under a given feedback level, simulating the worst-case scenario for the laser. A PRBS length of $2^7 - 1$ was used with a $2.2 V_{\text{peak-to-peak}}$ amplitude to record the eye diagram and estimate the BER output at 28 Gbps. The length of the external cavity, L_{ext} , is constant, on the order of several tens of meters, implying a long-delay optical feedback regime.

In what follows, the experiments are all performed at room temperature. Figure 3.22 shows the spectrum evolution of the single-modes DFB under different levels of feedback. Here, the optical feedback strength f_{ext} defined as the ratio between the reflected power and the emitting power, normalized to the maximal level, thus ranging from 0% to 100%. The maximum attainable feedback amount corresponds to -12dB. We can observe the transition from the feedback-free DFB spectrum to the onset of chaotic instability, characterized by a reduction in coherence and spectral broadening of the laser linewidth. Starting from a feedback level of -15 dB, the spectrum broadens significantly, ultimately reaching the coherence collapse state. Figure 3.23 illustrates the evolution of the onset of CC as a function of injection current above the threshold. For comparison purposes, we also fabricated 400 μm long and 2.472 μm grating tooth width DFB laser based on the same quantum well (QW) structure, with a coupling coefficient κL of 3.2 (orange curve). The coupling coefficients were measured from the stopband observed in the optical spectrum. The resistance to external optical feedback, as well as the onset of CC in DFB lasers are closely linked to their coupling coefficients κ . The laser with a higher coupling factor was found to be more resistant to feedback at the same injection current, since the guided propagation mode is better coupled with the index grating. However, the laser with $\kappa L = 3.2$ becomes multimode at high injection currents due to the appearance of lateral modes, whereas the laser with $\kappa L = 2$ (blue curve) remains single-mode even when high polarization currents are injected, thus achieving the same feedback rates demonstrating high feedback tolerance while having a better f_r , as seen previously. Helms and Petermann proposed a linear relationship that establishes a connection

between the onset of coherence collapse and the injection current when feedback is generated by a long external cavity [39], a behavior that is also observed in our measurements shown in Figure 3.23. We also note that the behavior of hetero-integrated DFB lasers is similar to that of traditional InP-QW DFB lasers when subjected to fibered feedback, indicating that as the injection current increases, the reinjection rate required to trigger chaotic operation also increases. For further comparison, we have plotted on the same graph the feedback rate necessary for the onset of coherence collapse, where applicable, for other components reported in the literature that have conducted similar studies. According to Figure 3.23, in the gray region, QD lasers (red) and high-Q-factor lasers (brown) are less sensitive to optical feedback, as reported in studies [33] and [40], respectively. In these studies, these lasers exhibited no performance degradation even under high optical feedback conditions (-9 dB and -14 dB, respectively) and did not enter the CC regime. Which confirms what we said above. For lasers based on InP [33] or III-V/Si from the previous fabrication run [47], coherence collapse typically occurs at a feedback level of approximately -20 dB, in contrast to our demonstration.

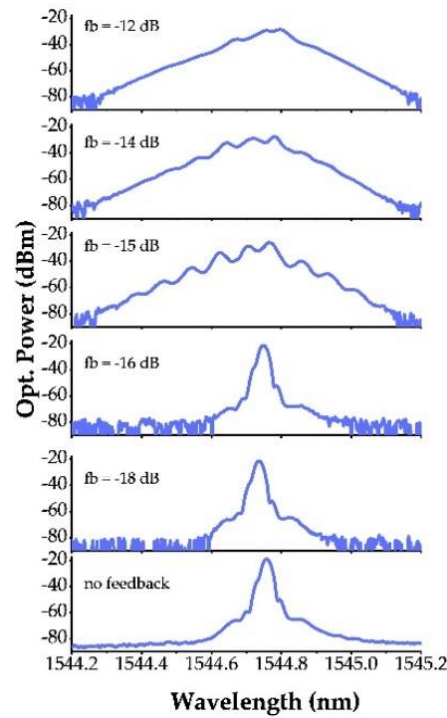


Figure 3.22: Comparison of measured optical spectra with different level of optical feedback.

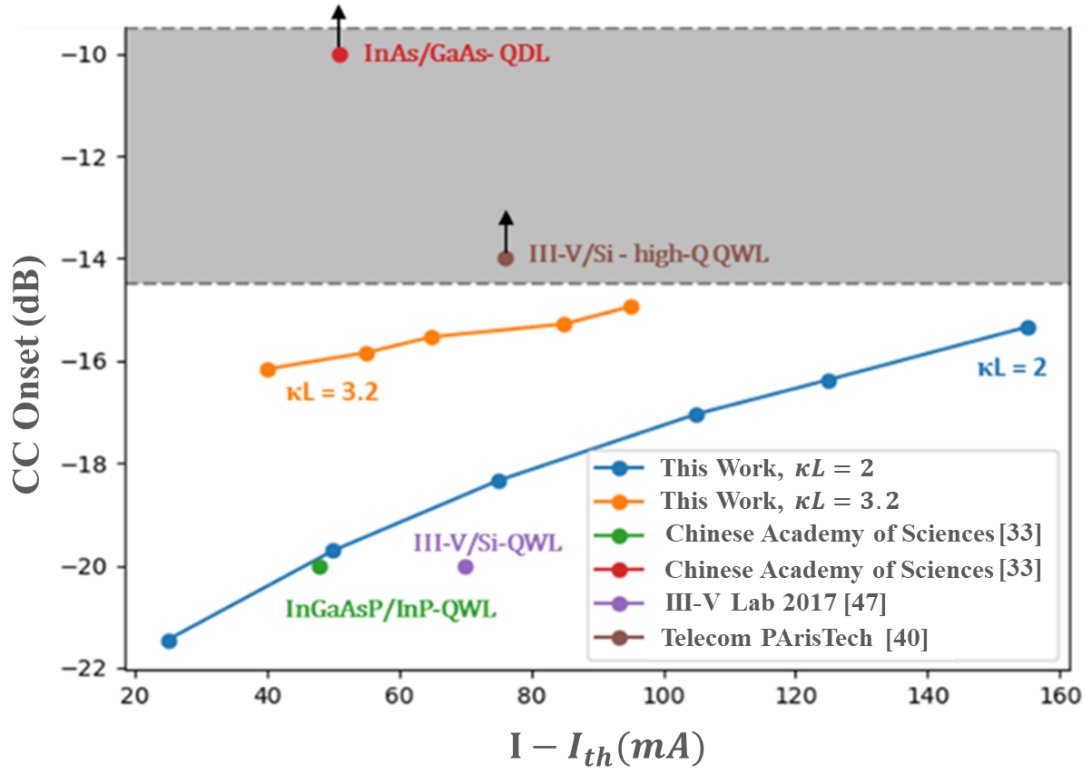


Figure 3.23: Onset of coherence collapse as a function of current for different DFB lasers.

Then, the modulation performance under feedback was conducted at room temperature. In these measurements, the QWL were biased at $7 \times I_{th}$, with an oscillation voltage set to $2.2 V_{peak-to-peak}$. The polarization of feedback signal was adjusted to ensure that all measurements were conducted under the polarization state where the BER performance of the lasers was the worst. The BER curves and eye diagrams under different feedback levels are shown in Figure 3.24 for B2B measurements. At the feedback free level, the laser maintaining a BER well below 10^{-9} , indicating error-free operation. However, as the feedback level increases to -32 dB, the effect of additive noise due to amplified modulated feedback within the laser cavity becomes apparent. This leads to a BER floor (orange circles) around 2×10^{-8} . Despite this, the power penalty at BER of 1×10^{-3} remains below 1 dB for -32-dB feedback and feedback free transmission, indicating that moderate feedback does not severely impair performance at this threshold. The BER performance deteriorates progressively with increasing feedback levels. As the feedback level increases from -32 dB to -22 dB, there is a relatively moderate degradation of the signal for these three cases. The BER floors appear around 2×10^{-8} , 1×10^{-7} and 1×10^{-6} for the feedback levels of -32 dB, -27 dB, and -22 dB, respectively. The BER curves (orange, green, and red) do not deteriorate significantly, and the power penalty at a BER of 1×10^{-4} remains below 1 dB for all three signals. This indicates that the CC regime has not yet been reached. However, as the feedback level is further increased beyond -17 dB (purple circles), there is a noticeable degradation in performance, indicating that the system is approaching the CC regime. This suggests that higher feedback levels lead to more pronounced noise and instability within the laser, pushing the system closer to the CC regime, where the laser's coherence and signal integrity are significantly compromised, confirming the observations made on the optical spectra (Figure 3.22). An improvement of eye diagram and

BER curve is expected using a NZ-DSF. Similar studies conducted on previous batches of III-V/Si DFB DML lasers at the III-V Lab provide valuable context. A. Gallet et al reported a transmission rate of 2.5 Gbps [47]. Additionally, S. Gomez et al. demonstrated a high-Q III-V/Si DFB achieving error-free transmission at 10 Gbps under optical feedback [48]. Furthermore, other research noted error-free transmission rates of 10 Gbps for InP-based DFB lasers, both QW and QD, achieving in back-to-back configurations. However, it is important to note that these studies were conducted at significantly lower modulation speeds compared to our work, which operates at 28 Gbps. Although error-free transmission was not attained under feedback conditions in our tests, all measurements remained below the HD-FEC threshold, except at -12 dB feedback. A key takeaway is that our component maintains an optimal balance between modulation speed, output power and resistance to feedback, underscoring its potential for high-speed communication applications.

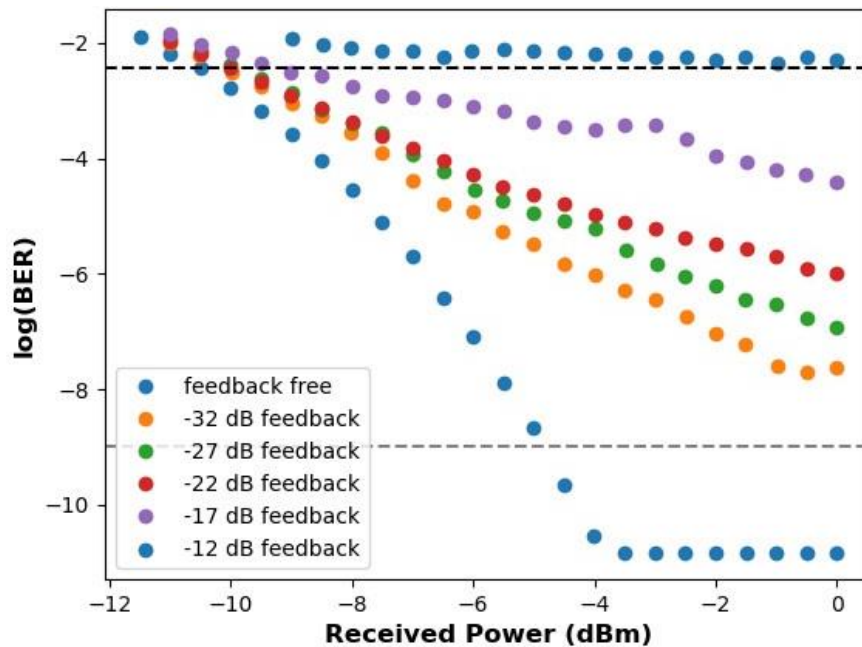


Figure 3.24: Bit-error-rate of 28 Gbps in back-to-back transmissions under various feedback levels.

In summary, the use of III-V heterogeneous integration on silicon photonic circuits opens new prospects for high-speed, low-cost, and energy-efficient optical communications. We successfully demonstrate the high optical feedback tolerance of DFB lasers for high-speed direct modulation. A direct NRZ-modulation data transmission at 28 Gbps under various feedback levels was performed to evaluate the DML, showing that the laser maintains robust performance with minimal power penalty even in the presence of significant optical feedback. Our experiments indicate that DFB lasers with a lower coupling coefficient $\kappa L = 2$ remain single-mode and stable under high polarization currents, achieving the same feedback tolerance as lasers with higher coupling coefficients, which become multimode. This makes them an ideal candidate for future high-speed, isolator-free optical communication systems. These results emphasize the viability of III-V-on-SOI DFB lasers for integration into photonic

circuits without the need for optical isolators, thereby simplifying design and reducing costs. The demonstrated stability at high data rates and resistance to optical feedback underscores the potential of these lasers for future high-speed, isolator-free optical communication systems.

3.5 Chapter Conclusion

In this chapter, we have investigated two types of heterogeneously integrated DFB lasers, each tailored for different application scenarios.

First, we demonstrate a low- κ , high-power and narrow linewidth ($\Delta\nu = 118$ kHz) distributed feedback laser based on a sampled Bragg grating (SBG-DFB) to achieve both reduced grating strength and lower intracavity losses. This enabled single-mode operation with high output power and a narrow spectral linewidth as low as $\Delta\nu = 118$ kHz. Such characteristics make the SBG-DFB laser highly attractive for applications where spectral purity and coherence are critical, for instance in coherent communication systems, microwave photonics, or sensing applications requiring narrow linewidth sources.

Second, we demonstrated a $\lambda/4$ phase-shifted DFB laser optimized for high-speed direct modulation. Thanks to its high intrinsic electro-optical bandwidth, reaching up to 21 GHz, this device enabled successful transmission of NRZ-OOK signals at multiple data rates in both back-to-back and transmission experiments over up to 5 km of SSMF. While some degradation is observed at higher symbol rates or longer distances, the laser shows strong potential for short-reach interconnects. Furthermore, advanced modulation formats such as PAM-4 at 50 Gbps were successfully transmitted, confirming the suitability of this architecture for next-generation datacom links. In addition, the device exhibited a remarkable tolerance to optical feedback, further reinforcing its relevance for high-speed IMDD transceivers.

Together, these results highlight the complementarity of the two approaches: the SBG-DFB laser targets narrow-linewidth applications where coherence is essential, while the $\lambda/4$ phase-shifted DFB laser addresses high-speed short-reach optical interconnects where bandwidth and modulation performance are the primary requirements.

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4. External modulation: Laser-modulator integration for access networks

In the previous chapter, we examined directly modulated lasers, we have put forward the dynamical behavior and we have observed and analysed the limitations in terms of bandwidth they impose. This chapter focuses on the integration of an EAM and a laser within a III-V-on-SOI heterointegration platform for WDM transmitters. We will first examine the EAM as a single object by studying the key parameters such as the optical modulation amplitude (OMA), the insertion loss or the EO bandwidth. Subsequently, we will integrate the previously studied DFB lasers (but this time, operating under direct current conditions) with these modulators. Then, to compensate for the insertion loss caused by the EAM, we will add an SOA.

In the remainder of the chapter, I will present the work done to integrate tunable lasers, EAMs and SOAs to demonstrate a widely tunable high-speed transmitter for access networks. I will analyze the performance of NRZ signals at data rates of 32 Gbps and beyond, investigating the BER and transmission metrics across various wavelengths and distances. The measurements indicate error-free transmission for a wide wavelength range in B2B configuration and with moderate power penalties—up to 2.8 dB for an 8 km transmission through a SSF.

Furthermore, we explore large-signal modulation experiments at 50 Gbps and 56 Gbps. The static and dynamic characteristics of the transmitter for different wavelength. Finally, the last section of this chapter will provide a brief discussion followed by a conclusion. The results discussed in this chapter have been published in [1], [2], and [3].

4.1 Introduction to External Modulation

External modulation is a technique used to modulate the optical signal from a laser using an external modulator, rather than modulating the laser source directly. This approach separates signal generation from signal modulation, thereby improving signal quality in optical systems. This helps mitigate undesirable effects such as noise, distortion, and frequency drift, which are more pronounced in direct modulation schemes. It is essential in high-speed systems where rapid and precise modulation is critical. Externally modulated lasers are crucial for high-speed optical communications, although they come with high fabrication and integration costs [4]. The integration of III-V materials with silicon photonics presents a promising pathway to create high-density, low-cost, and scalable devices for the next generation of transceivers [5, 6]. Two primary types of modulators are employed in external modulation: EAMs and MZMs. The Mach-Zehnder modulator operates by splitting the signal into two paths, with an electrical signal applied to one or both arms of the modulator. This alters the refractive index, leading to constructive or destructive interference when the beams are recombined, thereby modulating the signal [7]. MZMs offer low noise and low distortion levels [8], but they are

larger in size, more complex to fabricate, and consume more power compared to EAMs [9]. In contrast, the EAMs utilize the principle of optical absorption, where applying an electrical signal increases the semiconductor's absorption band edge, thus modulating the intensity of the signal [10]. EAMs are more compact and easier to integrate with lasers [11]. They consume less power than MZMs and are highly suitable for high-speed communications, making them the preferred choice for compact and energy-efficient systems [12]. For optical data encoding, III-V electro-absorption modulators that leverage the quantum-confined Stark effect (QCSE), are ideal candidates. Unlike Mach-Zehnder modulators, EAMs operate at lower drive voltages, do not require complex traveling-wave electrode designs, and exhibit lower insertion loss [13, 14]. Their key strength lies in the potential for integration with lasers, various integration methods, such as butt-joint integration [10], selective area growth (SAG) [15], and quantum well intermixing [16], have been used to create reliable and efficient externally modulated lasers. However, these techniques require sophisticated process control and increase costs due to the III-V regrowth.

To simplify device fabrication and reduce costs, a shared active layer stack of multiple quantum wells has been proposed for the laser, the EAM, and the SOA [17]. In this context, the use of a scalable technology becomes crucial to further reduce the cost-per-chip. The integration of III-V materials on a silicon photonics platform at wafer scale offers significant advantages over competing technologies. Leveraging advanced CMOS manufacturing facilities, III-V-on-silicon photonic integrated circuits can be fabricated on 200 and 300 mm wafers with mass production capabilities [18]. Additionally, these circuits benefit from the wafer-scale testing and packaging solutions developed within the silicon photonics community [19]. Integrating multiple laser sources with a modulator enables their use for various applications. In the following, we will examine the operating principle of a lumped-EAM developed on our heterogeneous III-V/Si platform. This modulator shares the same III-V structure as the components previously described. We will focus on optimizing its parameters to achieve a balance between performance and fabrication simplicity.

4.2 Electro-absorption modulator

4.2.1 Operation principle

Electro-absorption refers to the modification of the optical absorption in a bulk material or quantum well structure through the application of an electric field. This effect, observable in the same materials used for semiconductor lasers, allows for the design of both amplitude and phase optical modulators. A change in the material's absorption leads to a variation in the intensity of the optical wave passing through it. The coupling of the phase and amplitude through the Kramers-Kronig relations also results in an associated fluctuation in the optical wave's phase. Electro-absorption effects are also evident in a P-I-N or P-N junction, where an electric field is established within the structure by applying a bias voltage.

4.2.1.1 *Electro-absorption in bulk materials: The Franz-Keldish effect*

The materials used in the design of EAM modulators are primarily III-V semiconductors on GaAs or InP substrates. The energy gap between the valence bands and conduction bands defines the threshold in terms of photon energy, that distinguishes a quasi-transparent region from an absorbing one. When a photon has energy greater than the bandgap E_g , it is

absorbed, transferring its energy to an electron in the valence band, which then transitions to the conduction band. This electro-absorption mechanism is exploited in a reverse-biased P-I-N junction, causing the EAM modulator to function similarly to a photodiode. Incident photons with energy greater than E_g generate excess electrons in the conduction band and holes in the valence band within the space-charge region of the junction. The application of an electric field results in a reverse current, known as photocurrent, which is due to the separation of electrons and holes before they can recombine in the space-charge region.

When an electric field is applied across the material along an axis perpendicular to the P-I-N junction interfaces, the periodicity of the crystal potential is disrupted along this axis, causing a tilting of the energy bands (Figure 4.1-a) [20, 21]. As a result, the material's absorption is altered due to the overlap of the wave functions of states in the valence and conduction bands, where the energy difference is smaller than E_g . These wave functions exhibit an evanescent component that extends into the material's bandgap, allowing electron transitions from the valence band to the conduction band at energies lower than the bandgap energy E_g . As the energy levels converge, the overlap of the two wave functions becomes minimal, and the probability of transition becomes negligible. Thus, electro-absorption enables a modification of the absorption curve, extending it to energies below E_g .

Figure 4.1-b illustrates the shift of the absorption curve toward longer wavelengths with the electric field applied across the interaction region with the optical field. This effect is known as the Franz-Keldysh effect. While effective, it is insufficient to provide the required contrast for amplitude modulation in optical telecommunications. In fact, the absorption band edge at zero field is not abrupt enough for a clear transition between the transparent and absorbing regions, resulting in non-negligible absorption in the "on" state, which significantly reduces contrast.

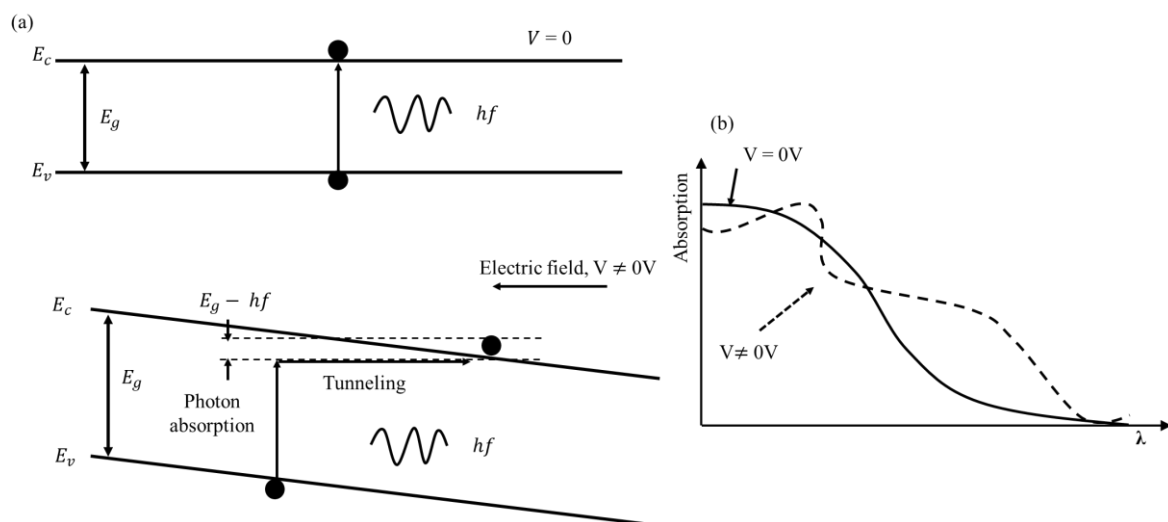


Figure 4.1: Franz-Keldysh effect: (a) semiconductor bandstructure (b) Absorption spectrum with and without an applied electric field (figure adapted from [21]).

4.2.1.2 Electroabsorption in quantum well materials: Quantum Confined Stark Effect

In quantum well (or quantum dot) devices, electro-absorption is governed by the QCSE [22-24]. When an electric field is applied perpendicular to the plane of the layers, the band structure undergoes deformation, causing the wells to take on a triangular shape (Figure 4.2-a). The discrete energy levels shift downward within the wells in proportion to the square of the applied electric field. Consequently, the energy of the charge carriers, and therefore the bandgap energy of the well, decreases, leading to a reduction in the transition energy and a shift towards longer wavelengths at the edge of the absorption band. The overlap of the wave functions diminishes, resulting in a reduction of the absorption peak at the band edge. Due to the two-dimensional confinement, we observe the exciton peaks. This can be represented by the coulomb interaction between an electron-hole pair. When an electric field is applied, the exciton energy shifts, leading to a change in the material's absorption edge [25]. Significant modifications in absorption towards longer wavelengths can be anticipated (Figure 4.2-b).

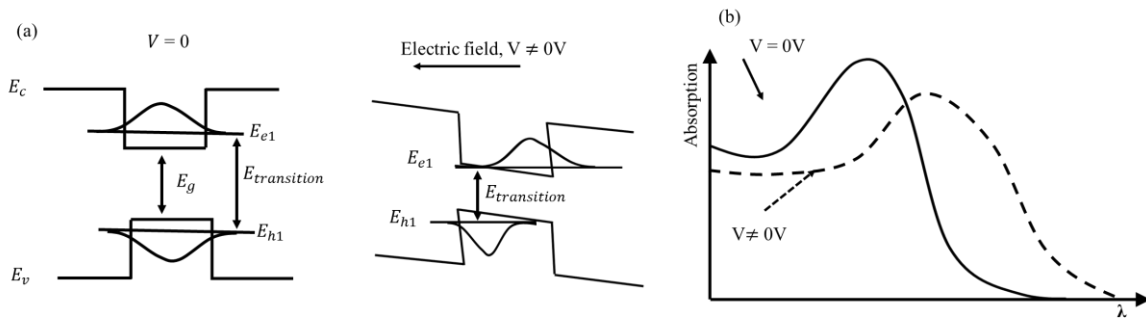


Figure 4.2: Quantum Confined Stark Effect: (a) semiconductor bandstructure (b) Absorption spectrum with and without an applied electric field (figure adapted from [21]).

Quantum well EAMs are sensitive to optical saturation when exposed to high optical power. Given the confinement by the barriers, the extraction of generated carriers is more challenging than in bulk materials. By filling the available states, these carriers hinder subsequent absorptions. Moreover, due to their larger effective mass, heavy holes are less efficiently swept by the applied field and tend to remain trapped within the wells. In comparison with MZMs, EAMs can be driven by very low voltages and can be much more compact (just a few tens of micrometers in length). This small size not only offers advantages in terms of power consumption and footprint but also facilitates laser integration to achieve a compact transmitter. However, the integration of EAMs and lasers, particularly on the III-V/Si platform, faces several challenges, one of which is optical losses. To achieve efficient mode coupling between the Si waveguide and the III-V modulator, the use of a high-performance mode converter—similar to those used in III-V/Si DFB lasers—is essential to minimize insertion loss. In the following sections, we will explore the optical design for a lumped III-V/Si EAM and discuss strategies to implement this laser-modulator integration, addressing the associated challenges.

4.2.2 Optical design

Devices were fabricated following frontside and backside SOI patterning, followed by direct bonding of III-V wafers. This heterogeneous platform developed by the III-V Lab is described

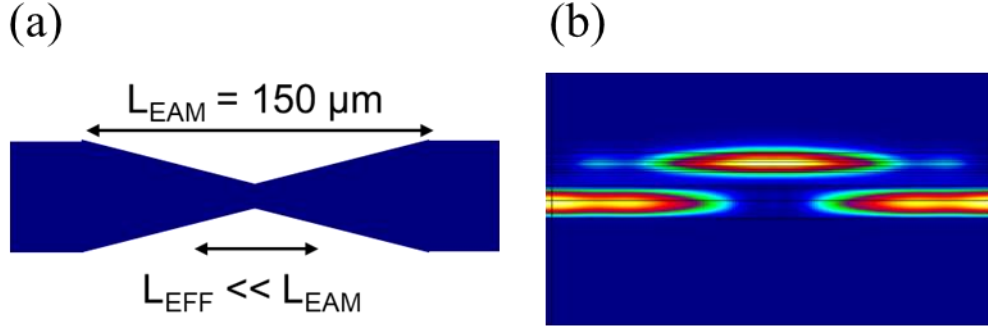


Figure 4.4: (a) Schematic top view of the "taperized" EAM, (b) Optical propagation of the optical TE mode through the EAM.

To better understand the structure, the figure 4.4-a presents a schematic of the "taperized" EAM, showcasing the two tapers side by side. The total length of the modulator, denoted as L_{EAM} , is a critical parameter for the device's performance. Although the mode is guided along the entire length of the modulator, it is only modulated when confined within the active III-V region, over a distance $L_{EFF} \ll L_{EAM}$, where the QWs are located, as shown in the simulation in Figure 4.4-b. This confinement is necessary to fully leverage the QCSE.

However, it is important to note that increasing the modulator length L_{EAM} reduces the electro-optic bandwidth because it raises the device capacitance and the associated RC time constant. Addressing bandwidth limitations due to capacitance is a crucial consideration. This RC-limited bandwidth is analyzed below (see Eq. (4 – 6)). Alternative solutions to overcome this limitation will be proposed in the following sections of the chapter.

4.2.3 Static extinction ratio & Insertion loss

The extinction ratio (ER) and insertion loss (IL) are important parameters to evaluate an optical modulator. The transmission coefficient, which determines the fraction of optical power transmitted through the modulator, is expressed as:

$$T(V) = e^{-\Gamma\alpha(V)L} \quad (4 - 1)$$

Where $\alpha(V)$ is the absorption coefficient that varies with the applied voltage, L is the total length of the modulator, and Γ is the optical confinement factor. The static extinction ratio (or on/off ratio) is defined as the output power at different bias points and at $V = 0$ V (Figure 4.6) the ER can be written as:

$$ER = \frac{P_{out}(V = 0)}{P_{out}(V)} = \frac{T(0)}{T(V)} \quad (4 - 2)$$

In decibels, it is expressed as:

$$ER(dB) = 10 \log_{10} \frac{T(0)}{T(V)} = 4.343 \Gamma \Delta\alpha L \quad (4 - 3)$$

Where $\Delta\alpha$ is the change in absorption coefficient with applied voltage. Thus, in principle, the magnitude of the extinction ratio can be made as large as desired by increasing the cavity length L (but trade off with modulation BW). The absorption coefficient α depends on the material properties, while the confinement factor Γ quantifies the fraction of the guided

modal power overlapping the active region. Let $U(x, y)$ denote the transverse modal-field profile of the fundamental TE-like mode. Using the normalized-field approximation (see Appendix A), Γ is given by [26] :

$$\Gamma = \frac{\int_{-\frac{\omega}{2}}^{\frac{\omega}{2}} \int_{-\frac{d}{2}}^{\frac{d}{2}} |U(x, y)|^2 dx dy}{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |U(x, y)|^2 dx dy} \quad (4 - 4)$$

The optical confinement factor can be increased by improving the overlap of the optical mode with the active region, for example, by narrowing the Si waveguide, as seen previously, or by increasing the active layer thickness (i.e., by adding QWs). Increasing the number of QWs enhances the optical mode confinement in the active region, thus improving modulation efficiency. However, this approach presents challenges in a photonic integration platform, where multiple components share the same III-V layer stack (such as lasers, SOAs). Adding QWs to improve the performance of EAMs could degrade the performance of other components, such as lasers, which typically require a lower optical mode confinement in the QWs to increase the output saturation power [27]. In our case, since we want to use the same III-V layer stack for multiple active components, we have adapted the Si waveguide width using adiabatic tapers where needed to control the optical mode confinement in the QWs, as explained above.

The total insertion loss is defined as the ratio of the output signal intensity to the input intensity, and it provides insights into the total losses in the system as a function of the applied DC bias voltage. Insertion loss refers to the amount of optical power lost as the signal passes through the modulator. These losses can arise from material absorption, scattering of the guided mode, and coupling loss.

At the "on" state, the insertion loss, denoted as IL_{on} , is the absorption loss in the active region when no external electric field is applied ($V=0$). It can be expressed as :

$$IL_{on} = \frac{P_{in} - P_{out}(V = 0)}{P_{in}} = 1 - e^{-\Gamma\alpha(V)L} \quad (4 - 5)$$

Since $\alpha(0)$ is always finite, a large cavity length L results in an exponential decrease in transmissivity $T(0)$. While theoretically, one could achieve an infinite ER using an infinitely long cavity, this would lead to nearly complete absorption of signal even in the "on" state, reducing signal transmission to near zero. In such a case, the insertion loss would approach 100%, which is undesirable. Therefore, the design must strike a balance between maximizing the extinction ratio and minimizing the insertion loss to maintain effective signal transmission during operation.

4.2.4 Measurement results

For EAMs, it is essential to achieve a high ER and a wide EO bandwidth. As previously mentioned, the ER is proportional to the length L of the modulator. The EO bandwidth of an EAM in lumped configuration is often limited by its RC bandwidth, defined by the series resistance R and the capacitance C of the active cavity. The relationship is given by :

$$f_{RC} = \frac{1}{2\pi RC} \quad (4-6)$$

The capacitance C of the EAM is proportional to the length L of the active cavity. The longer the length of the modulator, the greater its capacitance, because the capacitance is related to the surface area of the junction in the active region. This relationship is given by the following equation :

$$C = \varepsilon \frac{A}{d} = \varepsilon \frac{L \times w}{d} \quad (4-7)$$

Where $A = L \times w$ is the area of the active cavity, with L being the length and w the width, ε is the permittivity of the material (or, in our case, the effective mean value of the III-V epitaxial layer) and d is the thickness of the active region. Thus, a larger length L results in higher capacitance C , which reduces the RC bandwidth f_{RC} and consequently the EOBW of the modulator.

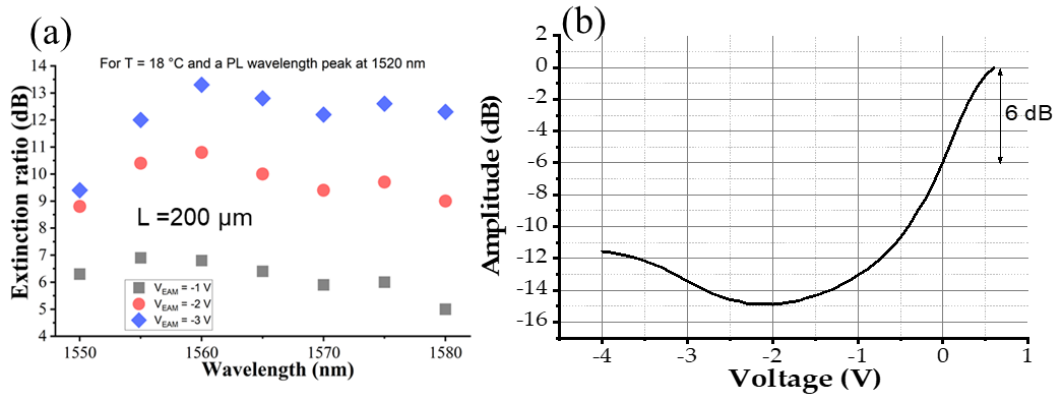


Figure 4.5: (a) The measured extinction ratio as a function of wavelength for different voltages; (b) Output power as a function of reverse bias for a 200 μm-long EAM at 1560 nm.

Figure 4.5-a shows a representative example that illustrates the ER as a function of the wavelength for a tapered EAM with a total length of 200 μm (75 μm for the input Si taper, 50 μm of straight, narrow Si waveguide and 75 μm of an output Si taper). The results are shown for three different reverse voltage values from -1 to -3 V. As can be observed, the ER has a maximum around of 1560 nm, resulting in values of 6.5 dB, 11 dB, and 13.5 dB, for each applied voltage. The figure 4.5-b presents the transmission power at 1560 nm, and the insertion loss at $V=0$. The measured IL is 0.027 dB/μm (excluding the grating coupler loss). This value is comparable to other III-V/SOI EAMs [28] and III-V monolithic modulators [29], and on average outperforms Ge-Si EAMs [30]

Measurements of the bandwidth were also conducted using the measurement setup described in Section 3.3.1, adapted for the modulator. The small-signal frequency response of the EAM was recorded at 20 °C. The obtained EO response of the modulator is presented in Figure 4.6-a. The measured 3 dB frequency bandwidths are around of 14 and 16 GHz for EAMs with lengths of 200 μm- and 150 μm, respectively. The low-frequency roll-off is attributed to the accumulation of charges in the Si substrate, a more detailed analysis is ongoing at the time of this writing. Although such values are quite modest compared to the state-of-the-art with

similar devices, it should be noted that the EO bandwidth slope is quite gradual, with a -6 dB bandwidth that extends to 27 GHz for the shortest EAM, which explains our ability to achieve open eye diagrams at 32 Gbps, as shown in Figure 4.6-b. As can be observed in figure 4.6-b and 4.6-c, the 200 μm -long EAM presents a better dynamic extinction ratio (DER), for the same data rate, than the one with $L = 150 \mu\text{m}$ at the expense of a much “blurred” eye diagram beyond 25 GHz with a large number of data points measured between the on- and off- states. On the other hand, the 150 μm modulator operates at higher bit rates due to its short cavity.

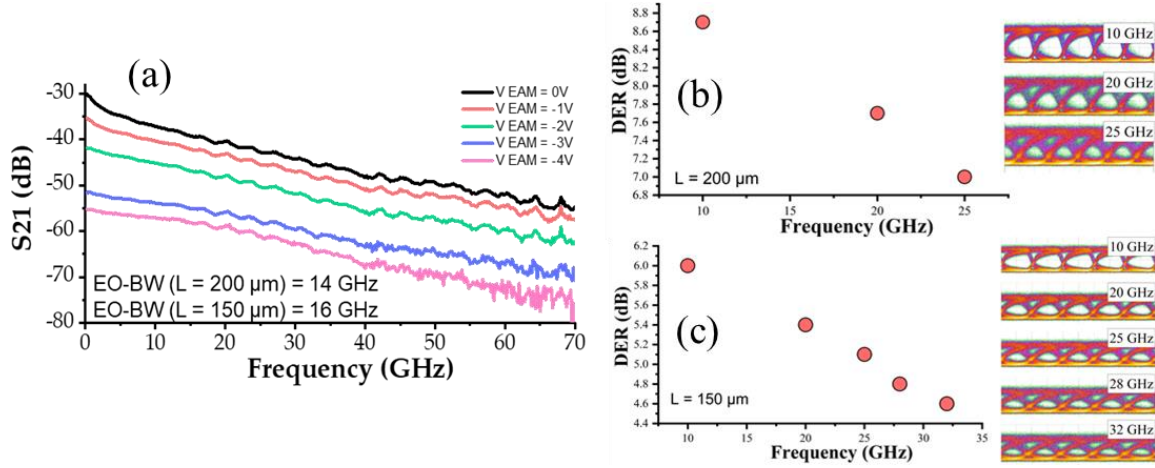


Figure 4.6: (a) Small-signal measurements for different bias reverse voltage, (b) and (c) the evolution of the DER versus the transmitted distance. Insets at the right shows the measured eye diagrams for the both 100 and 200 μm -long EAM.

We will now focus on evaluating the power consumption of our modulator, using the following equation adapted from [33]:

$$P \times B = \frac{1}{4} \frac{V_{p2p}^2}{Z_{in}} + I V_{bias} \quad (4 - 8)$$

Here, B denotes the transmission bit rate, V_{p2p} is the peak-to-peak drive voltage, and Z_{in} is the input impedance at the modulator's input port. V_{bias} and I are the applied bias voltage and the average DC current measured during operation (from the photocurrent), respectively and they present the DC power consumption. For measurements detailed above, the photocurrent was approximately 0.55 mA. The input impedance Z_{in} was measured at around 50 Ω , the applied bias voltage was 2.5 V, with a peak-to-peak drive voltage V_{p2p} of 3.5 V at a bit rate of 32 Gbps. The energy consumption at 32 Gbps was estimated around 1.9 pJ/bit for this 150 μm electro-absorption modulator. These values are well in-line with the ones previously reported for III-V/Si EAM [33].

It is worth to remark that although these EAMs present dynamic properties that are not the best in their class in terms of operation speed, they show an interesting balance between versatility, maximum saturable optical power, modulation efficiency and compatibility with laser integration in a single process. First, they can be integrated completely isolated from the laser, in a different III-V mesa, which reduces enormously the thermal and electrical cross-talk, contrarily to the designs where the EAM is integrated in the same III-V waveguide than the laser [31]. In addition, multiple EAMs can share a single laser source, as one can imagine using

a parallelized architecture based on a single laser source that feeds a bar of 8 or 16 EAMs, which can be of interest to increase the total bit rate. Aside from that, the EAM cross-section proposed here has a better resilience to the injected output power as it is based on a relatively thick III-V waveguide (and therefore, a de-confined optical mode) in contrast to the membrane approach [32] where the small effective mode area limits the maximum injected input power in the modulator to avoid saturation. Finally, it is worth to note that since we want to integrate the laser with the same III-V epilayer, the EAMs here presented have only 6 QWs, which is a relatively low number compared to the literature [33], which compromises the product modulation efficiency $\times$ EO bandwidth.

In this section, I presented the operation of EAM, we then addressed the optical design of these modulators. Additionally, we examined the measured performances, including the static extinction ratio, insertion loss, and modulation speed. Subsequently, we will integrate these EAMs with the DFB lasers characterized in the previous chapter to study their combined characteristics.

4.3 DFB Laser + Modulator + SOA

In this section, I present an integrated C-band Externally modulated Laser (EML) consisting of DFB laser and an EAM on the III-V-on-Si platform. This system is further enhanced by co-integrating an SOA to compensate for the losses introduced by the modulator. The impact of the SOA on the overall performance of the device will also be analyzed. All three components—the DFB laser, EAM, and SOA—are integrated using the same III-V epitaxial stack, as shown in Figure 4.7. It is important to note that although the various components share the same epitaxial stack, they have different InP mesas.

In scenarios involving multiple active components, like in our case, it is often desirable to use advanced fabrication techniques such as the SAG, the multi-die bonding or the butt-joint regrowth (in monolithic platforms) to tune the optical properties of each building block accordingly. As mentioned earlier in this chapter, such techniques are complex, requiring sophisticated process control and significantly increasing fabrication costs. For this reason, the co-integration of the three components using a single III-V epitaxial stack offers a simplified and cost-effective solution that merits attention.

Several demonstrations of EMLs have already been realized by integrating III-V materials on silicon photonic platforms. M. Shahin et. al. reported on 80 Gbps NRZ-OOK modulation in electro-absorption modulated DFB lasers on Si by utilizing the III-V adiabatic taper as an EAM [34]. Despite the promising results, this approach has several inherent downsides such as a non-negligible electrical and thermal cross-talk between the laser and the EAM or the limited number of modulators that can be fed by a single laser (only two, one from each side of the laser). Alternatively, T. Hiraki et. al. reported on an InGaAlAs membrane EMLs integrated on Si with over 67-GHz bandwidth [35]. They demonstrated NRZ modulation at 100 Gbps with a Dynamic Extinction Ratio (DER) of 3.8 dB. In this case, they used a passive Si waveguide to link the laser and the modulator, avoiding the electrical or thermal cross-talk. However, the use of such thin membranes with a lateral current injection prevents them from exploiting the QCSE in the QWs of the EAM. Furthermore, this approach requires a III-V regrowth step for the P- and N-doped areas, which increases the cost-per-chip and introduces an additional

technological challenge. Both above-mentioned demonstrations report modest on-chip output power values of -4.4 dBm [34] and 2 dBm [35], respectively. In fact, there is little work on the integration of SOAs in III-V-on-Si EMLs to recover the insertion loss introduced by the EAM and even provide further amplification [36].

4.3.1 Building block design

A schematic view of the PIC assembly is given in figure. 4.7. The assembly consists of a $\lambda/4$ DFB laser, an EAM and an SOA arranged from left to right. To facilitate wafer-level testing using optical fibers, a vertical grating coupler was designed at the output of the SOA. Additionally, a second vertical grating coupler was positioned at the left-hand side of the laser's output to assess its performance, taking advantage of its symmetric design.

Each device incorporates two adiabatic tapers to effectively couple the optical mode from the silicon waveguide to the III-V waveguide and vice versa, minimizing electrical or thermal cross-talk between devices. The width of the silicon waveguide is varied along the circuit to optimize the optical mode confinement in the QWs for each device. The fabrication process involved frontside and backside SOI patterning, followed by direct bonding of the III-V wafer (similarly to that described in section 2.4.3). The III-V epilayer stack is similar to the one described in Chapter 2, with a 200 nm-thick N-doped InP layer followed by a 50 nm-thick SCH, 6 Al-based QWs, another 50 nm-thick SCH and finally a thick (2 μm) P-doped InP layer.

The DFB laser has a length of $L = 600 \mu\text{m}$, with an optical mode confinement in the QWs of $\Gamma_{QWs} = 8\%$ and a grating strength of $\kappa = 30 \text{ cm}^{-1}$. The SOA features a length of $L = 1100 \mu\text{m}$ and maintains the same optical mode confinement of $\Gamma_{QWs} = 8\%$. For the tapered EAM, the length is $L = 150 \mu\text{m}$, with a maximum optical mode confinement of $\Gamma_{QWs} = 15\%$ in the center of the modulator.

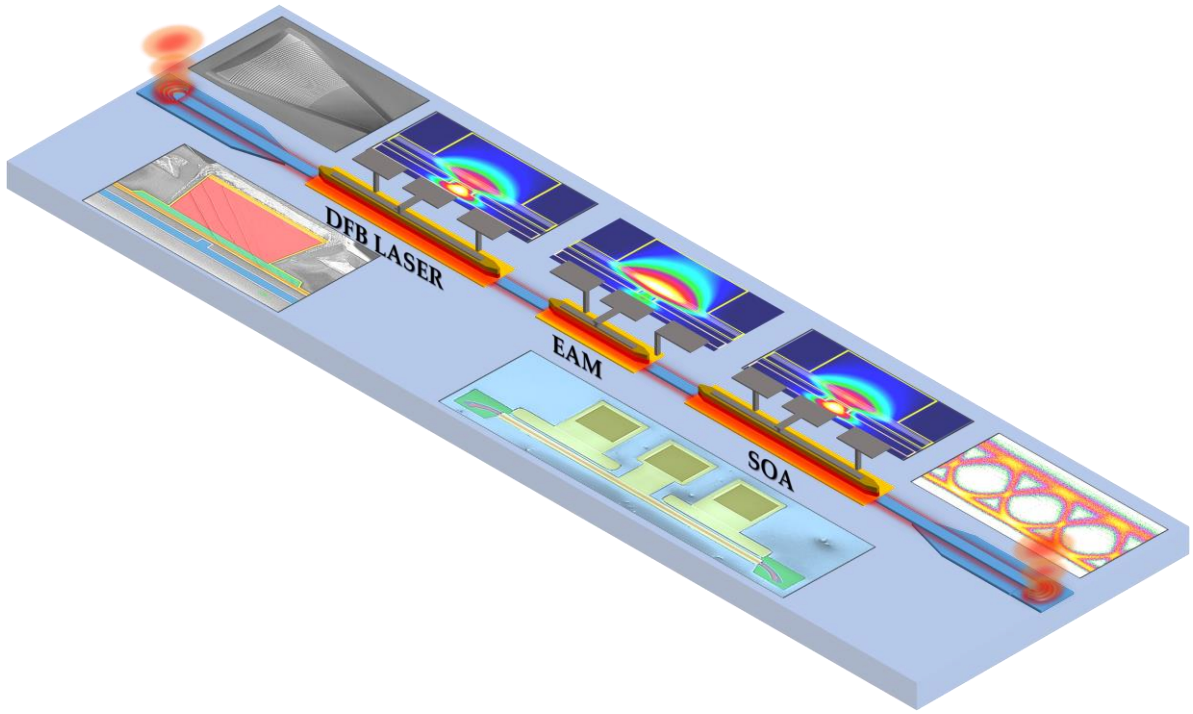


Figure 4.7: 3D schematic view of the III-V/Si monoepitaxial high-speed transmitter. Insets depict 2D FDM cross-section optical simulations for individual components, a measured eye diagram at 32 Gbps in OOK-NRZ and coloured SEM images of the device cross-section, the DFB laser's top view and SEM image of fiber grating coupler.

4.3.2 Performance Evaluation

4.3.2.1 Static characterization & Small-signal measurement

For C-band operation, the laser wavelength is centered around 1544 nm, while the EAM absorption band edge is located at a lower wavelength range, around of $\lambda = 1520$ nm. As the reverse voltage applied to the EAM increases, the absorption band edge shifts to longer wavelengths and begins to absorb the incoming optical signal, as described in section 4.2. The measurements were performed at a temperature of 20°C. Figure 4.8 shows the optical power of the integrated EML at different laser bias currents as a function of the voltage applied to the EAM. An ER greater than 17 dB was achieved for an injected current of $I = 50$ mA in the DFB and a reverse voltage of -1.25 V in the EAM. The optical absorption is gradually shifted towards higher reverse voltages for increasing injected currents. This phenomenon is attributed to an evident thermal drift in the laser for increasing current values that provokes a red shift of the Lasing wavelength. As a consequence, the reverse voltage needs to be increased to move further (in wavelength) the absorption band edge to obtain an equivalent ER than for the previous current. This phenomenon is described in the inset of figure 4.8-a, where a superposition of the absorption band edge (for a given reverse voltage) and the different lasing spectra is observed, showing a remarkable red shift of ~ 0.4 nm when increasing the injected current from 50 mA to 90 mA. Bearing in mind that the typical thermal coefficient in these lasers is around of 0.08 nm/°C, we can assume that the local temperature increase in the laser for such current range is $\Delta t \sim 5^\circ\text{C}$. Furthermore, the local heating in the laser has also an impact on the insertion loss, as seen in figures 4.8-b and 4.8-c. Indeed, as the lasing wavelength moves away from the absorption band edge for higher injected currents, we also observe a decrease of the insertion loss in the EAM. In spite of that, the modulated output power is still low for these transmitters (between 0 and 2 dBm) and consequently I

have also studied transmitters that integrate EMLs with SOAs to boost the output power (always using the same epitaxy and fabricated jointly with the DFB and the EAM).

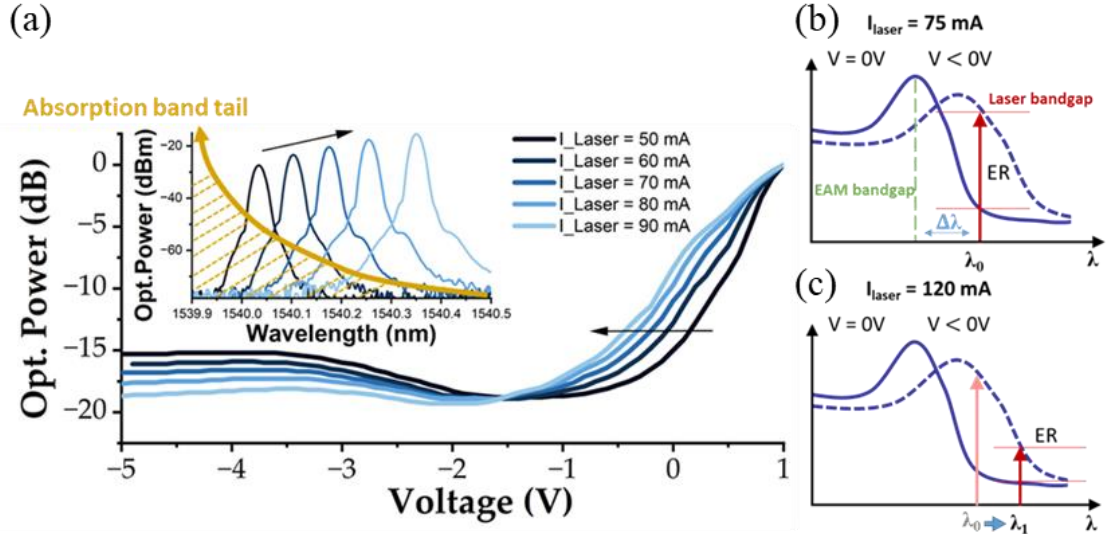


Figure 4.8: (a) Transmitted optical power as a function of the reverse voltage in the EAM for different injected current values in the DFB laser. The inset shows the superposition of the absorption band edge (golden arrow) with the lasing spectrum for increasing injected current values. (b) and (c) QCSE: absorption for different bias currents as a function of wavelength.

Figure 4.9-a presents the on-chip output power of the EML+SOA transmitter as a function of the injected current in the DFB for different bias current levels in the SOA (I_{SOA}) and at a reverse-biased voltage of $V_{\text{EAM}} = -2.5$ V in the EAM. This voltage represents the optimal polarization point for transmission measurements, which will be described later on. The DFB laser exhibited a lasing threshold current of $I_{\text{th}} = 10$ mA, with a consistent increase in output power up to $I_{\text{Laser}} = 90$ mA. At the SOA's output, a four-fold increase of the on-chip power is observed when increasing the I_{SOA} from 40 mA (Opt. Power = 4 mW) to 100 mA (Opt. Power = 16 mW). Higher I_{SOA} values were avoided to prevent the SOA from lasing. The inset of Figure 4.9-a shows the corresponding optical spectra, obtaining a single-mode emission centered at $\lambda = 1540$ nm with a SMSR over 45 dB. The continuous-wave (CW) output power as a function of V_{EAM} for an $I_{\text{Laser}} = 60$ mA and an $I_{\text{SOA}} = 100$ mA is observed in Figure 4.9-b. An extinction ratio over 7 dB is shown. Figure 4.9-c shows the measured electro-optical bandwidth (S21 parameter) of the EAM for $I_{\text{Laser}} = 60$ mA and $I_{\text{SOA}} = 100$ mA, denoting a roll-on at low frequency attributed to the optical gain saturation in the SOA and a 3 dB-bandwidth of 22 GHz.

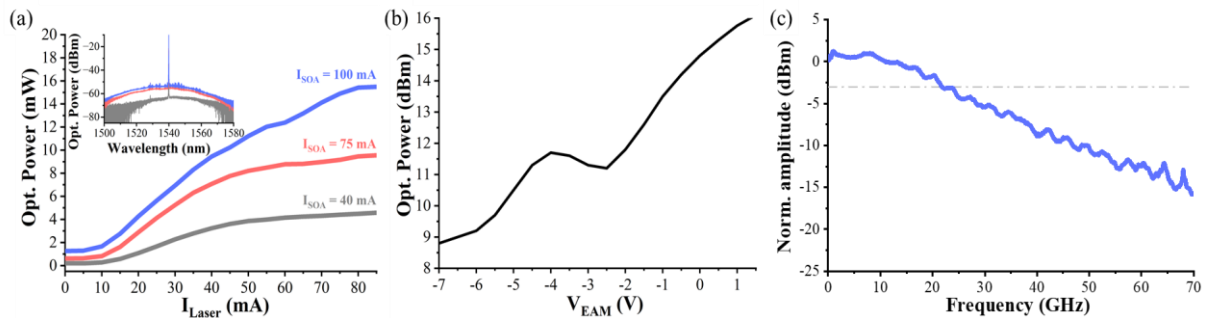


Figure 4.9: (a) On-chip output power versus laser current for a V_{EAM} of -2.5V and varying I_{SOA} . Inset illustrates the corresponding optical spectra. (b) Output power versus voltage in the EAM. (c) Measured EO bandwidth for an $I_{\text{Laser}} = 60$ mA and an $I_{\text{SOA}} = 100$ mA.

4.3.2.2 Bit error rate and transmission measurements for 32-Gbps non-return-to-zero signals

The previous section has shown the main static and small-signal analysis of externally-modulated lasers. A demonstrator with an integrated SOA booster has also been developed to compensate for the insertion loss in the EAM. In this section, I will present the measurements that have been performed in such components to further analyse their dynamic performance. For that, we will use again the experimental setup presented in the previous chapter (figure 3.12) to measure the eye diagram and the BER as a function of the transmitted signal through a SSMF.

Figure 4.10 shows the measured BER in an EML (no SOA booster) as a function of the bit rate (at 28 and 32 Gbps) for a back-to-back configuration and after 2 km transmission. The blueish area indicates the range of values for which the transmission is considered error-free (i. e. for $BER < 10^{-9}$). As can be observed, the transmitter shows an error-free transmission at 28 Gbps in B2B with a small penalty (< 1 dB) for a 2 km transmission. At a higher bit rate of 32 Gbps, we observe a small degradation but still a very competitive values, showing the 7% hard decision FEC limit at received power values below -14 dB. The inset presents the measured eye diagram at 32 Gbps (B2B), showing a DER of 4 dB. As can be observed, the eye diagram shows a balanced crossing point with little chirp and almost no overshoot compared to the measured signals for the DML (see previous chapter).

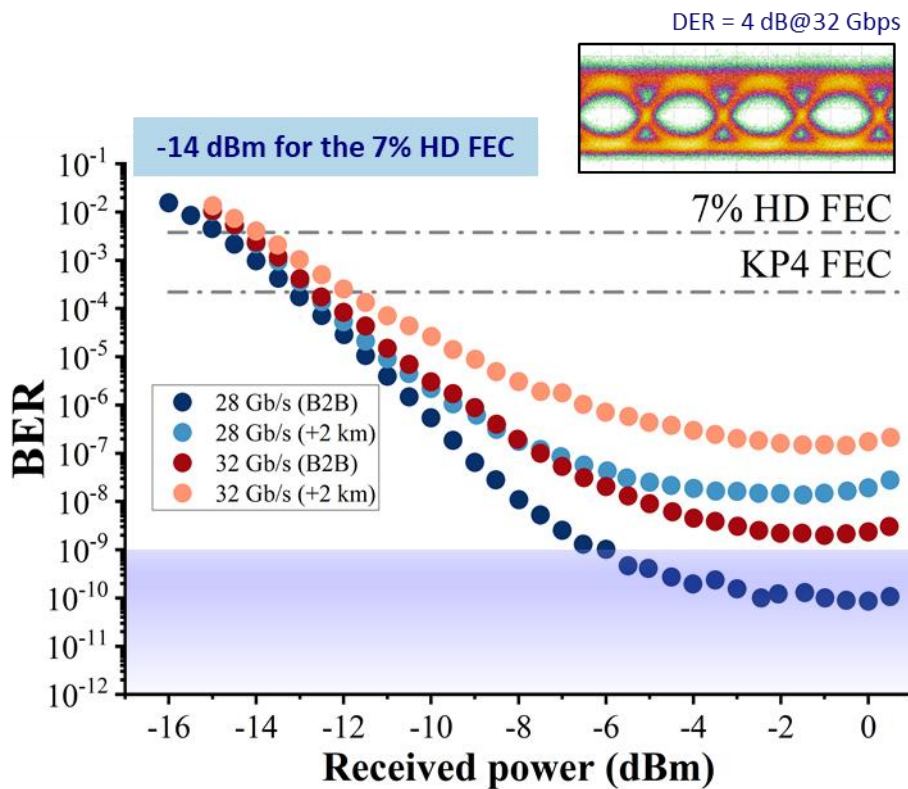


Figure 4.10: Measured BER without SOA as a function of the transmitted distance and bit rate for an OOK-NRZ signal. The inset shows a typical eye diagram at 32 Gbps.

Similar measurements were performed in the amplified EML containing the SOA booster. In this case, I was interested in the effect of the SOA in the transmission and particularly in the degradation caused by the amplified back-reflections. For that, I measured the BER for three

different values of injected currents in the SOA ($I_{SOA} = 40$ mA, blue diamonds, $I_{SOA} = 75$ mA, magenta dots and $I_{SOA} = 100$ mA, grey squares) and for an $I_{Laser} = 90$ mA and a $V_{EAM} = -2.5$ V with a voltage swing of $V_{pp} = -3$ V. Results are shown in Figure 4.11-a. The eye diagram is also shown in the inset, revealing a DER of 6 dB. A BER lower than the 7% HD-FEC limit for a received power of -14 dB and a saturation of the BER = 3×10^{-9} at 0 dB was measured for an $I_{SOA} = 40$ mA (blue diamonds). As can be observed, the increase of the injected current in the SOA up to an $I_{SOA} = 75$ mA (red dots) introduces an additional power penalty of 2 dB in the received power, reaching the 7% HD-FEC limit at -12 dB, and a saturation of the minimum achievable BER at 8×10^{-4} . For an $I_{SOA} = 100$ mA (grey squares), the additional power penalty reaches 6 dB with regard to the first measurement (blue diamonds) and the BER saturates at 5×10^{-3} , although still remains below the 7% HD-FEC. These results denote the impact of the integrated SOA on the EML, suggesting that it introduces a non-negligible optical feedback that degrades the BER, especially under high current injection. Despite this, biasing the SOA under a moderate current injection enables to recover the insertion loss at the EAM and introduces an additional amplification while having a minimal impact on the BER measurements. For comparison purposes, figure 4.11-a also shows the BER measurements in the EML transmitter without any integrated SOA at the output, obtaining similar results, as can be observed by direct comparison of the black crosses with the blue diamonds. Figure 4.11-b shows the comparison of the measured BER and eye diagrams for the SOA-EML transmitter in B2B configuration and after transmission through a 2 km- and 5 km-long SSMF (filled symbols) for an injected current in the SOA of 40 mA. As seen, a negligible power penalty is observed when comparing the B2B measurement (blue diamonds) with the +2 km transmission (violet stars). For the +5 km transmission, a 0.8 dB penalty in the received optical power at the 7% HD-FEC limit was recorded, with a BER that saturates at 5×10^{-7} for a 0 dBm of received optical power. For comparison, we can also see the BER measurement of the EML without the SOA booster for B2B and + 2Km transmission. Comparing the EML with the EML+SOA measurements, we observe a higher degradation of the BER for high power values in the receiver between the B2B and the + 2 km transmission that puts forward the role of the SOA. Furthermore, it is worth to note that we could not even measure the BER for a 5 km transmission in the EML due to the large number of errors found, contrarily to the EML+SOA, where we measured a BER 10^{-6} for 0 dBm in the receiver.

Looking at the measured eye diagram, we observed deformed rise and fall times and a gradual decrease of the DER from 6 dB to 4.6 dB with the transmitted distance. The origin of such degradation after 5 km transmission can be attributed to a combination of optical feedback, loss and chirp introduced by the SOA. Longer transmission distances are expected by using optical links that partially compensate the chirp, such as the NZ-DSF.

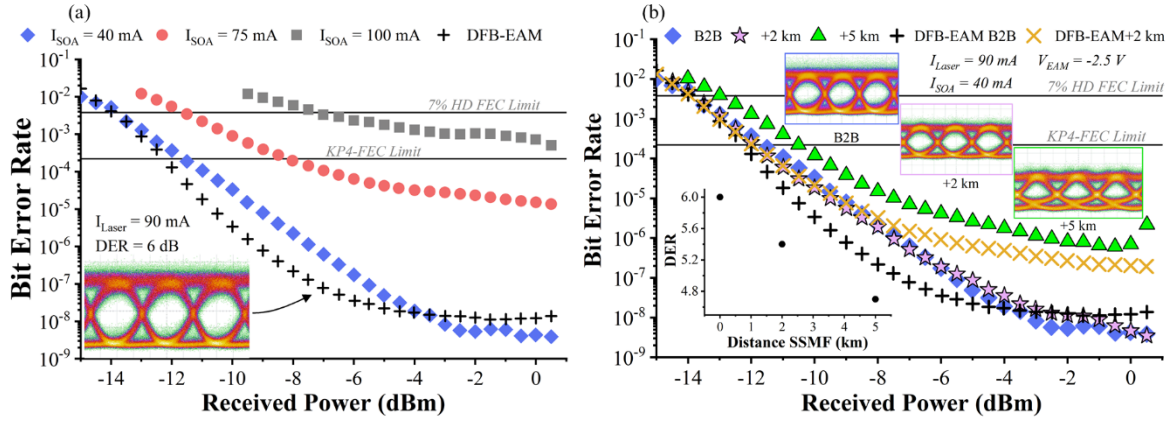


Figure 4.11: (a) Measured BER at 32-Gbps in back-to-back configuration as a function of the received power for different current injection values in the SOA. An $I_{Laser} = 90$ mA and a $V_{EAM} = -2.5$ V were used in all cases. The inset shows an eye diagram for an $I_{SOA} = 40$ mA. (b) Measured BER transmission at 32-Gbps for the SOA-integrated EML in back-to-back configuration (blue diamonds) and over transmission through a SSMF. The equivalent BER measurement is also shown for an EML. Insets at the right and at the bottom show the measured eye diagrams and the evolution of the DER versus the transmitted distance for the SOA-EML transmitter, respectively. Black horizontal lines mark the KP4-FEC limit, at 3.8×10^{-3} , and the 7% HD-FEC limit at 2.2×10^{-2} .

These results show the impact of the SOA on the high-speed transmission measurements of the transmitter. Several conclusions can be drawn. It is evident that the SOA introduces a non-negligible distortion in the transmitted signal that degrades the BER. Although a quantitative analysis has not yet been performed as it requires complementary measurements in independent components, we have been able to determine that for low-moderate injected current values in the SOA, the optical amplification of the emitted optical power is still beneficial as it enables to transmit further the optical signal with “little” degradation compared to the EML. However, for higher injected currents, we have observed a fast degradation of the transmission that can be attributed to the amplified back-reflection from the grating coupler.

In this section, we presented III-V-on-Si EMLs with/without co-integrated SOAs that deliver up to 16 mW (12 dBm) of on-chip output power with a 3 dB bandwidth of 22 GHz and provided 32-Gbps NRZ modulation with a DER of 6 dB in the eye diagram. I demonstrate a BER lower than the 7% HD-FEC limit for a received power of -14 dBm in B2B and a power penalty of 0.8 dB after transmission through 5 km of SSMF. I also revealed the impact of the optical feedback introduced by the SOA on the BER measurements. These results have put forward the potential of III-V/Si EMLs to be used as high-speed transmitters for metro or access networks. However, we still need to improve the operation speed even further to compete with other existing technologies. Furthermore, there is a need for tunable transmitters able to operate at different wavelengths to cope with the new access architectures based on reconfigurable networks.

For all that, we have also investigated a different compact EML architecture that integrates the laser, the EAM and the SOA in a single III-V waveguide, without any optical transition between them. Such design enables to reduce the footprint of the EAM considerably while it maximizes the optical mode confinement in the QWs already at the input of the device (contrarily to tapered EAMs, where little confinement is obtained at the input, where the mode is still highly confined in the Si waveguide). Such reduction in footprint will certainly

have a positive impact on the EO bandwidth as the total device capacitance will be reduced. In addition, instead of integrating a DFB laser, we have now used a widely tunable laser to demonstrate a reconfigurable high-speed transmitter in the C-band. The next section will provide further details on the design and characterization performed in this component.

4.4 Widely Tunable Laser + Modulator + SOA booster

As explained before, tunable laser diodes are of interest for the new reconfigurable metro/access networks and for modern communication systems in general. In this section, we will explore the integration of a III-V/Si tunable laser based on ring resonator vernier filters with a modulator and an SOA, aimed at designing a transmitter capable of meeting the demands of DWDM metropolitan and access networks. The first part introduces ring resonator-based tunable lasers, describing their operating principles and fundamental characteristics. In the second part, we demonstrate a compact transceiver integrating an externally-modulated widely tunable laser with a co-integrated SOA booster, capable of operating at 56 Gbps in NRZ and delivering on-chip output power of up to 5 mW. The transmitter has an optical bandwidth of 15 nm, covering wavelengths from 1542 to 1557 nm, and achieves a DER greater than 4.5 dB across all wavelengths. We also present BER measurements at 32 Gbps, showing error-free transmission in a B2B configuration and transmission over distances of up to 10 km through SSF.

4.4.1 Tunable Laser

Tunable III-V lasers on silicon consist of a Sagnac mirror at the back, a filtering system with two resonant rings that allow to select a specific wavelength, a III-V/Si transition, a gain section in AlGaInAs with multiple quantum wells, another III-V/Si transition, and an output Bragg mirror (see figure 4.12). In the following sections, we will introduce each of these elements.

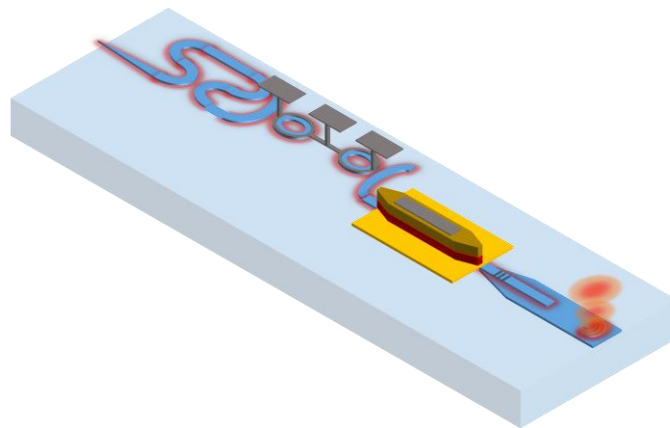


Figure 4.12: 3D schematic view of the tunable laser.

4.4.1.1 Ring-based Vernier filter

The filtering system is based on the Vernier effect, which involves the periodic alignment of two graduations whose free spectral ranges (FSRs) are not exactly the same. Therefore, if we have two resonator rings with slightly different spectral responses, there will be one wavelength where their resonances perfectly align. Tuning the laser is achieved by applying a localized temperature variation on top of the rings using heaters. When current is injected

into a resistor placed on top of the ring, the refractive index changes due to the thermo-optic effect. As a result, the spectral response of the filter shifts, leading to a new resonance wavelength at a different value. The temperature variation also induces thermal expansion of the material, which alters the physical length of the rings and consequently shifts their resonance wavelengths. This effect can be summarized by the following equation [37]:

$$\Delta\lambda = \Delta\lambda_L + \Delta\lambda_T = \alpha_W \frac{n_{eff}}{n_g} \lambda \Delta T + \frac{\sigma_T}{n_g} \lambda \Delta T \quad (4 - 9)$$

Where α_W is the thermal expansion coefficient of the material, n_g is the group index in the rings, λ is the resonance wavelength at ambient temperature, and σ_T is the variation of the effective refractive index with temperature.

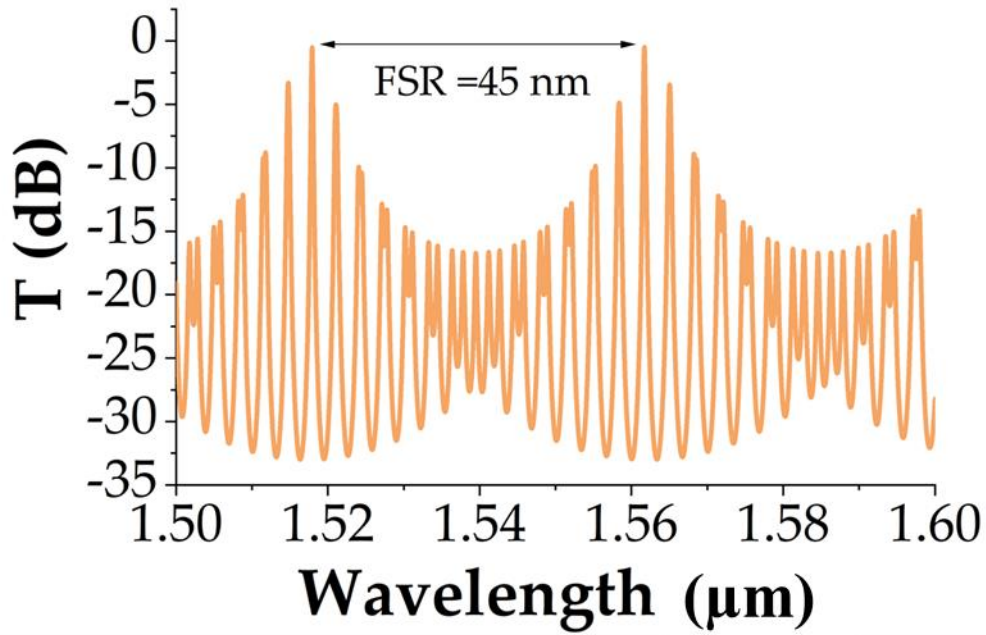


Figure 4.13: Simulated transmission spectrum of a double ring resonator laser.

If we tune the ring with the smaller FSR, the coincidence wavelength shifts toward longer wavelengths. The opposite occurs for the ring with the larger FSR. The combined FSR is given by:

$$FSR_{Vernier} = \frac{FSR_{R1} \times FSR_{R2}}{FSR_{R1} - FSR_{R2}} \quad (4 - 10)$$

The directional coupler has been optimized to exhibit minimal wavelength dependence. Simulations of the passive double ring, shown in figure 4.13, featuring a Vernier FSR of 45 nm. To have a laser effect, an optical cavity is required and therefore essentially two mirrors to close the cavity and therefore have optical feedback. In silicon photonics, two types of mirrors are typically used: Bragg mirrors and Sagnac mirrors.

4.4.1.2 Bragg mirror

Bragg mirrors are optical structures consisting of alternating layers of materials with different refractive indices, forming a periodic structure. Their working principle relies on the constructive interference of signal waves reflecting at each interface between the layers. These mirrors are valuable components, providing well-controlled optical feedback while remaining highly compact and relatively simple to model [38]. However, their precise fabrication is crucial to accurately reproduce the periodicity of the grating with minimal deviations, which poses a significant technological challenge. Figure 4.14-a shows an SEM image of a Bragg mirror etched into a rib waveguide structure on silicon. The periodic patterns, alternating between Si and SiO₂, form a diffraction grating with a period of 242 nm. This Bragg grating functions as a partially reflective mirror, essential for providing optical feedback within the laser cavity.

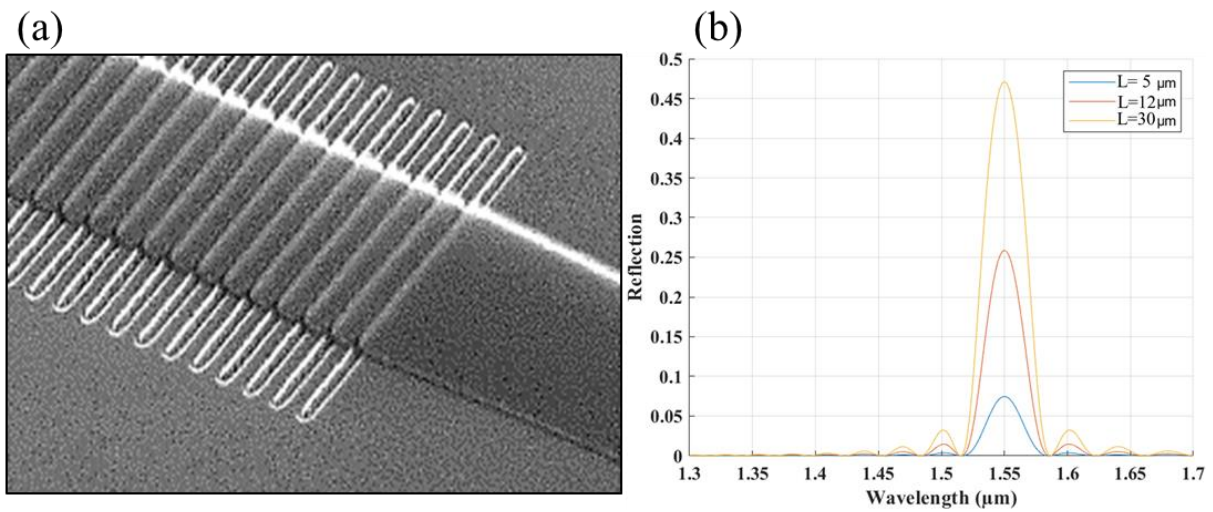


Figure 4.14: (a) Scanning electron microscope images of a Bragg grating (b) Reflectivity simulation for different length.

Figure 4.14-b shows the reflectivity curves as a function of wavelength for three different mirror lengths: 5 μm, 12 μm, and 30 μm. As the mirror length increases, the reflectivity becomes higher. In our case, the 12 μm Bragg mirror provides approximately 25% reflection, which is suitable for creating optical feedback in the tunable laser cavity. The other side of the laser cavity is closed by a Sagnac mirror, offering a total reflection of 100%.

4.4.1.3 Sagnac mirror

The design of the Sagnac mirror is based on a directional coupler looped back on itself (see insert in figure 4.15). The principle of this type of mirror involves forming a loop in the waveguide through evanescent coupling, which allows the mode to loop back. By knowing the coupling length and coupling distance between waveguides, the theoretical reflectivity of the mirror can be precisely determined. This type of mirror offers several advantages compared to Bragg mirrors:

- The mirror's reflectivity is easily controlled by adjusting the coupling length, which influences the loop's reflectivity coefficient.

- It is much less sensitive to temperature variations than a Bragg mirror, which can expand and thereby alter its reflection properties.
- It is relatively simple to fabricate on a circuit, as it involves only a single waveguide looped back on itself. In contrast, Bragg gratings require more precise fabrication techniques, often using expensive and time-consuming e-beam lithography steps. The Sagnac mirror can be produced using more accessible and less expensive photonic technology.

Since only one optical output is needed in the intended application, the mirror closest to the rings has 100% reflectivity to keep all the optical power in the cavity, while the right-side mirror has a lower reflectivity to allow for a sufficient optical feedback and output signal coupling to the EAM simultaneously.

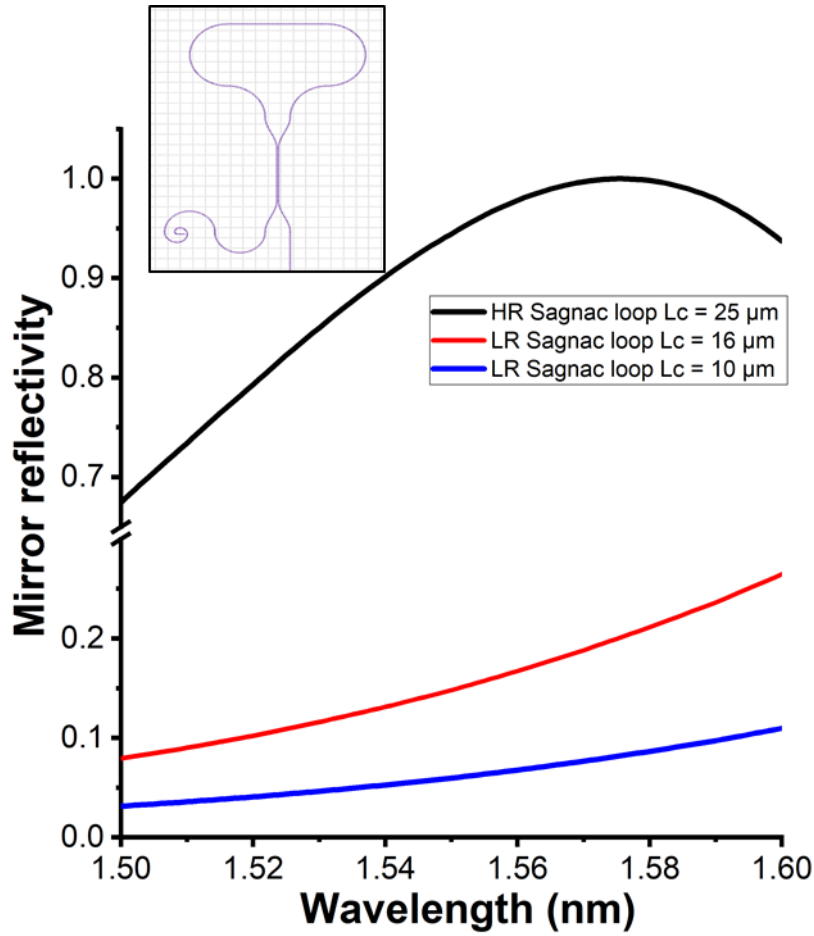


Figure 4.15: Mirror reflectivity versus wavelength for Sagnac loop mirrors with varying coupler lengths (L_c). The black curve represents the high reflectivity (HR) Sagnac loop with $L_c = 25 \mu\text{m}$, while the red and blue curves correspond to low reflectivity (LR) Sagnac loops with $L_c = 16 \mu\text{m}$ and $L_c = 10 \mu\text{m}$, respectively. The inset shows the layout of the Sagnac loop structure.

Figure 4.15 presents the simulated reflection spectrum for a Sagnac high-reflectivity (HR) mirror, alongside two other designs where the coupling length of the directional coupler is varied. Specifically, the designs have coupling lengths of 10 μm, 16 μm, and 25 μm for the total reflection Sagnac configuration. The directional couplers in all designs are separated by 300 nm. As expected, the HR Sagnac mirror with the longest coupling length (25 μm) exhibits the highest reflectivity, achieving near-total reflection across the entire wavelength range. In

contrast, the mirrors with shorter coupling lengths (10 μm and 16 μm) show lower reflectivity, making them suitable for applications requiring partial feedback. The reflectivity of all Sagnac mirrors shows some wavelength dependence, with slightly higher reflectivity at longer wavelengths. This is likely influenced by the dispersion characteristics of the waveguide and the coupling region.

4.4.1.4 Static characterization

The static characteristics of the tunable lasers were measured. Although it is not the same laser integrated into the transmitter, it features a very similar design. There may be differences in terms of threshold, output power, and tunability. The measurements were carried out using a multi-electrode current source to supply current to the laser, along with two standard Keithley 2400 current sources to control the current injected into the heaters. All measurements were conducted at a controlled temperature of 20°C. The experimental setup is the same as that used for characterizing the DFB lasers. Figure 4.16 presents the measurements of power and voltage for a laser as a function of the injected current (L-I-V), along with an evaluation of the tunability of the emission wavelength for each component. This evaluation involves measuring the spectrum of a laser while varying the current through its heaters in specified increments. The goal is to create a 2D mapping of the achieved wavelength as a function of the current applied to each heater, targeting a given SMSR (Figure 4.16-b).

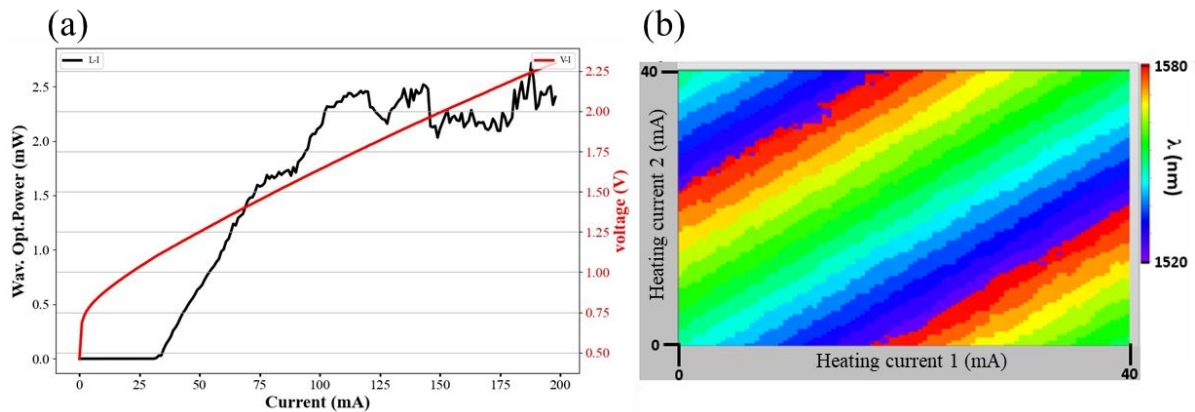


Figure 4.16: (a) L-I-V characteristic of the tunable laser and (b) an example of 2D mapping spectrum of the laser for various currents injected into the heater.

A laser threshold of 32 mA and a maximum waveguide-coupled output power of 2.7 mW for an injected current of $I=190\text{mA}$ were obtained. The kinks observed on the P-I curve of figure 4.16-a correspond to longitudinal mode hopping in the cavity, caused by localized Joule heating in the gain section as the current injection increases. For the V-I curve of the laser's gain section. We measure a differential resistance of 7.6 Ω , in agreement with the values reported elsewhere in similar devices [39]. Figure 4.16-b displays an example of the 2D mapping of the laser's tunability across the entire range of currents for both heaters, with a fixed laser current of 60 mA. Now that we have introduced ring resonator-based tunable lasers, detailing their operating principles and fundamental characteristics, we will shift our focus to the second part of this discussion. Here, I will present a compact transceiver that integrates an externally modulated, widely tunable laser with a co-integrated SOA booster.

4.4.2 Design and Implementation of the transmitter

4.4.2.1 Device Description

Building on the components previously described, we will now co-integrate a tunable laser with an EAM and a SOA. Figure 4.17 (from [1]) illustrates the completed III-V-on-Si wafer (3 inches) with a zoomed-in view showcasing a simplified 3D schematic of the heterogeneously integrated III-V-on-Si transmitter. As noted in Section 4.2.2, the minimum dimensions imposed by the adiabatic tapers in the modulator impact the bandwidth of the previous design. One way to circumvent this limitation is to avoid using adiabatic tapers. Generally, adiabatic couplers are employed for optical mode transfer between devices (laser, EAM, and SOA) to guide the optical signal from the III-V waveguide to the silicon waveguide. However, in this transmitter, the optical mode remains confined within the III-V waveguide, eliminating the need for couplers. This design choice reduces parasitic capacitance, increases bandwidth, and results in a more compact device. Electrical isolation between components is achieved through ion implantation followed by etching of the InGaAs p-contact (see the SEM image in Figure 4.17).

Several demonstrations of tunable lasers with EAMs have been achieved using III-V materials on silicon photonic platforms. For instance, in work by J. Zhang et al. [40], an externally modulated, widely tunable transmitter was fabricated using micro-transfer-printing technology. This approach integrated a tunable laser with a silicon MZM, achieving 40 Gbps NRZ-OOK modulation in a B2B configuration. Operating over a wavelength range of 1539 to 1573 nm, the device demonstrated a fiber-coupled average power of -9 dBm.

Since we employed direct wafer bonding to define the III-V-on-Si platform, all three co-integrated components utilize the same III-V epitaxial stack and the same InP mesa. However, this approach also presents certain constraints, particularly when closely integrating different components within a single waveguide that may require distinct III-V epitaxial stacks for optimal performance, as is the case in this work. Despite these challenges, we can achieve a favorable trade-off and utilize a single III-V epitaxy to demonstrate on-chip transmitters that provide adequate optical power and low driving voltage while enabling high-speed operation and a good ER, as will be discussed further. While the use of a single III-V epitaxial stack simplifies the integration of multiple components, it is crucial to also consider the underlying SOI platform, which plays a pivotal role in enabling efficient optical signal routing and coupling. In this work, the SOI platform contains a 500 nm crystalline Si layer and has a 2 μ m buried oxide (BOX). The wafer is then etched to define two different Si waveguide heights, 300 nm for the Vernier filters, the Sagnac mirrors and the vertical grating couplers and 500 nm for the Si waveguide interposers located below the III-V waveguides to enable an efficient optical mode transfer. The device shares the same cross section as the DFB laser described previously. It is composed of a III-V epitaxial structure consisting of multiple layers, including a P-I-N junction composed of P-doped InP, an AlGaInAs MQWs active region flanked by SCH, and an N-doped InP layer. These InP-based layers are grown using MOVPE. The III-V active waveguides are defined after direct bonding of III-V epitaxial layers onto a pre-patterned SOI wafer, followed by device metallization. A DVS-BCB passivation cladding ensures superior lateral optical confinement and avoids the lateral current leakage under high injection. This

configuration offers an effective mode confinement in the QWs, resulting in reduced laser threshold current and an efficient OMA [38].

Devices were fabricated following the process flow detailed in section 2.4.3. To compensate for the insertion loss introduced by the modulator and to provide further optical amplification, SOAs can be integrated at the outputs of the transmitters. A vertical grating coupler integrated at the SOA's output facilitates wafer-level testing using optical fibers (see the SEM image in Figure 4.17). The double-pass ring-based Vernier filters are designed with two racetrack resonators coupled to a bus waveguide, each having respective FSR of 2.52 nm and 2.68 nm, resulting in a Vernier FSR of 43.56 nm. The transmission coefficient between the bus waveguide and a single ring is approximately 0.93, defining the ratio between the transmitted and incident fields in the bus waveguide.

Optical feedback in the tunable laser cavity is provided by a Bragg mirror (length = 12 μm and period = 242 nm) etched into a Si rib waveguide at the laser's output (around 25% reflection) and a Sagnac mirror at the laser's back side (100% reflection). Both components are depicted in the SEM images of Figure 4.17. We have chosen 20% of reflectivity to increase the laser's output power. By doing so, we also improve the OMA in the modulator (provided that we do not saturate it, which is not the case in this work). The EAM features a length of $L = 100 \mu\text{m}$ and an optical confinement of $\Gamma_{QWs} = 13.7\%$, while the SOA has a length of $L = 920 \mu\text{m}$ and a lower Γ_{QWs} of 10.95% to enhance output saturation optical power. By reducing the quantum confinement in the SOA, the output saturation power (P_{sat}) is increased : $P_{sat} \propto \frac{1}{\Gamma_{QW}}$. For the laser, we have $L = 670 \mu\text{m}$, $\Gamma_{QWs} = 13.5\%$, and $\Gamma_{Bragg} = 0.19\%$. These parameters provide an optimal balance, enabling the efficient simultaneous operation of the laser, EAM, and SOA.

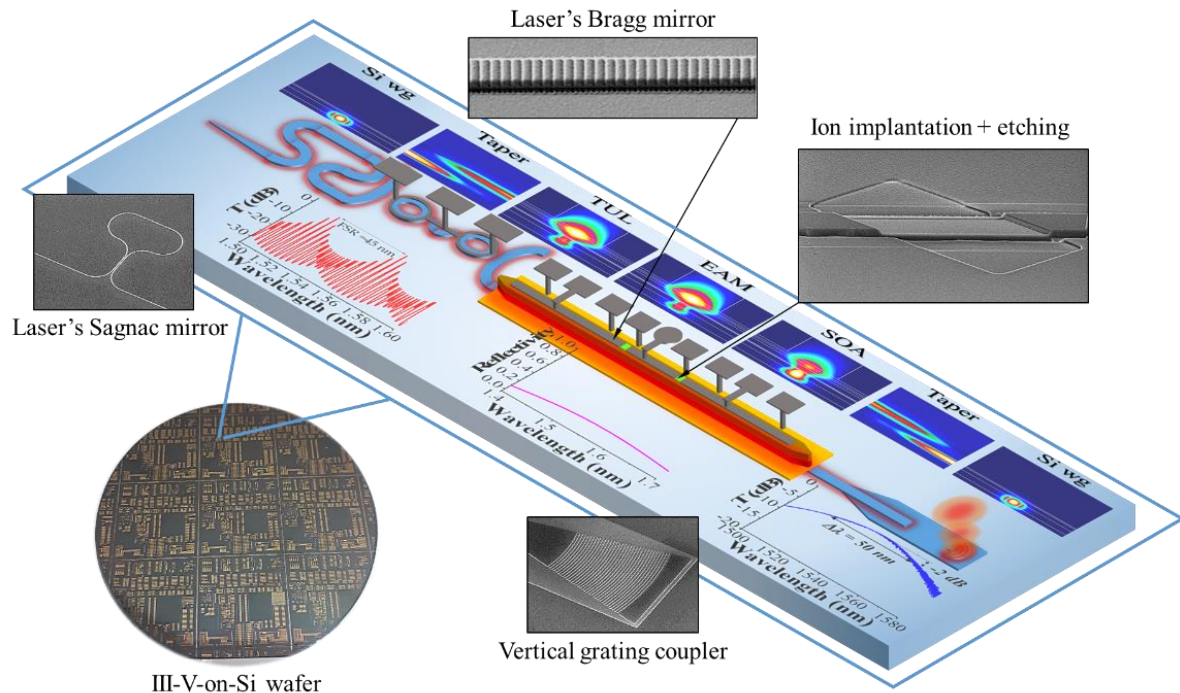


Figure 4.17: Picture of the finalized wafer with a zoomed view showing the schematic structure of the heterogeneously integrated transmitter. From left to right: the highly reflective Sagnac mirror (SEM image); the Vernier filter; the III-V gain section of the laser; the low reflective laser's Bragg mirror (SEM image); the III-V section of the EAM and the SOA booster separated by electrically isolated areas (SEM image); and the fiber grating coupler (FGC) for fiber coupling (SEM image). The inset curves below show the simulated Vernier filter transmission for a wavelength range from $\lambda = 1500\text{--}1600\text{ nm}$ and the simulated reflectivity and transmission spectrum of the Bragg mirror. The upper insets display several 2D FDM cross-section simulations of the TE optical mode for each component.

4.4.3 Performance Evaluation

4.4.3.1 Experimental setup

The experimental setup for evaluating the dynamic characteristics of the externally-modulated widely tunable laser co-integrated with the SOA booster (Device Under Test, DUT) is very similar to the one described in chapter 3. The device is mounted on a stage equipped with a TEC for temperature stabilization. To drive the transmitter module, a standard electrical source (Keithley 2400) is used along with either an RF signal for small signal analysis or with a pseudorandom binary sequence (PRBS) generator, connected to a two-stage variable-gain RF-linear amplifier (SHF S807 C) with a wide bandwidth. The optical signal generated is collected at the SOA side from the vertical grating coupler using a lensed optical fiber to minimize back-reflections. To prevent optical feedback from the setup, an optical isolator is included. An optical switch (1×4) facilitates the selection of different optical paths. The first path is connected to a power meter and an optical spectrum analyzer to evaluate the static performance. The second output signal is used to investigate the small-signal modulation response. This is achieved by applying a DC bias current and a low-amplitude RF signal generated by a VNA, and then analyzing the output power using an optoelectronic converter (LCA : Agilent N5242A/N4375B). The third optical path in the setup is utilized for transmission measurements. The output is connected to an EDFA and a TBPf to amplify and filter the received signal accordingly. On the receiver side, a high-speed photodiode (XPDV-2320R) is used. The amplified signal is sampled by a high-bandwidth Keysight storage oscilloscope with a bandwidth of 70 GHz (Agilent 86118A). The final path is designated for BER measurements.

using the Anritsu MP1800A Signal Quality Analyzer, a multi-channel BERT. The signal is first amplified and filtered before passing through a VOA (Agilent 81577A) and a high-speed photodiode (XPDV-2320R) for these measurements. It is worth to mention that, although we have not performed an in-depth yield analysis, the results presented here below are representative for several devices that have been measured at different sites of the wafer.

4.4.3.2 Static characterization

The widely tunable laser, EAM, and SOA co-integrated system were mounted on a temperature-controlled platform set at $T = 20\text{ }^{\circ}\text{C}$. Figure 4.18-a displays the on-chip output power as a function of the laser's injected current and for various values of injected current in the SOA when the EAM section is not biased. The output optical power is collected by a fiber grating coupler with a lensed fiber with a waist diameter of $3\text{ }\mu\text{m}$, which introduces 3-4 dB of coupling losses. Several kinks in the L-I curve are observed, which we attribute to mode hopping due to the passive-active phase mismatch between the III-V gain section in the laser and the Vernier filter spectral response. A consistent increase in the L-I curve is observed for consecutive values of the injected current in the SOA, with a maximum of approximately 5 mW for an injected current of $I_{\text{SOA}} = 80\text{ mA}$ in the SOA and for an injected current of $I_{\text{Laser}} = 100\text{ mA}$ in the laser's gain section. Beyond this value of I_{SOA} , the optical power quickly saturates and steadily decreases for higher SOA current values, indicating the onset of nonlinear and/or thermal effects, as evident in Figure 4.18-a (see the red and orange curves). The measurements were performed for one laser emission wavelength around $\lambda = 1551\text{ nm}$. In addition, a decrease of the laser's current threshold is observed for higher I_{SOA} values, which has been attributed to the optical feedback. Figure 4.18-b represents the measured V-I response of the laser, from which we extracted a differential resistance of $7.5\text{ }\Omega$ for $I_{\text{Laser}} = 100\text{ mA}$, consistent with values reported elsewhere in similar devices [41]. In the inset, the V-I response of the thermal heaters is presented, showing a similar series resistance of approximately $R \approx 175\text{ }\Omega$. Each thermal heater tunes the laser emission wavelength in steps of $\sim 2.7\text{ nm}$ towards an opposite side of the spectral range, inducing a red shift (or a blue shift) to attain a total wavelength range of $\sim 40\text{ nm}$.

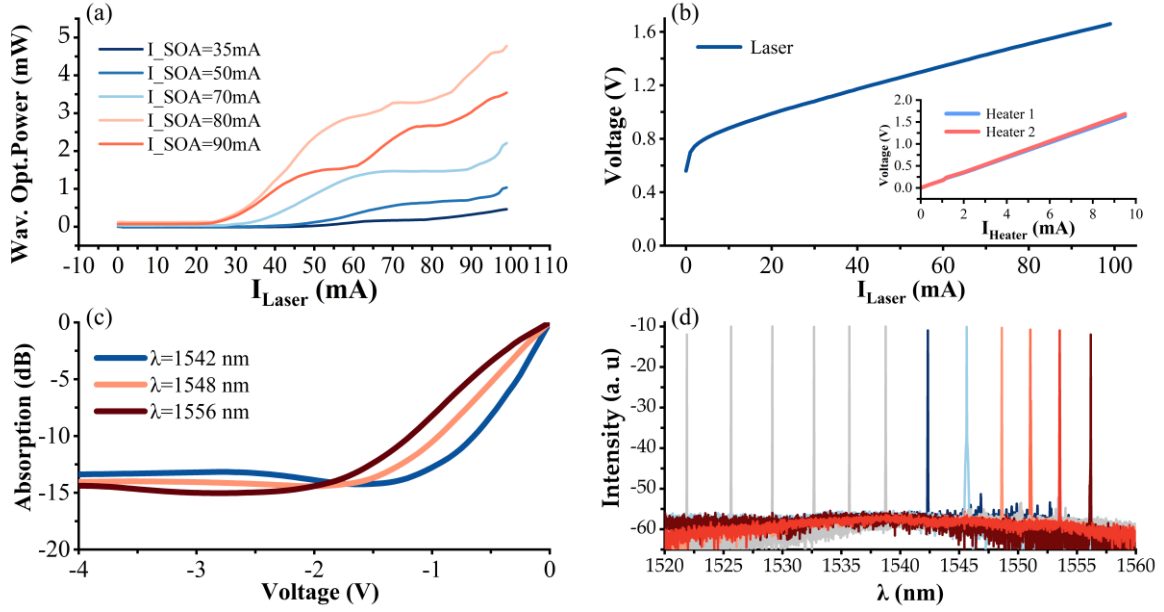


Figure 4.18: (a) L-I characteristic of the tunable laser measured for different I_{SOA} values. (b) The measured V-I response of the laser and the thermal heaters (inset). (c) Measured optical absorption as a function of the reversed bias voltage in the EAM for different wavelengths. (d) Superimposed optical spectra for different operation modes in the widely tunable laser, showing the range of wavelengths in which the EAM can operate (coloured lines).

The static extinction ratio characteristic of the EAM has been explored. Illustrated in Figure 4.18-c, the optical absorption of the EAM was measured at different wavelengths, varying with the applied reverse-bias voltage. Throughout the measurements, injection currents of 60 mA and 80 mA were maintained in the gain sections of the tunable laser and the SOA booster, respectively. Initially, the absorption increases steadily with the voltage until reaching saturation around -2V, indicating that the component has entered a regime of strong depletion. For longer wavelengths, a higher reverse bias is required to achieve the same optical absorption, suggesting that we are moving away from the modulator's absorption band edge, as expected. The maximum ER also varies with wavelength, peaking at 16 dB at 1556 nm and declining to 14.5 dB at 1542 nm. In Figure 4.18-d, a superposition of the measured lasing spectra for different driving currents in the thermal heaters is presented. The gray spectra represent the laser wavelengths for which the EAM is not transparent in the off state. The colored spectra correspond to the wavelength range where the EAM absorbs and provides a good extinction ratio, comparable to the values reported in the literature for III-V/Si modulators [28]. The wavelength tuning range spans over 15 nm, from 1542 nm to 1557 nm. Throughout this range, a side-mode suppression ratio (SMSR) of at least ~ 50 dB is maintained.

4.4.3.3 Small signal analysis

Subsequently, we conducted a small signal frequency response measurement of the EAM using a Vector Network Analyzer. The equipment was previously calibrated following the vendor's protocol. These measurements were performed at a single wavelength ($\lambda = 1551$ nm) and at a constant current of 80 mA for the SOA and 70 mA for the laser, at a temperature of 20°C. Figure 4.19 shows the EO bandwidth of the integrated EAM for the three available modulator lengths to assess the length's impact on bandwidth. All measurements were

conducted with a bias voltage of -3 V. The gray curve shows the EO response of the high-speed photodiode (XPDV-2320R). We observed a very slow decay of the AM signal for all three EAMs, presumably due to an interface charge accumulation in the silicon substrate. The absence of a clear cut-off frequency, with a slope of at least -10 dB/decade, has conducted us to define the EOBW at -6 dB, as shown in the inset of Figure 4.19. We observed a maximum bandwidth at approximately -3V before saturation occurred. By reducing the modulator length from 200 μm to 100 μm , we nearly doubled the bandwidth, as expected. This resulted in an increase from 20 GHz for 200 μm to 38 GHz for 100 μm . A shorter modulator section has a lower RC constant, although it also reduces the extinction ratio. In our case, the measured ER for the most compact EAM ($L = 100 \mu\text{m}$) is good enough for 56 Gbps modulation, as it will be shown later on.

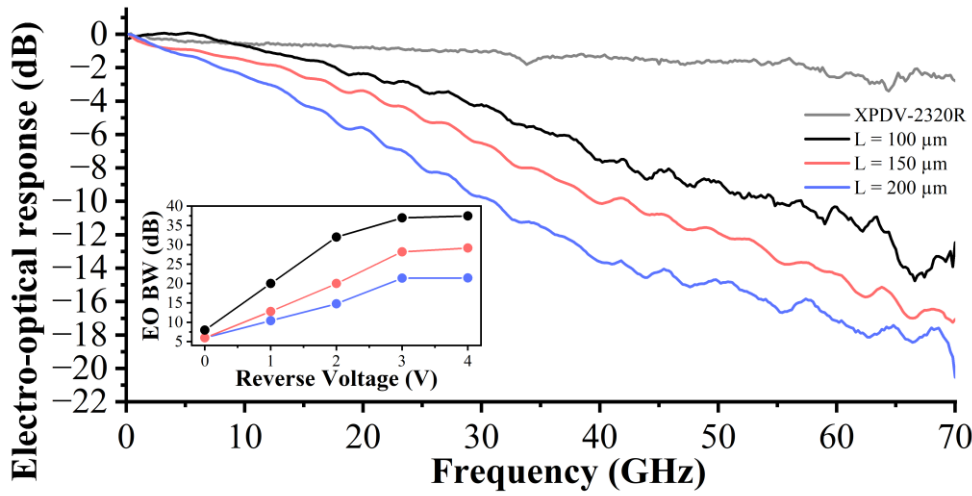


Figure 4.19: Measured small-signal EO response for the high-speed photodiode used for the VNA calibration (gray curve) and for various lengths of the integrated EAM ($L = 100 \mu\text{m}$, $150 \mu\text{m}$ and $200 \mu\text{m}$). The inset displays the measured EOBW at -6 dB as a function of the bias voltage.

4.4.3.4 Bit error rate and transmission measurements for 32-Gbps non-return-to-zero signals

The data transmission measurements were conducted at various wavelengths covering the entire optical bandwidth of the transmitter ($\sim 15 \text{ nm}$), with a step of $\sim 3 \text{ nm}$ (corresponding to the Vernier's coarse wavelength tuning), using 32 Gbps NRZ signals with a pattern length of $2^{31}-1$ at 20°C . The propagation distances span from 0 (B2B) to 10 km, with intervals of 2 km, employing SSMF. During the experiments, the current values in the gain sections of the laser, the SOA booster and the bias voltage of the EAM were slightly adjusted for each wavelength and distance to minimize the BER, remaining around of 70 mA, 80 mA, and -2.5 V, respectively. The swing voltage (V_{pp} , peak-to-peak voltage) was fixed at 3 V for all measurements. The modulated waveguide-coupled output power varies between 2 and 3 mW in function of the transmitted wavelength. Figure 4.20 illustrates the obtained BER measurements after different transmission distances as a function of the received optical power at six different wavelengths: 1542 nm, 1545 nm, 1548 nm, 1551 nm, 1554 nm, and 1557 nm. The corresponding eye diagram can also be observed in the inset. We noted a decrease in the DER from 8.5 dB for the optimum transmission wavelength at $\lambda = 1554 \text{ nm}$ to 6.5 dB (for $\lambda = 1557 \text{ nm}$) and 4.5 dB ($\lambda = 1542 \text{ nm}$), which correspond to the wavelengths at each side of the transmitter's optical bandwidth. It is worth to remark that the optical

bandwidth is limited, at the blue wavelength side of the optical spectrum, by the absorption band edge of the EAM (lower wavelength from which the EAM is transparent), and at the red wavelength side, by the longest attainable wavelength in the widely tunable laser, which is delimited by the optical gain wavelength span of the gain section. This fact can be easily observed by looking at the spectra of Figure 4.18-d, where we can see that the lasing wavelength at $\lambda = 1557$ nm is almost at the band edge of the optical gain spectrum. Error-free propagation was achieved for most tested wavelengths in the B2B configuration and for transmission distances up to 2 km. Figure 4.20-e depicts the measured BER and eye diagrams at a wavelength of 1554 nm. A negligible power penalty was observed when comparing the B2B measurement (dark blue) with the +2 km transmission (light blue). For B2B testing, a received power sensitivity of approximately -5.5 dBm was achieved for a BER of 10^{-12} , which corresponds to the minimum measurable value of our equipment. Notably, there was a power penalty of 0.5 dB, 1 dB, and 2.8 dB in the received optical power at the 7% HD-FEC limit, respectively, when compared with the 4 km, 6 km, and 8 km transmissions under B2B conditions. We observed increasing degradation with transmission distance, which can be attributed to a combination of optical feedback, loss, and chirp. Longer transmission distances are expected using optical links that partially compensate for chirp, such as non-zero dispersion shifted fibers (NZ-DSF).

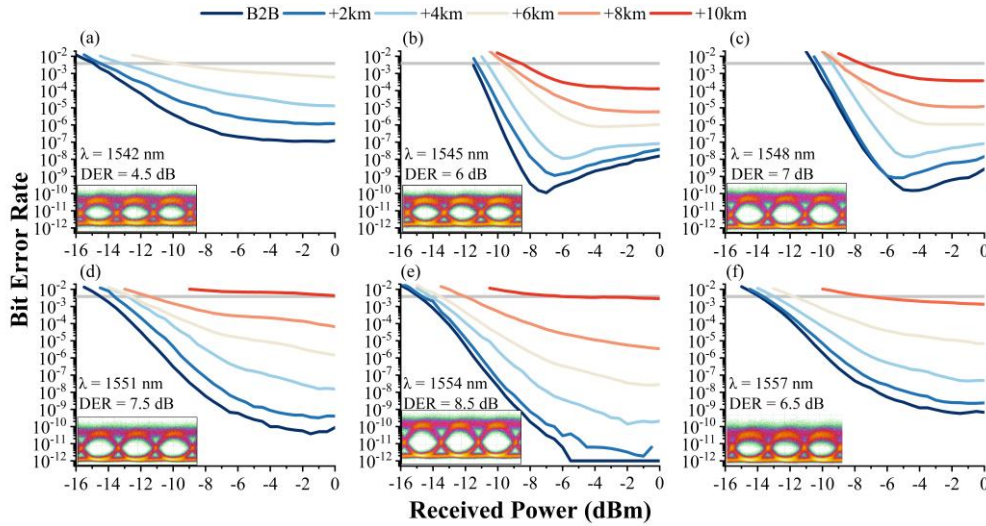


Figure 4.20: Measured BER at 32 Gbps as a function of the received power for each wavelength over distances ranging from B2B to 10 km with a step of 2 km. The inset shows the eye diagrams for each wavelength, and the gray horizontal lines mark the HD-FEC limit.

The EAM and the SOA having opposite chirp values, a fine tuning of their driving voltage and current values enables for a partial compensation of the chirp, enabling longer transmission distances than in directly modulated lasers (DMLs) or externally-modulated lasers without SOA boosters (EMLs) [42]. It is worth mentioning that, for all wavelengths, the power penalty is smaller than 0.5 dB at the 7% HD-FEC limit when comparing B2B transmission with the +2 km transmission, indicating good data transmission performance.

4.4.3.5 Transmission measurements for 50 Gbps and beyond

We evaluated the large-signal intensity modulation of the III-V/Si widely tunable transmitter through data transmission experiments using a NRZ-OOK signal with a $2^{31}-1$ PRBS pattern length. The DC bias current operating points for the tunable laser and the SOA section were

established at 65 mA and 70 mA, respectively. The reverse bias voltage of the EAM is set at -2.5 V, while the swing voltage is fixed at 3 V for all measurements. Figure 4.21-a illustrates open eye diagrams for a B2B configuration at 1550 nm, demonstrating 50 Gbps modulation speed with a dynamic extinction ratio of 4.7 dB. Similar ER values were obtained for other wavelengths after optimization of the operating point for each device in the transmitter.

At last, the 56 Gbps NRZ large signal modulation was conducted using a $2^{31}-1$ PRBS pattern length at the same wavelength. The swing voltage remains constant, and the bias voltage applied to the EAM is approximately -2.2 V. The bias current for the tunable laser and the SOA were set at 65 mA and 66.8 mA, respectively, representing the optimal values for measuring open eye diagrams at such bit rate. Figure 4.21 (b) and (c) show the input electrical signal and the output optical eye diagrams respectively, which reveals a dynamic extinction ratio of 4 dB. The electrical eye diagram starts showing distortion introduced by the RF amplifier, which shows a gradual performance degradation beyond 50 GHz limit of our measurement setup and equipment. As a consequence, the optical eye diagram shows a slightly distorted signal with some jitter, although it still provides clearly discernible openings. It is worth to remark that we did not use any electrical amplification or post-data treatment at the receiver, nor equalization for all the measurements, which could significantly enhance the component's performance. Additionally, as the EAM modulator is an RC-limited device, further bandwidth enhancement can be achieved by reducing its length even more, although we will have an additional penalty on the extinction ratio. Still, this high-speed and widely tunable III-V/Si transmitter has shown good dynamical properties up to 56 Gbps in the C-band, providing a good starting point to develop high-speed and compact silicon photonic transmitters that can be used in high-capacity optical networks.

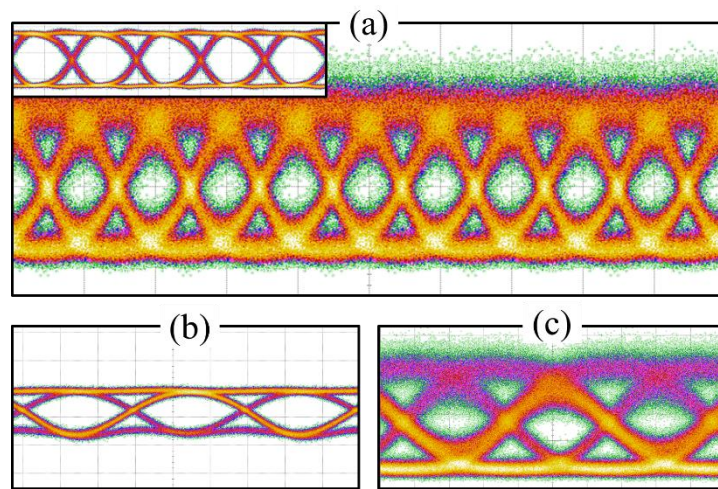


Figure 4.21: Optical and electrical eye diagrams measured from an NRZ modulation signal with a PRBS pattern length of $2^{31}-1$ in a back-to-back configuration at different modulation speeds: (a) 50 Gbps, with the inset displaying the electrical eye diagrams. (b) and (c) showcase the electrical and optical eye diagrams for 56 Gbps.

In this study, we demonstrated III-V-on-Si C-band transmitters based on externally-modulated widely tunable lasers with integrated semiconductor optical amplifiers. By using ring-based Vernier filters, the laser exhibits a wide single-mode wavelength tuning range of approximately 40 nm with stable monomode operation and an SMSR of at least 50 dB, achieving up to 5 mW of on-chip output power. Through static characterization, we investigated the performance of the EAM, revealing an extinction ratio that varies with

wavelength, peaking at 16 dB at 1556 nm. Small signal analysis of the EAM demonstrated a 6 dB cut-off frequency bandwidth of 38 GHz. Bit-error-rate measurements confirmed error-free transmission at 32 Gbps in a B2B configuration for several wavelengths, and transmission over various distances was conducted with moderate power penalties, not exceeding 2.8 dB for a transmission of 8 km through SSMF. Furthermore, large-signal intensity modulation experiments demonstrated the transmitter's capability to achieve transmission rates of up to 56 Gbps non-return-to-zero with a dynamic extinction ratio of 4 dB in the eye diagram in a B2B configuration. These results underscore the potential of III-V/Si integrated transmitters for high-speed and compact silicon photonic applications, paving the way for future developments in wavelength division multiplexing applications.

4.5 Chapter Conclusion

In this chapter, I investigated the static and dynamic characteristics of externally-modulated lasers (EMLs) that utilise lumped III-V/Si electro-absorption modulators (EAM) that contain QWs. I have focused on the assessment of the trade-off between a scalable and mature fabrication and a competitive performance by using a single III-V epitaxy integrated on a Si PIC by direct bonding. The modulators were fabricated using the same epitaxial stack than those used for lasers or SOAs, with a focus on optimizing optical confinement in each component. Initially, a DFB laser was integrated with an EAM and an SOA, utilizing a tapered EAM component. Although the lumped EAM has to be as compact as possible to limit the impact of its capacitance to provide a high EO bandwidth, the need for an adiabatic transition between the Si and the III-V waveguide imposes a relatively long design that affects the EAM's performance. To address this, we aimed to reduce the modulator's length while maintaining the optical mode in the III-V waveguide therefore minimizing the use of tapers. Next, we integrated a widely tunable laser with the EAM and SOA using the same mesa-InP. Static performance analysis of the laser was conducted, followed by transmission measurements of the complete transmitter using NRZ signals at 32 Gbps across various wavelengths and distances, up to 10 km. These measurements confirmed the emitter's ability to maintain a low BER, demonstrating error-free propagation in B2B configurations for several wavelengths, with power penalties peaking at 2.8 dB for 8 km. Further testing at 50 Gbps and 56 Gbps confirmed the transmitter's robustness, achieving dynamic extinction ratio of 4 dB. The results highlight the potential of III-V/Si integrated transmitters for high-speed and compact optical transmission systems, indicating promising directions for future innovations in WDM applications and high-capacity optical networks requiring intensity-modulated direct detection (IM-DD) schemes.

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5. Conclusions & Perspectives

5.1 Conclusions

This work focuses on developing high-speed transmitters designed to meet the growing bandwidth demands of optical communications. The integration of III-V materials on silicon photonic circuits introduces new possibilities for high-speed, low-cost, and energy-efficient optical communication systems. Our objective was to design a compact, high-speed III-V/Si transmitter operating in the C-band, intended as a cost-effective solution for dense wavelength division multiplexing (DWDM) in metropolitan and access networks. This thesis primarily involves modeling, designing, and measuring these devices.

To achieve this, I initially investigated III-V-on-silicon DFB laser properties with double-side patterning on the SOI waveguide, incorporating backside Bragg grating. This part focused on optimizing the structure design and optical coupling for low- κ laser, which significantly enhanced bandwidth performance and stability, providing improved control over noise.

The second part concentrated on direct modulation and its limitations, which I addressed through transmission experiments and dynamic analyses on DFB lasers with backside Bragg grating. I tested two types of DFB lasers: a long-cavity device ($L = 880 \mu\text{m}$) with a sampled Bragg grating (SBG-DFB) for more uniform optical power distribution, and a shorter $\lambda/4$ DFB device with a cavity length of $400 \mu\text{m}$. The SBG-DFB, featuring a low coupling factor ($\kappa < 40 \text{ cm}^{-1}$), achieved a narrow linewidth of $\Delta\nu = 118 \text{ kHz}$, single-mode operation with a side-mode suppression ratio (SMSR) over 45 dB, an on-chip output power of 22 mW (13.4 dBm), and a wall-plug efficiency of 17%. Dynamic characterization showed an intrinsic 3 dB modulation bandwidth of 14 GHz. Modulation at 25 Gbps was successful in both back-to-back (B2B) and 2 km standard single-mode fiber (SSMF) transmission.

The $\lambda/4$ -DFB laser with a shorter cavity exhibited an on-chip output power of up to 12 mW, a threshold current of 26 mA, and $\kappa = 54 \text{ cm}^{-1}$. This device operated in single mode across a wide current range, achieving a high SMSR exceeding 50 dB and an intrinsic modulation bandwidth of 21 GHz at -3 dB. It supported NRZ-OOK signal transmission at 32 Gbps in B2B and 2 km transmission over SSMF. To further enhance modulation bandwidth, I tested 4-level Pulse Amplitude Modulation (PAM-4) at 50 Gbps. Optical feedback tolerance was also evaluated, with direct NRZ-modulation data transmission at 28 Gbps under varying feedback levels. The results showed robust performance with minimal power penalty, even under significant feedback, maintaining BER below the HD-FEC threshold at 28 Gbps, except at -12 dB feedback.

Subsequently, I explored external modulation to address the need for higher bandwidth, focusing on laser-EAM integration for high-density access networks. We opted to use the same

epitaxial stack as those used for lasers and SOAs, with optical confinement optimized for each component. Mode transfer between components was achieved with an adiabatic taper, and an SOA was integrated to offset modulator losses. The externally-modulated laser co-integrated with an SOA delivered up to 16 mW of on-chip output power at $\lambda = 1540$ nm. The transmitter supported 32-Gbps NRZ modulation with a significantly higher dynamic extinction ratio (DER) of 6 dB in the eye diagram. Transmission through 5 km of SSMF resulted in a 0.8 dB power penalty at the receiver and a DER of 4.6 dB. We also assessed the impact of SOA-induced optical feedback on BER measurements, achieving a BER below 7% HD-FEC at moderate bias values and received power of -14 dBm.

Lastly, I integrated a widely tunable laser with the EAM and SOA, using the same mesa-InP design. The tunable laser achieved a wavelength range of approximately 40 nm, with the transmitter covering a 15 nm range and a maximum output power of 5 mW. Transmission measurements of the complete transmitter using NRZ signals at 32 Gbps across various wavelengths and distances up to 10 km demonstrated error-free propagation in B2B configurations, with power penalties peaking at 2.8 dB for 8 km. Further testing at 50 and 56 Gbps confirmed the transmitter's robustness, achieving a DER of 4 dB.

The following table presents the results achieved in this thesis. These results highlight the potential of III-V/Si integrated transmitters for high-speed and compact optical transmission systems, indicating promising directions for future innovations in WDM applications and high-capacity optical networks requiring intensity-modulated direct detection (IM-DD) schemes.

Transmitters	Ref	Wav. Power (mW)	DC Voltage and P2P Voltage (V)	Extinction ratio (dB)	Bit rate (Gbps)
DML — SBG-DFB —	[1]	22	$I = 100\text{mA} / 2$	3	25
DML — DFB —	[2]	12	$I = 140\text{mA} / 3.5$	3.7	32
LASER+EAM — DFB — EAM —	[3]	6	-2.5V / 3.5	4	32
LASER+EAM+SOA — DFB — EAM — SOA —	[3]	16	-2.5V / 3.5	6	32
TUL-EAM-SOA — TuL — EAM — SOA —	[4]	5 Wavelength range of 15 nm	-2.5V / 3	From 4.5 to 8.5	56

Table 5.1 : Review of the principal experimental achievements.

5.2 Perspectives

As this thesis has focused on the development of components operating in the C-band, future work could extend these developments to enable operation in the O-band. The transition to the O-band offers several promising advantages, particularly for applications in short-reach and data center communications, where the O-band's reduced chromatic dispersion in SSMF can significantly enhance signal quality over short distances.

Further future work will further examine the impact of optical/electrical feedback on the performance of DFB lasers. Indeed, by introducing optical feedback, one can enhance the frequency selectivity of the laser, thereby reducing the spectral linewidth. Furthermore, optical feedback can mitigate intensity noise and enhance the frequency stability of the laser, which can positively affect the overall system performance [5], [6].

An extension of the feedback effect is the dual feedback approach, which involves applying both optical and electrical feedback to the laser. In this configuration, a portion of the optical signal is introduced into the laser cavity while simultaneously adjusting the device's supply voltage. Dual feedback aims to leverage the intricate interaction between optical and electrical feedback to achieve improved performance.

The configuration of the desired system is schematically depicted in Figure 5.1. A theoretical study was conducted by members of our team to model this dual-feedback system [7], modifying the dimensionless Lang-Kobayashi equations [8]. The developed formalism provides quantitative insights into the dynamics of this dual-feedback system, and the stability condition is established based on the criterion of the first Hopf bifurcation.

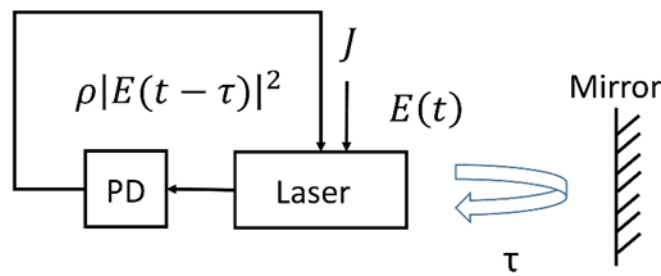


Figure 5.1 : Scheme of the desired dual-feedback system (from [7])

The laser is excited by two sources: a constant current J and a current proportional to the delayed electric field from a distant mirror $\rho|E(t - \tau)|^2$, τ where represents the external-cavity round-trip time, and ρ (in s^{-1}) is a time-independent coefficient. This approach can provide even more precise control over the spectral and temporal properties of the laser, paving the way for new optimization possibilities.

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6. Résumé en Français

6.1 Introduction

Les réseaux de télécommunications optiques constituent l'épine dorsale de l'Internet moderne, ils sont aujourd'hui essentiels à l'infrastructure numérique mondiale, répondant à un besoin croissant de transmission de données à très haut débit. Historiquement, les communications à longue distance utilisaient des câbles coaxiaux et des systèmes radioélectriques, mais ceux-ci étaient limités en bande passante et en distance. L'introduction des fibres optiques dans les années 1970, combinée à l'invention du laser à semi-conducteurs [1], a révolutionné le secteur en permettant des transmissions sur des milliers de kilomètres avec des pertes extrêmement faibles ($\approx 0,2$ dB/km autour de 1550 nm) [2]. Aujourd'hui, l'explosion des usages (vidéo 4K, cloud computing, intelligence artificielle, 5G/6G, LiDAR, objets connectés) entraîne une croissance exponentielle du trafic de données—multiplié par environ 12 en dix ans [3]. Ce phénomène, souvent comparé à une "loi de Moore photonique" [4], impose de développer des systèmes plus rapides, plus compacts, moins coûteux et plus économes en énergie.

Un système de communication optique se compose typiquement d'un émetteur : laser + modulateur (intégré ou externe) ; d'un canal optique : une fibre optique mono-mode pour longue distance, avec éventuels amplificateurs (EDFA) [5] et compensateurs de dispersion ; d'un récepteur : photodétecteur. Les réseaux optiques se structurent selon trois niveaux :

- Réseaux longue distance ("long-haul") : câbles transocéaniques au-delà de 6 000 km ;
- Réseaux métropolitains et régionaux : quelques centaines de km ;
- Réseaux d'accès : 20–40 km.

Dans ce dernier segment (les réseaux d'accès), les architectures de type Passive Optical Network (PON) sont privilégiées [6] grâce à son architecture simple et à ses coûts réduits. Les principaux standards actuels, normalisés par l'ITU-T, sont :

- GPON (2,5 Gbit/s ↓, 1,25 Gbit/s ↑) ;
- XG-PON (10 Gbit/s ↓, 2,5 Gbit/s ↑) ;
- NG-PON2 (jusqu'à 40 Gbit/s grâce au multiplexage WDM + TDM) ;
- 50G et 100G PON [7].

Les travaux de recherche portent sur le 100G-PON et au-delà, mais ces débits sont encore au stade de R&D pour l'accès. En parallèle, dans le cœur de réseau et le backbone, les systèmes optiques cohérents atteignent déjà 100G, 400G et 800G par canal, grâce au multiplexage en

longueurs d'onde (DWDM) et aux modulations avancées, mais ces technologies ne s'appliquent pas directement au PON.

Au cœur des émetteurs optiques, les lasers à rétroaction distribuée (DFB) occupent une place centrale, leur réseau de Bragg intégré sélectionne finement la longueur d'onde d'émission, assurant un fonctionnement monomode stable, une bonne efficacité énergétique, et un coût maîtrisé [8]. Notamment grâce à la photonique sur silicium (SiPh) qui se distingue par sa compatibilité avec les procédés CMOS et sa capacité à intégrer des composants passifs compacts [9].

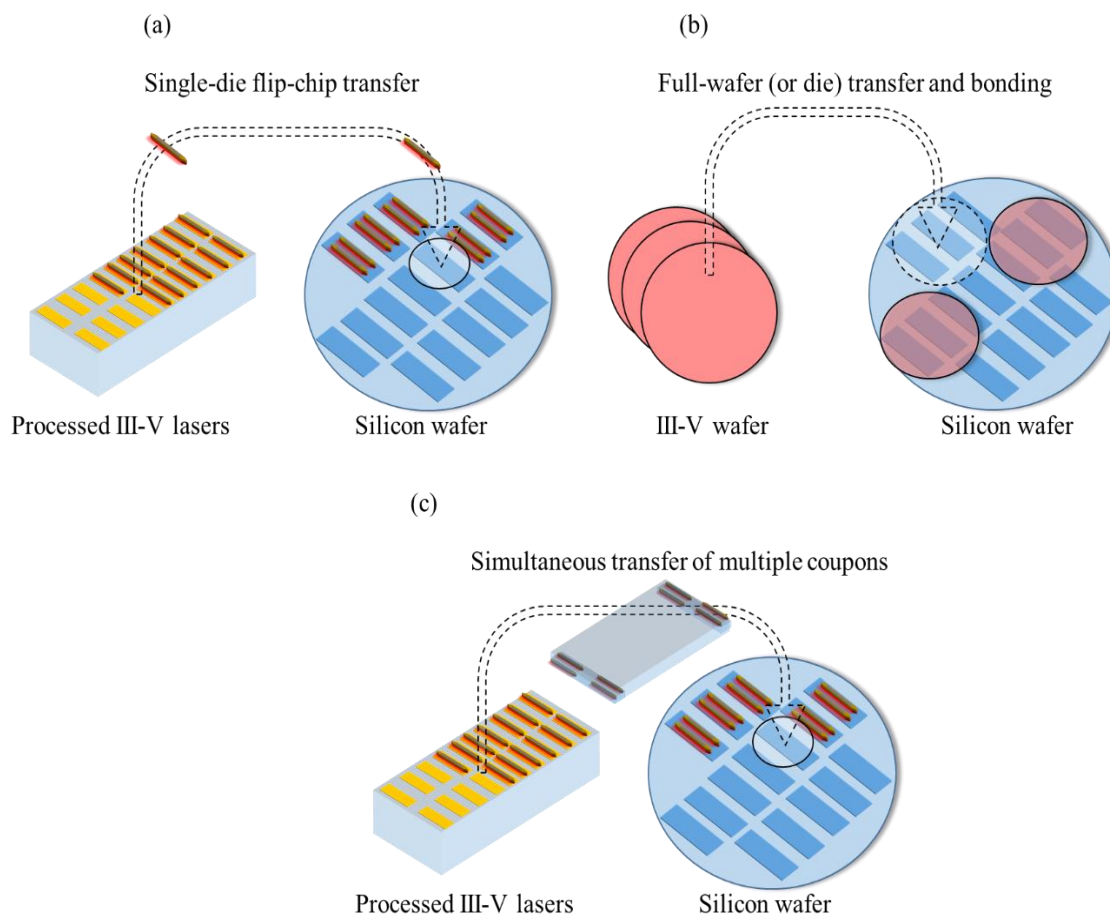


Figure 6.1 : Schéma illustrant les trois techniques d'intégration des composants photoniques : (a) montage par flip-chip, (b) collage direct et (c) micro-transfert par impression.

La photonique sur silicium exploite la maturité industrielle de la plateforme CMOS ainsi que le contraste élevé d'indice de réfraction entre le silicium et son oxyde, offrant ainsi des dispositifs compacts et performants. Par ailleurs, la transparence optique du silicium pour les longueurs d'onde supérieures à 1 100 nm en fait un support privilégié pour les applications en télécommunications. Cependant, en raison de sa bande interdite indirecte, il nécessite l'apport de matériaux III-V à bande interdite directe, Figure 6.1, comme l'InP ou l'InGaAsP, pour générer de la lumière [10]. Dans cette optique, la plateforme hétérogène III-V sur silicium représente une approche particulièrement pertinente pour la co-intégration monolithique de

sources laser et de leurs circuits de pilotage électroniques, contribuant à l'amélioration des capacités et de l'efficacité des réseaux optiques de nouvelle génération. L'intégration peut se faire par flip-chip bonding, micro-transfer printing, ou par collage direct (Direct bonding, utilisé dans cette thèse) du III-V sur silicium. Dans l'intégration hétérogène, le mode optique est partagé entre la couche III-V et le guide d'onde en silicium via le couplage évanescent, créant un supermode.

Les stratégies de modulation constituent un autre axe de différenciation. La modulation directe (Directly Modulated Lasers : DML) reste une solution attractive en raison de sa simplicité, de sa compacité et de sa faible consommation énergétique. Cependant, sa bande passante intrinsèque, limitée par la fréquence d'oscillation de relaxation, demeure relativement faible. Diverses approches comme la résonance photon-photon (PPR), le detuned loading, l'injection optique ou la modulation push-pull permettent d'étendre ces limites. La modulation externe (Externally Modulated Lasers : EML), en revanche, offre une bande passante supérieure, un meilleur rapport d'extinction et un chirp réduit, au prix d'une intégration plus complexe.

Les avantages et inconvénients des différentes plateformes d'intégration sont résumées dans le tableau suivant :

Plateforme	Laser intégré	Ø comp	Maturité	Diamètre wafer (mm)	Rendement	Fabless
InP	Oui	++	+++	76	++	Oui
Silicon Photonics	Non	+++	+++	200–300	+++	Oui
III-V/Si (direct bonding/ flip-chip)	Oui	+++	++	200–300	/	Non
Micro-transfer printing	Oui	+++	+	200–300	/	Non
III-V/Si (direct growth)	Oui	++	+	200–300	/	Non
LNOI	Non	++	+	150–200	+	Non

Tableau 6.1 : Analyse des avantages et des limites des principales plateformes d'intégration photonique (Ø comp. : dimension des composants)

Ce travail de thèse s'inscrit dans la recherche d'émetteurs III-V/Si haute vitesse adaptés aux réseaux métropolitains et d'accès en DWDM. L'objectif est de démontrer des dispositifs compacts, efficaces et compatibles avec une production à grande échelle, capables d'atteindre des débits supérieurs à 50 Gbps tout en minimisant la consommation énergétique. La suite du manuscrit s'articule autour de deux grandes familles de dispositifs :

- Les lasers directement modulés,
- Les sources à modulation externe (EML), combinant laser, modulateur électro-absorption (MEA, ou EAM en anglais) et amplificateur optique intégré (SOA), pour une amélioration de la bande passante et des performances de transmission.

L'objectif principal de la thèse est la modélisation, la conception, et la caractérisation expérimentale ces différentes sources, tout en proposant des solutions d'intégration compatibles avec les exigences industrielles des réseaux d'accès de prochaine génération.

6.2 Lasers à rétroaction distribuée (DFB)

Dans cette partie, j'ai étudié et conçu des lasers DFB intégrés sur plateforme silicium sur isolant (SOI) par intégration hétérogène de matériaux III-V. Ces lasers sont essentiels dans les systèmes optiques à détection directe (IM-DD) utilisés en réseaux d'accès à haut débit, car ils offrent une émission monomode stable, une largeur spectrale réduite et la possibilité d'être modulés directement à haute vitesse. La longueur d'onde d'émission est fixée par la période du réseau, qui doit coïncider avec la bande de gain du matériau actif (InP/InGaAsP ou InP/InGaAlAs dans le cas étudié). Les lasers DFB sont privilégiés dans les réseaux WDM et PON pour leur pureté spectrale et leur capacité de modulation directe à haute vitesse [11].

Le comportement dynamique des DFB peut être décrit à partir des équations de taux couplées électron-photon, reliant la densité de porteurs, la densité de photons et le gain optique [8]. Ces équations incluent les recombinaisons radiatives, Auger et de surface, ainsi que les pertes internes et la durée de vie photonique. Elles permettent de déterminer la fréquence de résonance et les limitations en bande passante.

En complément, la théorie des modes couplés (CMT) est utilisée pour modéliser la propagation des ondes avant et arrière dans la cavité et pour calculer le facteur de couplage en fonction de la géométrie du réseau. Ce couplage est caractérisé par un facteur noté κ , exprimé en cm^{-1} , et principalement déterminé par la variation de l'indice de réfraction au sein du milieu. Le facteur de couplage du réseau de Bragg, généralement exprimé sous la forme du produit κL où L est la longueur de la cavité, constitue un paramètre essentiel pour la caractérisation des lasers DFB. Ce paramètre joue un rôle central dans la stabilité monomode, il est donné selon la formule suivante [12] :

$$\kappa L = \frac{\pi \Delta n}{\lambda} L \quad (6 - 1)$$

L'intégration hétérogène III-V/Si a été réalisée par collage direct des couches actives sur un circuit photonique silicium pré-structuré. Le guide d'onde silicium (500 nm) est encapsulé dans du SiO_2 (voir Figure 6.2-a), et la géométrie (largeur W_{Si}) est ajustée pour contrôler la répartition modale. Deux approches sont étudiées : le réseau gravé côté supérieur et le réseau gravé côté arrière (backside grating). Ce dernier permet de réduire le facteur de couplage et

de limiter l'effet de *longitudinal spatial hole burning* (LSHB), permettant ainsi l'utilisation de cavités plus longues tout en conservant un fonctionnement monomode.

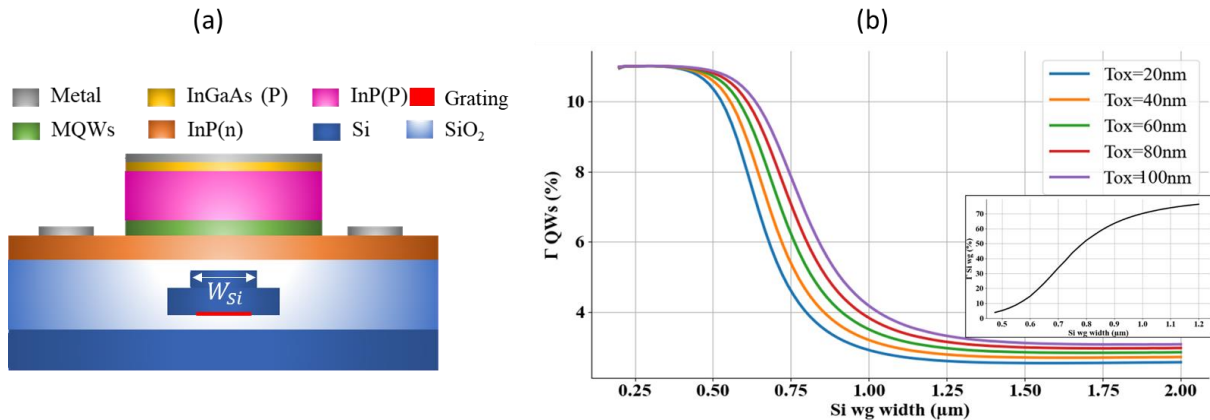


Figure 6.2 : (a) Structure DFB III-V sur silicium : coupe transversale. (b) Facteur de confinement simulé dans le puits quantique (QW) en fonction de la largeur du guide d'onde en silicium (W_{Si}) pour différentes épaisseurs de la couche de collage. L'encart montre l'évolution du facteur de confinement dans le guide silicium en fonction de W_{Si} pour une couche de collage de 20 nm.

L'optimisation du facteur de confinement optique dans les puits quantiques étant essentielle pour atteindre de hautes vitesses de modulation directe, différentes largeurs de guide silicium et épaisseurs de couche de collage ont été évaluées par simulation (Figure 6.2-b). Ces paramètres influencent directement la répartition du mode entre le guide actif III-V et le guide silicium, ainsi que les pertes internes associées.

Afin de limiter les pertes dues à l'absorption inter-valence dans la couche InP(P) tout en maintenant un bon recouvrement du mode avec la zone de gain, une épaisseur de couche de collage de 20–25 nm a été retenue. Par ailleurs, l'analyse comparative de plusieurs largeurs de guide silicium a montré que l'augmentation de W_{Si} accroît le confinement dans le silicium et réduit le rapport (Γ_{QWs}/Γ_{Si}). Ainsi, une largeur de guide de 0,825 μm a été choisie pour la conception du DFB, car elle permet de réduire ce rapport d'un facteur trois par rapport à la largeur de 0,7 μm , optimisant ainsi le compromis entre efficacité optique et pertes internes.

Pour améliorer les effets du LSHB, un réseau de Bragg échantillonné (Sampled Bragg Grating, SBG) est proposé dans la configuration du backside grating. Cette approche segmente le réseau en zones actives et inactives, créant une répartition plus homogène de l'énergie optique. Les simulations (Figure 6.3) montrent une atténuation des pics d'intensité aux extrémités et une augmentation du rendement global.

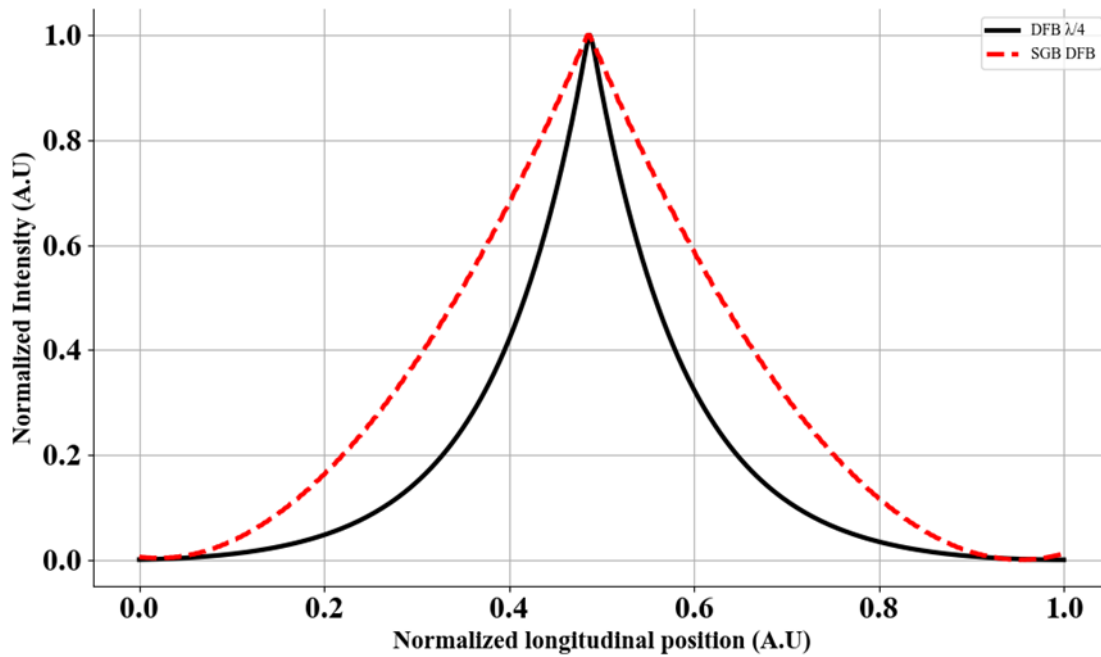


Figure 6.3 : Distribution longitudinale simulée de l'intensité optique normalisée dans la cavité pour deux conceptions de lasers DFB : un DFB à déphasage $\lambda/4$ (courbe noire) et un DFB à réseau de Bragg échantillonné (SBG-DFB, courbe rouge en pointillés)

De manière générale, pour un laser DFB intégré de façon hétérogène, le coefficient de couplage κ dépend fortement de la largeur du guide silicium (W_{Si}), puisqu'il est proportionnel au confinement du réseau.

À l'issue de cette phase de conception et d'optimisation, trois variantes de lasers DFB ont été retenues pour l'évaluation :

- DFB $\lambda/4$ avec réseau gravé en partie supérieure du guide silicium,
- DFB $\lambda/4$ avec réseau gravé en partie inférieure (backside grating),
- DFB à réseau de Bragg échantillonné (SBG-DFB) également en configuration backside grating.

La Figure 6.4 compare l'évolution du coefficient de couplage κ en fonction de la largeur du guide silicium pour ces trois conceptions. Le premier design correspond à une architecture classique de DFB $\lambda/4$ III-V sur silicium avec réseau gravé au-dessus du guide. Le second reprend la même structure mais avec le réseau gravé sous le guide, tandis que le troisième correspond à la variante SBG-DFB en backside grating.

Les résultats montrent que la conception SBG-DFB permet d'obtenir un facteur de couplage κ plus faible, même pour des largeurs de guide silicium élevées ($\kappa \leq 50 \text{ cm}^{-1}$), tout en conservant un compromis favorable entre gain optique et pertes internes grâce à l'ajustement de W_{Si} . Les simulations de distribution modale 2D confirment cette tendance : dans le DFB $\lambda/4$ classique, le mode optique présente un recouvrement important avec la couche InP(P), entraînant des pertes internes élevées, alors que dans le SBG-DFB, le mode est davantage confiné dans le guide silicium, ce qui réduit significativement ces pertes.

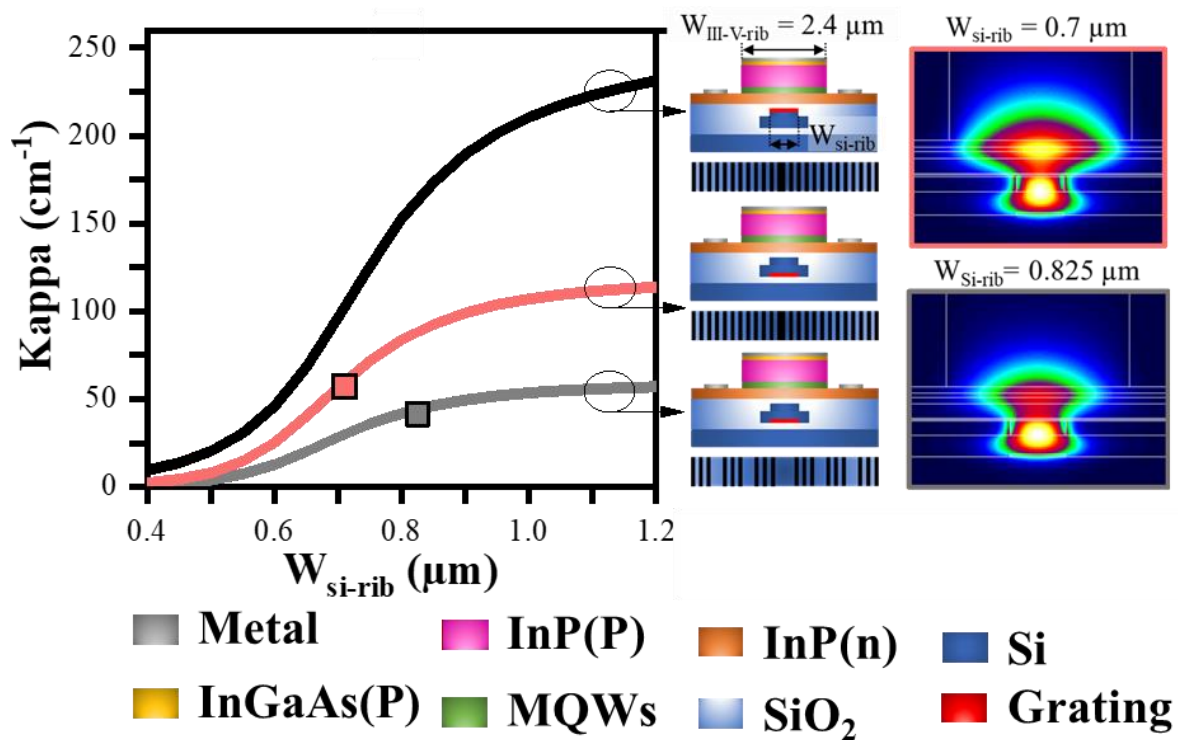


Figure 6.4 : κ simulé en fonction de W_{Si} pour la conception classique (courbe noire), la conception avec réseau en backside (courbe magenta) et la conception à réseau de Bragg échantillonné en backside (courbe gris clair). À droite, la simulation de la distribution modale optique 2D est présentée pour deux conceptions de lasers avec des valeurs de W_{Si} différentes.

Ce chapitre a établi les bases conceptuelles nécessaires à la réalisation de lasers DFB III-V/Si haute performance. Les choix de conception — réduction du facteur de couplage κ , recours à des réseaux de type SBG et optimisation des dimensions du guide — ont été guidés par la recherche d'un compromis entre bande passante, stabilité spectrale et efficacité énergétique.

Les deux architectures ayant leur réseau de Bragg gravé sur backside du guide Si feront ensuite l'objet d'une caractérisation expérimentale détaillée. Celle-ci portera notamment sur le facteur de couplage κ , les caractéristiques électriques et thermiques, les caractéristiques dynamiques, ainsi que la comparaison entre les structures $\lambda/4$ -DFB et SBG-DFB. Ces résultats seront présentés dans le chapitre suivant, consacré à l'étude des performances en modulation directe à haute vitesse.

6.3 Lasers directement modulés à haute vitesse (DML)

Les lasers directement modulés III-V/Si représentent une solution compacte et efficace pour les transmissions IM-DD à haut débit, en supprimant le besoin de modulateurs externes. Leur performance est cependant limitée par le chirp, la bande passante et la dispersion en fibre. L'objectif de ce chapitre est d'analyser la caractérisation statique et dynamique de DFB III-V/Si fonctionnant en bande C.

Deux architectures principales ont été considérées :

- DFB $\lambda/4$,
- SBG-DFB.

Dans la suite, les résultats expérimentaux sont présentés séparément pour ces deux structures, d'abord pour le DFB $\lambda/4$, puis pour le SBG-DFB.

DFB $\lambda/4$

La courbe L-I-V (Figure 6.5) obtenues montrent un seuil de 26 mA à 20 °C, une puissance couplée dans le guide supérieure à 13 mW à 140 mA. Les spectres optiques confirment une émission monomode avec un taux de réjection des modes secondaire supérieur (SMSR) à 50 dB sauf autour de 90 mA où un mode hopping transitoire a été observé. Le facteur de couplage a été déduit de la largeur de la stopband autour du pic principal et est d'environ 54 cm^{-1} . La résistance différentielle, déduite des courbes V-I, elle est d'environ $6,5 \Omega$ pour une cavité de $400 \mu\text{m}$. Sa valeur diminue avec l'augmentation de la longueur de cavité (voir Figure 3.4).

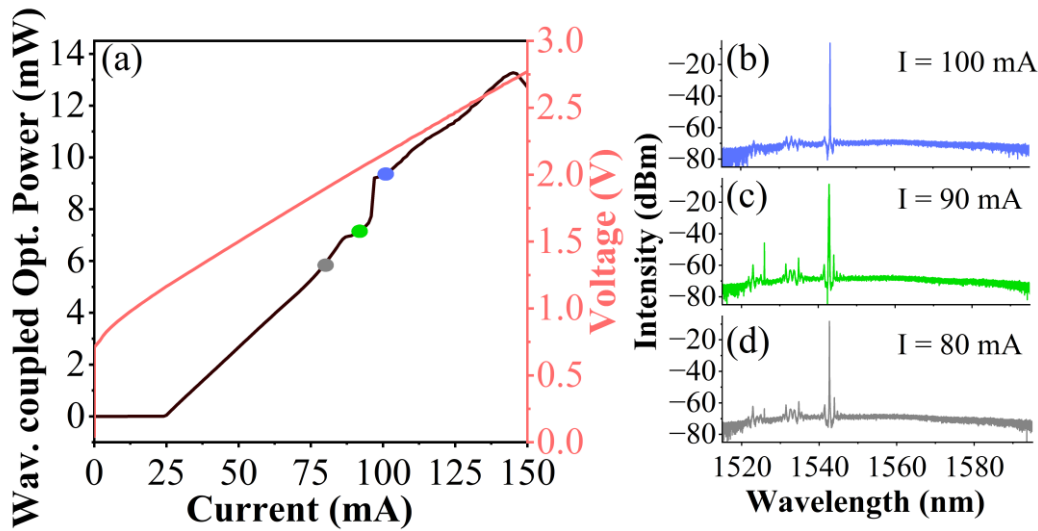


Figure 6.5 : Courbes L-I-V et spectres optiques d'un laser DML pour différents courants de polarisation.

SBG-DFB

Le dispositif étudié possède une longueur de cavité de $880 \mu\text{m}$. À partir de la largeur de la stopband dans le spectre optique (Figure 6.6), un facteur de couplage $\kappa = 37 \text{ cm}^{-1}$ a été extrait. Les lasers SBG-DFB ont été caractérisés dans les mêmes conditions expérimentales que le laser $\lambda/4$ -DFB.

Comme le montre la Figure 6.6, le laser SBG-DFB démontre un courant de seuil de 35 mA. La puissance optique couplée dans le guide atteint plus de 20 mW par sortie, confirmant la symétrie du champ intracavité. L'efficacité murale totale (WPE), atteint une valeur maximale de 17 %, et la résistance différentielle extraite des courbes V-I est de $4,5 \Omega$ (Figure 6.6-b). Le spectre optique reste monomode sur l'ensemble de la gamme de température testée (20–45 °C), avec un SMSR supérieur à 45 dB. La température caractéristique T_0 a été évaluée à 52 °C à partir des courbes L-I mesurées à différentes températures (Figure 6.6-d), en bon accord avec la littérature [13]. L'impédance thermique a été estimée à $Z_{th} = 25 \text{ °C/W}$ par la

méthode d'athermalisation (Figure 6.6-e), et le décalage spectral en fonction de la puissance dissipée est de 4 nm/W.

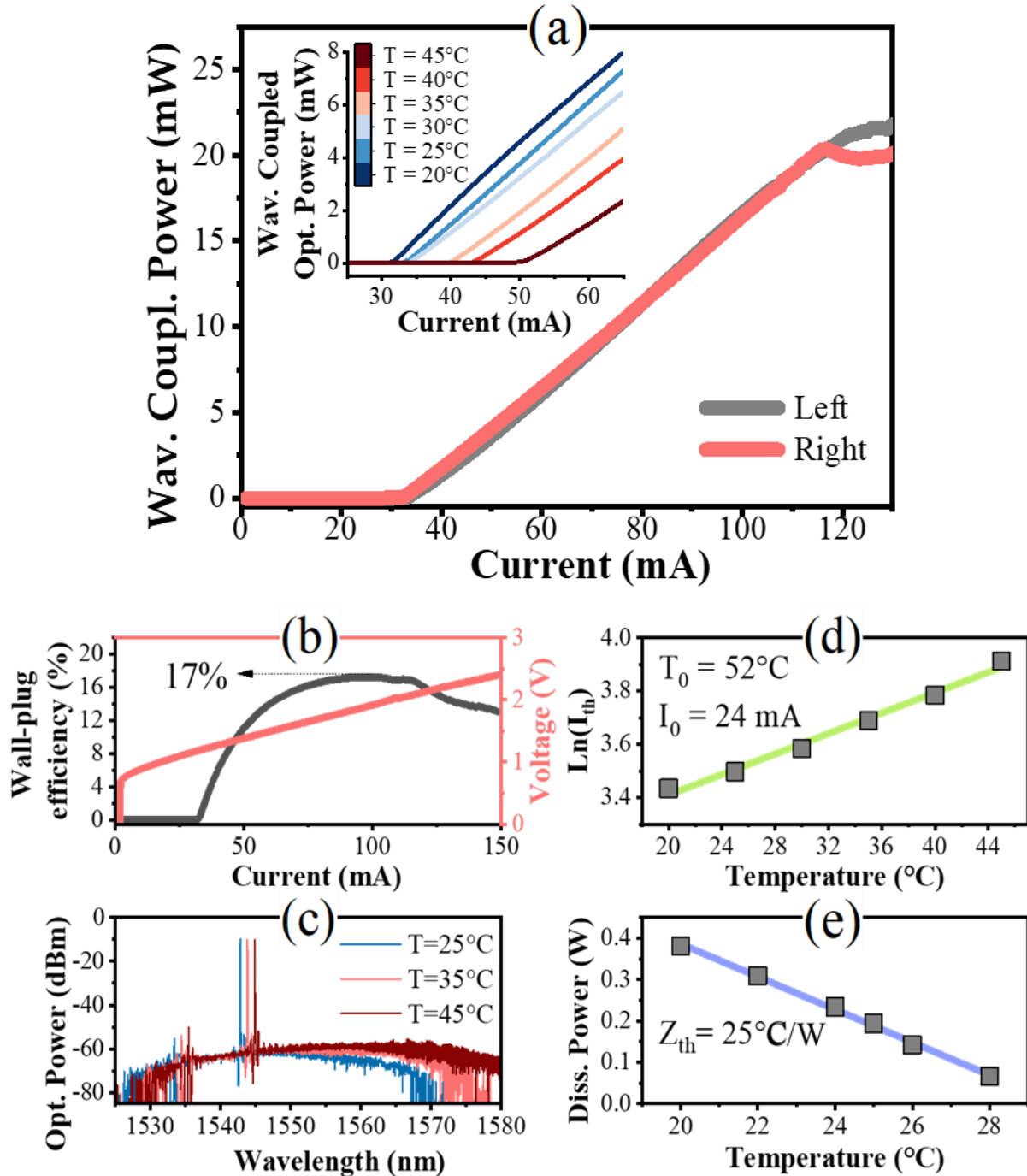


Figure 6.6 : (a) Caractéristique L-I mesurée respectivement sur la sortie gauche (courbe grise) et sur la sortie droite (courbe magenta) du laser SBG-DFB. L'encart montre l'évolution du courant de seuil I_{th} en fonction de la température. (b) Efficacité murale totale (somme des puissances gauche et droite) et tension mesurée en fonction du courant injecté. (c) Spectres optiques obtenus à différentes températures. (d) Évolution du logarithme naturel de I_{th} en fonction de la température de la platine TEC. (e) Puissance dissipée en fonction de la température de la platine TEC pour une longueur d'onde d'émission $\lambda = 1544 \text{ nm}$.

La largeur de raie spectrale a ensuite été mesurée par la méthode d'auto-hétérodyne. L'évolution de la raie suit la loi de Schawlow–Townes modifiée [14] :

$$\Delta\nu = \frac{v_g^2 \cdot h\nu \cdot n_{sp} (1 + \alpha_H^2) \ln\left(\frac{1}{R_m}\right)^2}{8\pi P \eta L^2} \quad (6 - 2)$$

Où v_g est la vitesse de groupe, R_m la réflectivité facette, η l'efficacité quantique, $h\nu$ l'énergie photonique, n_{sp} le facteur d'émission spontanée et α_H le facteur d'Henry. Une largeur de raie minimale de 118 kHz a été mesurée à 110 mA (Figure 6.7-a). Au-delà de 120 mA, un ré-élargissement est observé, attribué à trois mécanismes principaux : les fluctuations d'émission spontanée des modes latéraux non pincés, les fluctuations locales de densité de courant dans la cavité et les perturbations de la rétroaction porteurs/photon qui favorisent le LSHB et la compression de gain [15].

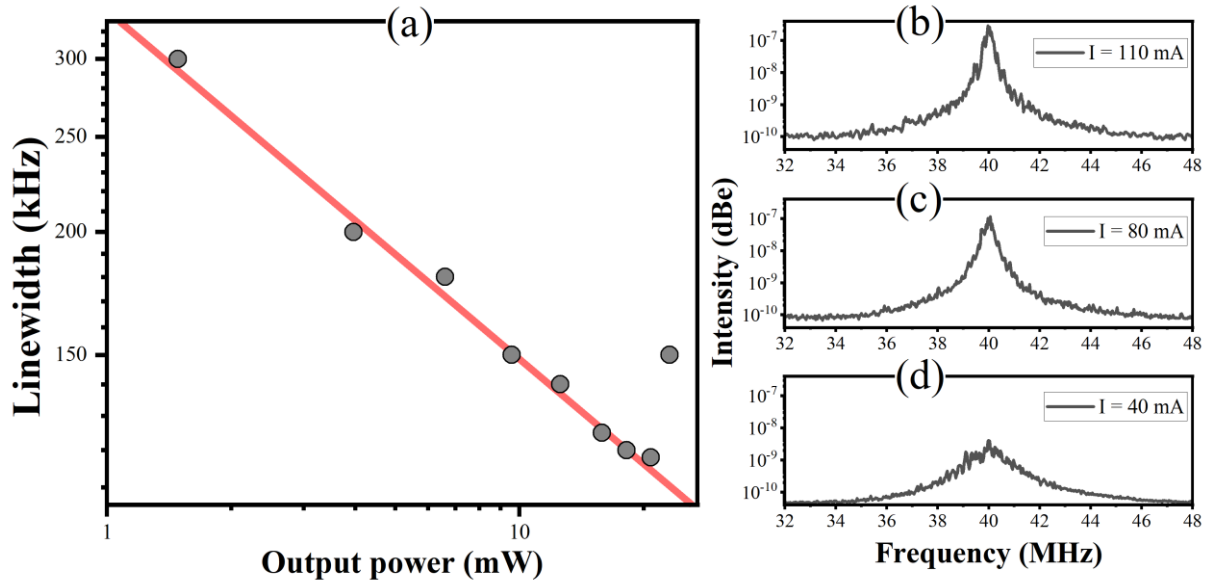


Figure 6.7 : (a) Évolution de la largeur de raie mesurée en fonction de la puissance de sortie. La ligne magenta correspond à l'ajustement linéaire des données expérimentales, en excluant la dernière valeur à $I=120$ mA. Spectres électriques mesurés sous un courant de polarisation de (b) 110 mA, (c) 80 mA et (d) 40 mA.

Les performances dynamiques ont été évaluées par la suite. La dynamique d'un laser modulé directement est gouvernée par la fréquence de résonance f_r , donnée par [11] :

$$f_r = \frac{1}{2\pi} \sqrt{\frac{v_g}{q} \Gamma \eta_i \frac{dg}{dN} \frac{I - I_{th}}{V}} \quad (6 - 3)$$

Où Γ est le facteur de confinement, $\frac{dg}{dN}$ le gain différentiel, V le volume actif et I le courant injecté.

La bande passante électro-optique (petit signal, S21) a été mesurée sur les deux architectures. Le SBG-DFB à longue cavité présente une bande passante intrinsèque de 14 GHz pour une fréquence de résonance $f_r = 9\text{ GHz}$, ce qui traduit l'influence de la durée de vie photonique plus élevée dans cette cavité allongée. À l'inverse, le $\lambda/4$ -DFB à cavité courte atteint une bande passante intrinsèque de 21 GHz avec $f_r = 14\text{ GHz}$, grâce à une durée de vie photonique plus faible.

Ces résultats en petit signal se traduisent dans les performances en transmission grand signal. Le $\lambda/4$ -DFB permet des transmissions NRZ en back-to-back (B2B) (0km) à 25 Gbps, 28 Gbps et 32 Gb/s, avec des taux d'extinction (ER) de 2.8 dB, 2.6 dB et 2.4 dB, respectivement. Pour une modulation à 28 Gbps le laser a démontré un fonctionnement sans erreur jusqu'à 1 km avec un taux d'erreur binaire (BER) inférieur à 10^{-10} . Pour une modulation à 32 Gbps, le laser transmet sous la limite de correction 10^{-3} FEC. Le SBG-DFB, quant à lui, a été validé en transmission NRZ en back-to-back à 25 Gbps avec un taux d'extinction de 3 dB.

Ces performances confirment que les deux structures sont compatibles avec les débits ciblés dans les réseaux IM-DD de prochaine génération. Toutefois, la stabilité spectrale, une largeur de raie étroite et une meilleure tolérance au spatial hole burning confèrent au SBG-DFB un avantage pour des applications où la robustesse spectrale est critique, tandis que le $\lambda/4$ -DFB à cavité courte conserve l'avantage d'une bande passante intrinsèque plus large et un potentiel en termes de vitesse de modulation. Dans la suite, nous concentrons donc notre analyse sur le laser $\lambda/4$ -DFB afin d'évaluer jusqu'où ses performances dynamiques peuvent être poussées, notamment en régime de transmission multi-niveaux PAM-4.

La modulation multi-niveau PAM-4 a ainsi été étudiée afin d'évaluer les performances du laser III-V/Si en régime de transmission à haut débit. Le banc expérimental mis en place (Figure 6.8) repose sur l'utilisation d'un convertisseur numérique-analogique (DAC) alimenté par deux séquences NRZ décorélées de longueur $2^7 - 1$. Après multiplexage et amplification RF, un signal électrique à quatre niveaux est généré et utilisé pour piloter directement le dispositif. Le laser est polarisé à 140 mA avec une tension de 3,5 V crête-à-crête. Le signal optique collecté est amplifié par EDFA, filtré par un TBPF, puis détecté par une photodiode rapide, avant enregistrement sur oscilloscope haute bande passante (70 GHz).

Les résultats montrent que la modulation PAM-4 à 25 Gbps est réalisée avec succès : les diagrammes optiques de l'œil présentent quatre niveaux bien distincts, confirmant la stabilité du signal (Figure 6.9-b). En revanche, à 50 Gb/s (25 Gbaud), une dégradation significative apparaît, principalement imputable aux limitations du DAC et à la distorsion introduite par l'amplificateur RF, comme en témoignent les diagrammes électriques (Figure 6.9-a). Le laser atteint également les limites de sa bande passante intrinsèque, ce qui contribue à la détérioration de la qualité du signal, bien que les quatre niveaux optiques restent discernables.

Il convient de souligner que ces essais ont été menés sans traitement numérique du signal (DSP) ni égalisation au récepteur. Avec l'implémentation d'un DSP adapté, la marge de bruit pourrait être significativement améliorée. Ces résultats démontrent néanmoins que les lasers

directement modulés III-V/Si sont compatibles avec des transmissions PAM-4 jusqu'à 50 Gbps, ouvrant des perspectives intéressantes pour les futurs systèmes optiques à très haut débit.

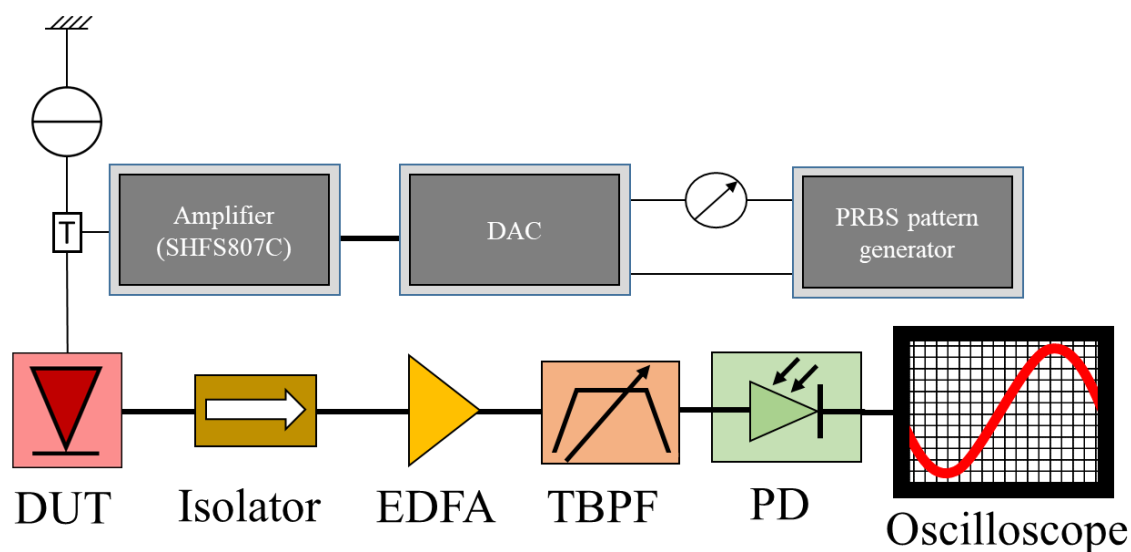


Figure 6.8 : Schéma du banc expérimental utilisé pour la modulation directe PAM-4.

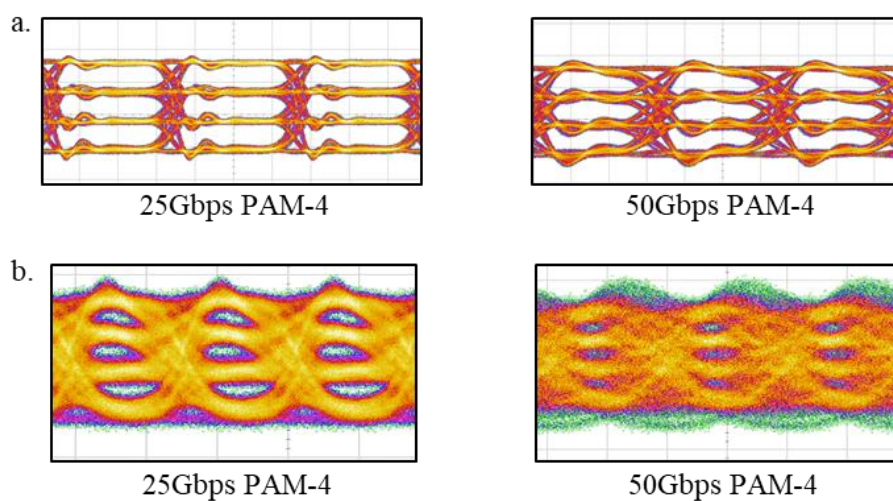


Figure 6.9 : (a) Diagrammes de l'œil électriques après le multiplexeur pour des signaux PAM-4 à 25 et 50 Gbps. (b) Diagrammes de l'œil optiques pour des signaux PAM-4 à 25 et 50 Gbps.

Afin de situer nos résultats par rapport aux travaux antérieurs, le Tableau 6.2 présente une comparaison de nos dispositifs DFB hétéro-intégrés avec l'état de l'art.

State of the art	I_{th} (mA)	WPE (%)	On-chip P_{out} (mW)	EOBW (GHz)	Modulation Speed (Gbps)
[16]	10.5	17	6	15	20
[17]	35	-	-	13	25
[18]	7.5	3	5	24@6dB	40
[19]	32	3.3	5	34 (Photon-photon ressonance : PPR)	43
[20]	2.9	-	0.82*	100 (PPR + SiC)	112
[21]	2.75	-	0.73*	108(PPR + SiC)	256 – PAM4
This work-SBG [22]	35	17	20	14	25
This work- $\lambda/4$ -DFB [23]	26	7	12	21	50 – PAM4

Tableau 6.2 : Comparaison des performances des lasers III-V/Si de cette thèse avec l'état de l'art.

Après avoir caractérisé les performances dynamiques des lasers III-V/Si en modulation NRZ et PAM-4, il est essentiel d'évaluer leur robustesse en conditions réelles de fonctionnement, où la présence de réflexions optiques parasites peut fortement dégrader la stabilité spectrale et la qualité du signal. La section suivante est donc consacrée à l'étude de la sensibilité aux rétroactions optiques externes et à l'analyse de la stabilité des DFB $\lambda/4$, en mettant en évidence les mécanismes menant à l'effondrement de la cohérence (*coherence collapse*, CC) et leur impact sur la transmission à haut débit.

Les lasers DFB hétéro-intégrés III-V/Si sont particulièrement sensibles aux réflexions optiques externes. Même un faible taux de rétroaction (< 1 %) peut induire une instabilité, un élargissement spectral, voire un effondrement de cohérence [24], [25]. Ce phénomène est critique pour l'intégration des lasers dans des circuits photoniques compacts, car l'utilisation d'isolateurs optiques reste difficilement compatible avec les procédés CMOS et augmente fortement les coûts. Le développement de lasers tolérants au feedback est donc une étape clé pour des systèmes photoniques isolator-free.

Le feedback appliqué expérimentalement est quantifié par :

$$f_{ext} = \frac{P_{réfléchie}}{P_{émise}} \quad (6 - 4)$$

Et le temps d'aller-retour de la cavité externe par :

* La puissance de sortie optique maximale couplée à la fibre.

$$\tau_{ext} = \frac{2n_{ext}L_{ext}}{c} \quad (6 - 5)$$

Avec n_{ext} l'indice de la cavité externe et L_{ext} sa longueur.

L'expérience (Figure 6.10) a été réalisée sur les lasers $\lambda/4$ -DFB à cavité courte. Le signal est partiellement réinjecté via un circulateur et un atténuateur variable, permettant de contrôler le taux de feedback jusqu'à -12 dB.

Les spectres mesurés (Figure 6.11) montrent une nette dégradation dès -15 dB de rétroaction, avec un élargissement de raie caractéristique de l'entrée dans le régime de CC.

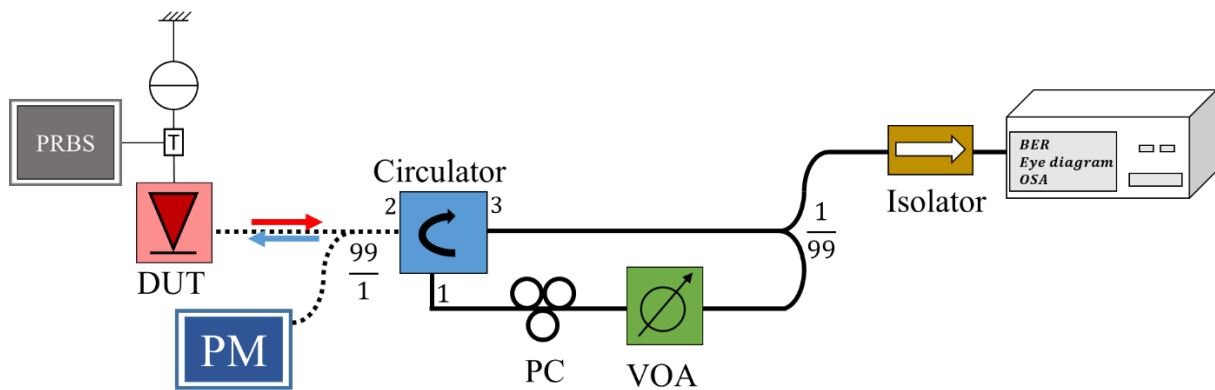


Figure 6.10 : Schéma du banc expérimental utilisé pour l'étude de la rétroaction optique.

En modulation NRZ à 28 Gbps, les mesures de BER et de diagrammes de l'œil (Figure 6.12) montrent un fonctionnement sans erreurs en absence de feedback et une dégradation progressive avec l'augmentation du niveau de rétroaction. Jusqu'à -22 dB, la pénalité de puissance reste inférieure à 1 dB à un BER de 10^{-3} . Au-delà de -17 dB, une forte dégradation apparaît, indiquant la proximité du régime de CC.

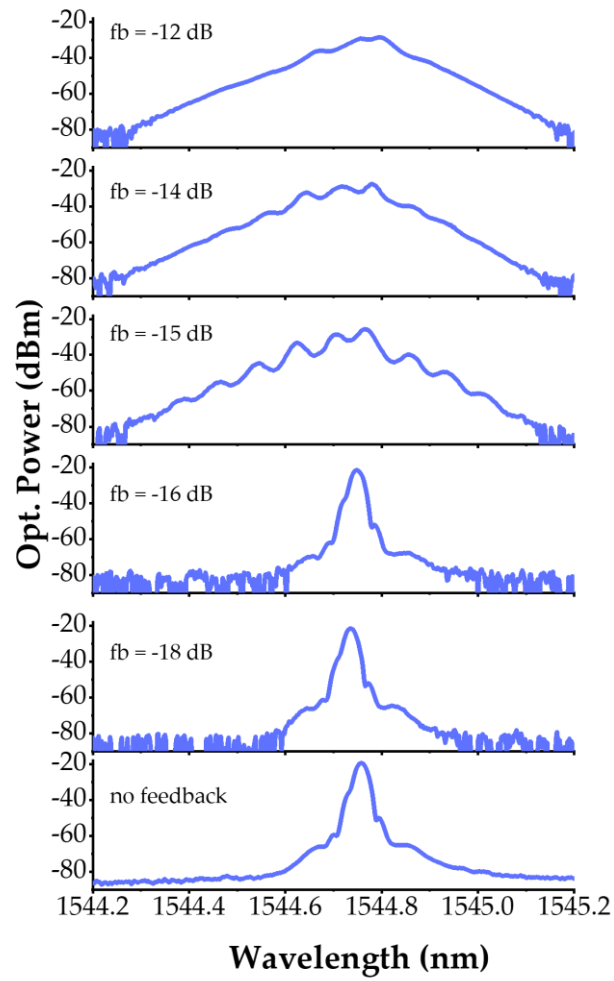


Figure 6.11 : Spectres optiques mesurés du laser DFB pour différents niveaux de rétroaction optique.

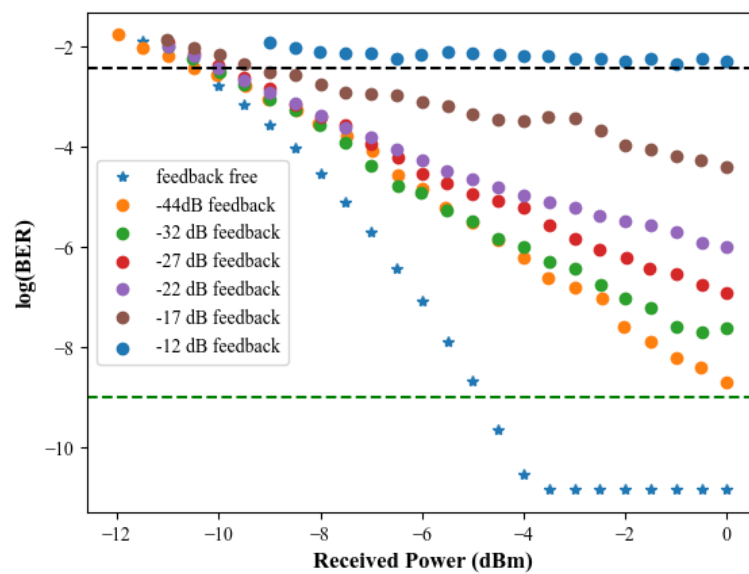


Figure 6.12 : Courbes de taux d'erreurs binaires) à 28 Gbps en configuration back-to-back sous différents niveaux de rétroaction optique.

Ces résultats démontrent que les lasers III-V/Si DFB à faible κL présentent un compromis favorable : ils restent monomodes et stables à fort courant, tout en tolérant des niveaux de feedback comparables aux DFB III-V/Si et InP classiques, mais à des vitesses beaucoup plus élevées (28 Gbps contre 2,5–10 Gbps dans les études précédentes [26], [27]). Ces caractéristiques confirment le potentiel des DFB III-V/Si pour des systèmes de communication isolator-free, combinant haut débit, stabilité et faible coût d'intégration.

Dans cette partie, j'ai démontré que les lasers DFB hétéro-intégrés III-V/Si peuvent être utilisés comme sources monomodes directement modulées à haut débit. Deux architectures ont été explorées : un SBG-DFB à faible facteur de couplage, puissance élevée et largeur de raie étroite ($\Delta\nu = 118$ kHz), et un DFB $\lambda/4$ déphasé présentant une large bande passante électro-optique intrinsèque (21 GHz).

La caractérisation dynamique a mis en évidence la capacité du DFB $\lambda/4$ à assurer des transmissions NRZ jusqu'à 32 Gbps. Le format PAM-4 a également été validé, avec des transmissions réussies jusqu'à 50 Gbps, démontrant la compatibilité de ces lasers avec des schémas de modulation avancés adaptés aux liens optiques courte portée.

Enfin, l'analyse de la tolérance aux rétroactions optiques a confirmé que ces lasers conservent un fonctionnement stable jusqu'à des niveaux de feedback significatifs, avec des pénalités de puissance limitées. Cette propriété, combinée à leur haut débit et leur pureté spectrale, souligne le potentiel des DFB III-V/Si comme candidats prometteurs pour des émetteurs compacts, peu coûteux et sans isolateur, dans des systèmes IMDD de nouvelle génération.

Ces résultats mettent néanmoins en évidence les limites intrinsèques de la modulation directe. Pour surmonter ces contraintes, le chapitre suivant est ainsi consacré à l'intégration de modulateurs optiques externes et amplificateur, afin d'explorer des architectures hybrides offrant un meilleur contrôle du chirp, une plus grande portée en fibre et une amélioration des performances globales.

6.4 Laser à Modulation Externe (EML) : DFB + EAM + SOA

La modulation externe repose sur la séparation entre la génération de lumière et la modulation du signal, réalisée par un composant dédié tel qu'un modulateur électro-absorption (EAM) ou un modulateur de Mach-Zehnder (MZM). Cette approche améliore la qualité du signal en limitant le bruit, la distorsion et la dérive fréquentielle inhérents à la modulation directe. Les sources modulées en externe (EML) se sont ainsi imposées comme des éléments clés des systèmes de transmission optique à haut débit. Toutefois, leur coût et leur complexité d'intégration freinent encore leur déploiement à grande échelle.

Dans cette perspective, l'intégration hétérogène de matériaux III-V sur silicium apparaît comme une solution prometteuse pour concevoir des dispositifs à la fois compacts, performants et compatibles avec les procédés CMOS, ouvrant la voie à une production de

masse. Ce chapitre s'intéresse à l'intégration de modulateurs optiques externes sur plateforme III-V/Si, et en particulier aux EAMs co-intégrés avec des lasers DFB et des amplificateurs optiques à semiconducteurs (SOA). L'objectif est de dépasser les limitations intrinsèques de la modulation directe et de développer des architectures hybrides adaptées aux réseaux IM-DD à très haut débit.

Les EAMs reposent sur le principe de l'électro-absorption, c'est-à-dire la modification de l'absorption optique d'un matériau semi-conducteur sous l'effet d'un champ électrique. Deux phénomènes majeurs sont exploités :

- L'effet Franz-Keldysh (FKE), observé dans les matériaux bulk, provoque un élargissement de la courbe d'absorption vers les grandes longueurs d'onde sous champ électrique.
- L'effet Stark quantique confiné (QCSE), observé dans les puits quantiques, déforme les niveaux d'énergie et modifie l'absorption de façon plus efficace, avec apparition de pics excitoniques.

Deux paramètres essentiels pour évaluer un modulateur optique sont l'extinction ratio (ER) et les pertes d'insertion (IL). Le coefficient de transmission optique, qui exprime la fraction de puissance transmise à travers le modulateur, s'écrit :

$$T(V) = e^{-\Gamma\alpha(V)L} \quad (6 - 6)$$

Où $\alpha(V)$ est le coefficient d'absorption modulable en fonction de la tension appliquée.

À partir de cette expression, l'extinction ratio (ER) est défini comme le rapport entre la puissance transmise à l'état $V = 0$ et à un état biaisé V :

$$ER = \frac{P_{out}(V = 0)}{P_{out}(V)} = \frac{T(0)}{T(V)} \quad (6 - 7)$$

Les pertes d'insertion (IL) quantifient la fraction de puissance perdue lors de la transmission. À l'état $V = 0$, elles s'écrivent :

$$IL_{on} = \frac{P_{in} - P_{out}(V = 0)}{P_{in}} = 1 - e^{-\Gamma\alpha(V)L} \quad (6 - 8)$$

Une cavité trop longue conduirait à une extinction élevée mais à des pertes d'insertion quasi totales, rendant la transmission inefficace. Le design doit donc assurer un compromis entre un ER élevé et un IL faible afin de garantir une modulation efficace. Pour dépasser ces contraintes et réaliser une source intégrée, nous avons associé un laser DFB $\lambda/4$, et l'EAM à un amplificateur optique à semiconducteurs. Cette architecture, permet de compenser les pertes introduites par le modulateur, rendant le dispositif mieux adapté aux transmissions à haut débit. Dans ce qui suit, nous présentons la conception et la caractérisation de tels émetteurs III-V/Si intégrant un DFB, un EAM et un SOA sur une même plateforme.

Tous les composants — DFB, EAM et SOA — sont réalisés sur la même épitaxie III-V, mais définis dans des mesas InP distinctes afin de limiter les couplages thermiques et électriques (Figure 6.13). Cette co-intégration sur une pile épitaxiale unique constitue une alternative plus simple et économique que les approches de type SAG ou regrowth [28], [29].

Afin de mieux situer le fonctionnement global du dispositif, il est utile de préciser les dimensions et paramètres optiques des différents blocs actifs. Le laser DFB présente une longueur de $600\text{ }\mu\text{m}$, un facteur de confinement optique dans les puits quantiques $\Gamma_{QWs} = 8\%$, ainsi qu'un coefficient de couplage $\kappa = 30\text{ cm}^{-1}$. Le modulateur électro-absorption a une longueur de $150\text{ }\mu\text{m}$, avec un confinement maximal de $\Gamma_{QWs} = 15\%$ dans la région active. Enfin, l'amplificateur optique à semiconducteurs mesure $1100\text{ }\mu\text{m}$ et partage le même facteur de confinement de $\Gamma_{QWs} = 5\%$ que le laser. Ces choix géométriques traduisent un compromis entre efficacité de modulation, puissance de sortie et limitation des pertes internes.

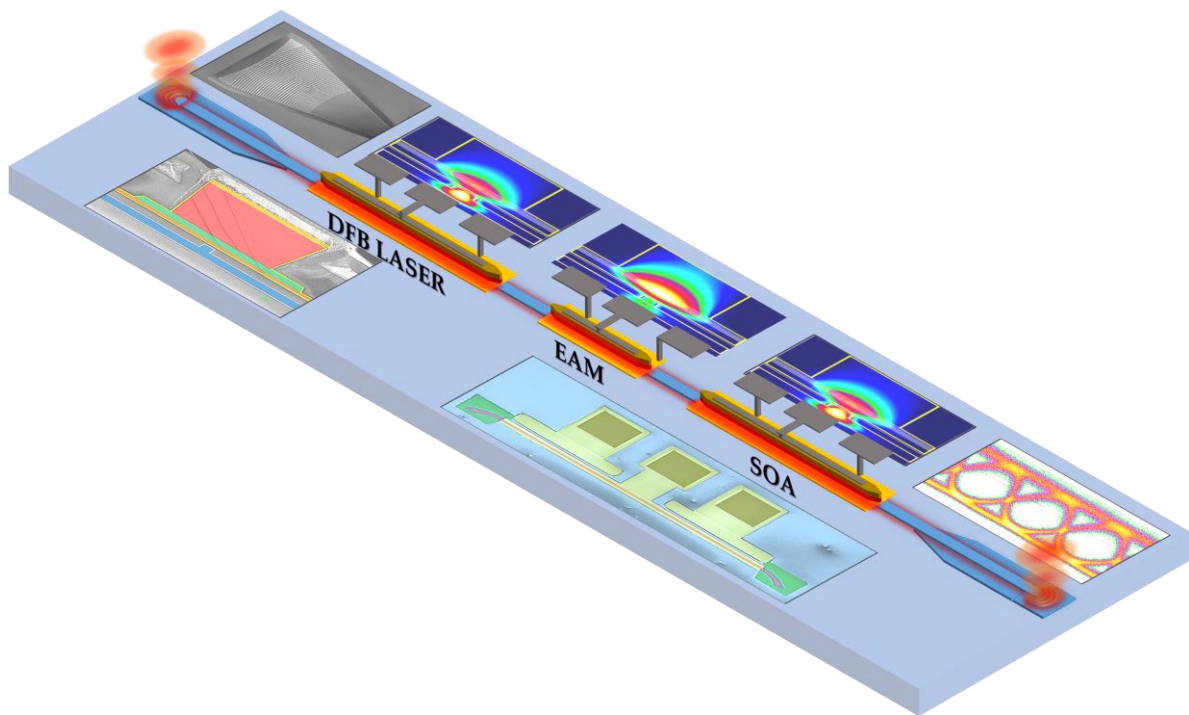


Figure 6.13 : Schéma 3D du transmetteur III-V/Si avec DFB, EAM et SOA. Les encarts montrent les simulations de confinement optique, un diagramme de l'œil à 32 Gbps, et des images MEB du guide et du coupleur.

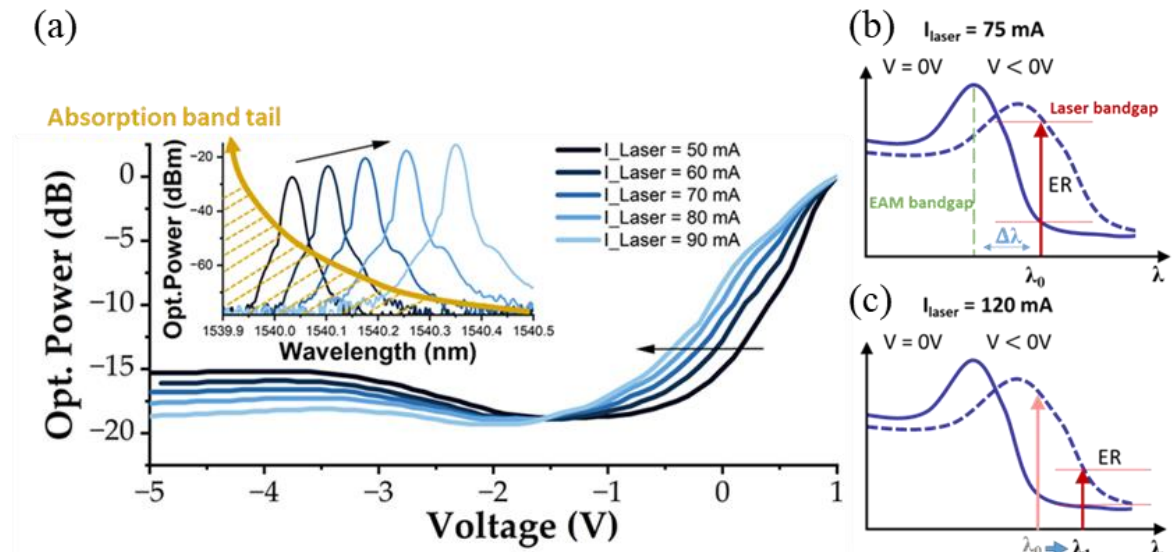


Figure 6.14 : (a) Puissance transmise en fonction de la tension appliquée à l'EAM pour différents courants du DFB. L'encart montre la superposition entre le bord d'absorption et le spectre laser (red-shift thermique). (b)-(c) Spectres d'absorption sous différents courants de polarisation.

L'émetteur fonctionne en bande C, avec une émission centrée autour de 1544 nm. Le bord d'absorption de l'EAM est situé à 1520 nm et se déplace vers les grandes longueurs d'onde lorsque la tension inverse appliquée augmente, conformément à l'effet électro-absorption (QCSE). Pour un courant de 50 mA injecté dans le laser et une tension de -1,25 V sur l'EAM, un rapport d'extinction supérieur à 17 dB a été mesuré. L'augmentation du courant d'injection provoque un red-shift du spectre laser d'environ 0,4 nm (Figure 6.14-a), ce qui correspond à une élévation locale de température estimée à 5 °C. Ce décalage spectral se traduit par une variation des pertes d'insertion de l'EAM, qui diminuent lorsque la longueur d'onde laser s'éloigne du bord d'absorption (Figure 6.14-b et Figure 6.14-c).

L'intégration du SOA permet ensuite d'amplifier significativement la puissance optique en sortie. Le laser DFB présente un courant seuil bas, de l'ordre de 10 mA, et délivre une puissance croissante avec le courant de polarisation du SOA. Avec le SOA polarisé à 100 mA, une puissance sur puce de 16 mW est atteinte, soit un gain par un facteur quatre par rapport à un fonctionnement sans amplification. Le spectre obtenu est monomode, centré à 1540 nm avec un SMSR supérieur à 45 dB (Figure 6.15-a). L'analyse électro-optique met en évidence une bande passante de 22 GHz (Figure 6.15-c), confirmant le potentiel de l'architecture pour des transmissions à haut débit.

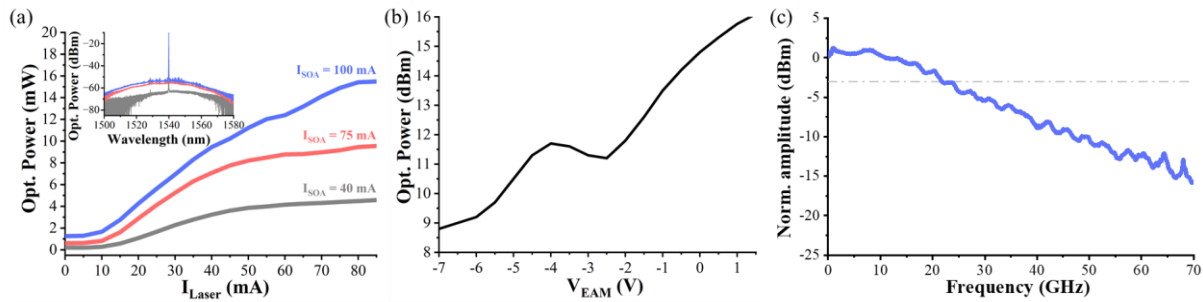


Figure 6.15 : (a) Puissance en sortie fonction du courant du DFB et du SOA (inset : spectre monomode). (b) Puissance transmise vs tension EAM. (c) Réponse S21 montrant une BW = 22 GHz.

Les mesures dynamiques en modulation NRZ-OOK ont ensuite été réalisées. Dans le cas de l'EML sans SOA, une transmission sans erreur a été obtenue jusqu'à 28 Gbps en configuration back-to-back ainsi qu'après 2 km de fibre monomode standard. À 32 Gbps, une légère dégradation apparaît mais le BER reste inférieur au seuil HD-FEC. Le diagramme de l'œil mesuré (Figure 6.16) montre une ouverture nette avec un rapport d'extinction dynamique de 4 dB.

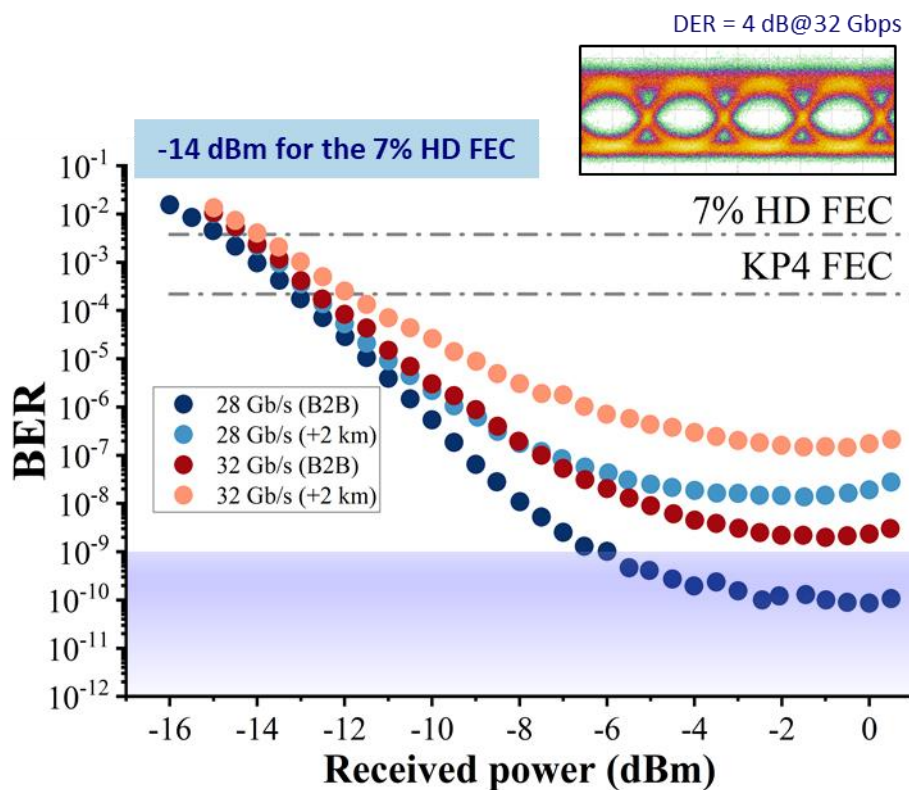


Figure 6.16 : BER mesuré à 28 et 32 Gbps en fonction de la distance (B2B et 2 km). L'encart montre un diagramme de l'œil à 32 Gbps.

L'intégration du SOA a permis d'augmenter la puissance disponible pour la transmission, mais elle a également introduit un effet de rétroaction amplifiée qui se traduit par une dégradation progressive du BER à forts courants d'injection dans le SOA. Néanmoins, pour un courant modéré de 40 mA dans le SOA, les performances restent stables avec un BER < HD-FEC et une puissance équivalente à celle obtenue avec l'EML seul. Les transmissions sur 2 km et 5 km de fibre confirment le bénéfice de l'amplification : une pénalité négligeable est mesurée après 2

km, et une pénalité de seulement 0,8 dB après 5 km, alors que l'EML sans SOA ne permettait pas de transmission correcte à cette distance (Figure 6.17). Les diagrammes de l'œil restent ouverts, avec un DER de 6 dB en B2B qui décroît légèrement à 4,6 dB après 5 km.

Ces résultats démontrent que l'architecture EML+SOA sur plateforme III-V/Si est capable de soutenir des transmissions NRZ jusqu'à 32 Gbps avec une bonne robustesse, tout en compensant les pertes du modulateur. L'amplificateur apporte un gain significatif de puissance, mais impose de travailler dans une zone de fonctionnement modérée pour limiter les effets de feedback optique et préserver la qualité du signal.

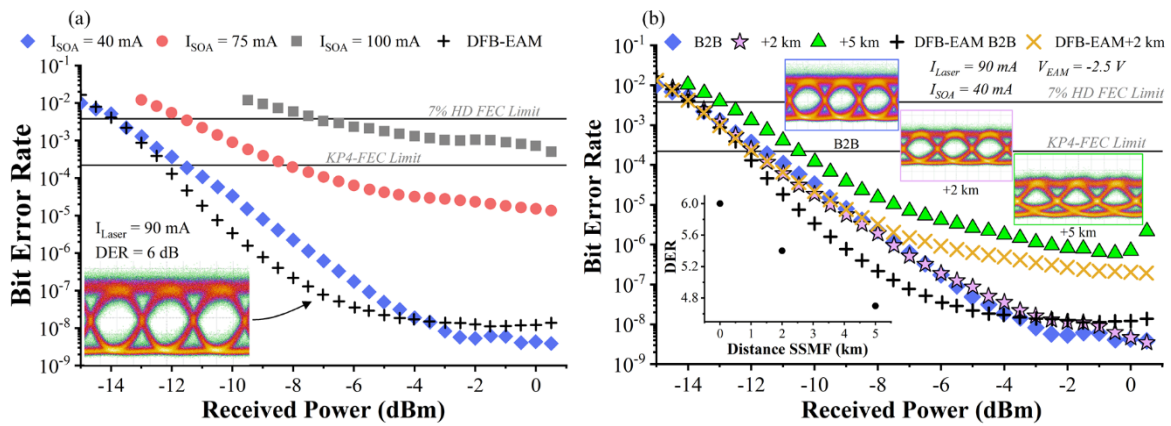


Figure 6.17 : (a) BER mesuré à 32 Gbps en B2B pour différents courants du SOA (inset : œil à DER = 6 dB). (b) BER comparatif après 2 km et 5 km de SSMF avec EML seul ou EML+SOA (encarts : diagrammes de l'œil et évolution du DER).

En résumé, l'intégration conjointe d'un laser DFB $\lambda/4$, d'un modulateur électro-absorption et d'un amplificateur optique à semiconducteurs sur la même plateforme III-V/Si a permis de réaliser un émetteur compact et performant. Les mesures ont montré un ER statique supérieur à 17 dB, une puissance de sortie atteignant 16 mW grâce au SOA, et une bande passante électro-optique de 22 GHz. Les transmissions NRZ ont été démontrées jusqu'à 32 Gbps, avec des performances stables après 2 km et une dégradation limitée à 0,8 dB après 5 km de fibre standard. L'ajout du SOA apporte ainsi un net bénéfice en termes de budget optique, mais au prix d'une sensibilité accrue aux effets de rétroaction lorsqu'il est polarisé à fort courant. Ces résultats placent l'architecture proposée au niveau des meilleures réalisations rapportées dans l'état de l'art, tout en conservant une complexité technologique réduite grâce à l'utilisation d'une seule épitaxie.

Néanmoins, pour répondre aux besoins des réseaux d'accès reconfigurables, la flexibilité spectrale des émetteurs devient un enjeu majeur. C'est pourquoi, au-delà de ces premiers résultats sur les EML à longueur d'onde fixes, nous avons exploré une architecture alternative intégrant un laser accordable avec EAM et SOA dans un seul guide III-V, afin de conjuguer haut débit, compacité et agilité spectrale. Cette approche est détaillée dans la section suivante.

6.5 Laser à Modulation Externe (EML) : Laser accordable + EAM + SOA

Après avoir démontré la faisabilité d'émetteurs III-V/Si intégrant un DFB, un EAM et un SOA sur la même épitaxie, l'étude s'est étendue à une architecture plus avancée, reposant sur un laser accordable. L'objectif est de combiner les avantages d'une modulation externe haut débit avec la flexibilité spectrale indispensable aux futurs réseaux optiques reconfigurables. Cette évolution répond à la demande croissante des réseaux optiques d'accès et métropolitains, qui nécessitent des sources à la fois haut débit et reconfigurables spectralement, compatibles avec les architectures DWDM flexibles.

Dans cette approche, le laser, le modulateur électro-absorption et l'amplificateur sont réalisés dans un même guide d'onde III-V, sans transition optique intermédiaire. Cette conception réduit fortement l'empreinte du dispositif et améliore le confinement optique dans les puits quantiques. Contrairement aux EAMs co-intégrés avec le DFB des sections précédentes, dont le design optique reposait sur des tapers adiabatiques assurant le transfert progressif du mode optique de la couche de silicium vers la couche III-V, la solution présentée ici adopte une approche plus compacte. Dans les EAMs à tapers, le confinement optique pouvait certes être optimisé en ajustant la largeur des guides d'onde en silicium, mais au prix d'une longueur de modulateur plus importante. Or, cette longueur est un paramètre critique puisqu'elle impacte directement la bande passante via l'augmentation de la capacité parasite de la jonction PIN. À l'inverse, la configuration monolithique considérée ici maximise dès l'entrée le recouvrement modal avec la zone active, réduisant la capacité parasite et permettant ainsi d'élargir la bande passante électro-optique. Toutefois, l'intégration monolithique entraîne un couplage thermique et électrique plus marqué entre les différentes sections, ce qui impose un contrôle précis des conditions de fonctionnement. Dans notre cas, l'isolation électrique entre composants est assurée par implantation ionique, suivie de la gravure du contact p-InGaAs (voir images MEB en Figure 6.18).

Dans notre cas, le procédé de collage direct (wafer bonding) a été utilisé, permettant à l'ensemble des composants (laser, EAM, SOA) de partager la même épitaxie III-V et le même mesa InP. Bien que cette approche impose certaines contraintes — en particulier l'impossibilité d'adapter précisément l'épitaxie de chaque composant — elle offre un compromis robuste, garantissant puissance optique, faible tension de commande et haut débit dans une architecture monolithique simplifiée.

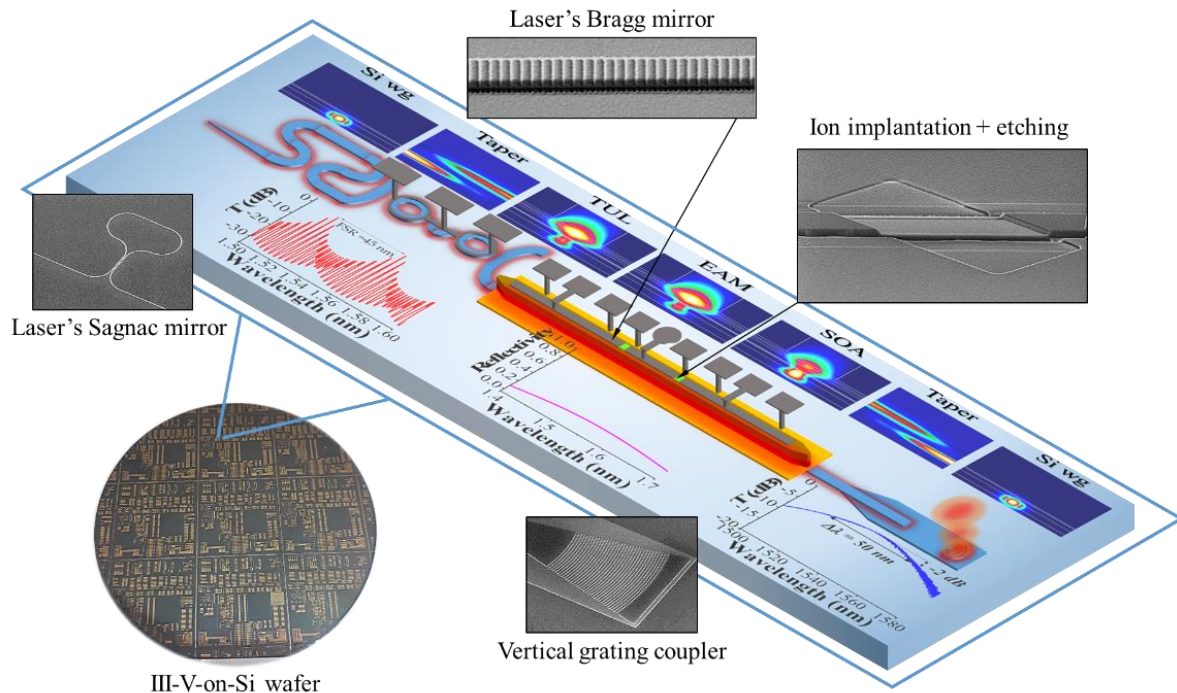


Figure 6.18 : Photographie du wafer final avec un zoom montrant la structure schématique du transmetteur intégré III-V/Si. De gauche à droite : miroir de Sagnac fortement réfléchitif (image MEB), filtre Vernier, section de gain III-V du laser, miroir de Bragg du laser faiblement réfléchitif (image MEB), section III-V du modulateur électro-absorption (EAM) et de l'amplificateur optique à semiconducteurs (SOA) séparés par des zones électriquement isolées (image MEB), et enfin coupleur à réseau pour l'insertion en fibre (image MEB). Les courbes en encart en bas montrent la transmission simulée du filtre Vernier dans la plage $\lambda = 1500\text{--}1600\text{ nm}$ ainsi que le spectre de réflectivité et de transmission du miroir de Bragg. Les encarts supérieurs présentent plusieurs simulations 2D par différences finies (FDM) de la distribution modale TE pour chaque composant.

Le laser utilisé est un laser accordable en bande C, il a été conçu avec une longueur de $670\text{ }\mu\text{m}$, un facteur de confinement modal dans les puits quantiques de $\Gamma_{QWs} = 13,5\%$ et un couplage de Bragg de $\Gamma_{Bragg} = 0,19\%$. L'EAM présente une longueur de $100\text{ }\mu\text{m}$ et un confinement optique de $\Gamma_{QWs} = 13,7\%$. SOA, d'une longueur de $920\text{ }\mu\text{m}$, a quant à lui été conçu avec un confinement plus faible, $\Gamma_{QWs} = 10,95\%$, afin d'augmenter la puissance de saturation en sortie ($P_{sat} \propto 1/\Gamma_{QWs}$).

Le laser utilise un miroir de Bragg ($12\text{ }\mu\text{m}$, période 242 nm , $\sim 25\%$ de réflectivité) en sortie et un miroir de Sagnac en fond de cavité ($\sim 100\%$ de réflectivité). Ce choix optimise la puissance de sortie et améliore le rapport modulation optique (OMA) à l'entrée de l'EAM, tout en maintenant un SMSR élevé. Le filtre Vernier est constitué de deux résonateurs en forme de racetrack, avec des espacements spectraux libres (FSR) de $2,52\text{ nm}$ et $2,68\text{ nm}$ respectivement, produisant un FSR effectif de $43,56\text{ nm}$. Ce filtrage assure la sélectivité spectrale et la stabilité de la longueur d'onde d'émission.

Cette architecture intégrée combine ainsi simplicité de fabrication (une seule épitaxie III-V), compacité (pas de tapers) et performances élevées (faible seuil laser, forte efficacité de modulation, amplification SOA). Les résultats statiques et dynamiques qui suivent permettront d'évaluer plus précisément le potentiel de ce transmetteur.

Le transmetteur a été testé sur une platine thermo-régulée à $T = 20\text{ }^{\circ}\text{C}$.

La caractéristique L-I du laser est représentée en Figure 6.19-a pour différentes valeurs de courant injecté dans le SOA, alors que l'EAM reste polarisé à 0 V. On observe plusieurs décrochages (« kinks ») dans les courbes L-I, attribués à des sauts de modes provoqués par un désaccord passif-actif entre la section active III-V et la réponse spectrale du filtre Vernier. Malgré cela, une augmentation régulière de la puissance optique est observée avec l'augmentation du courant injecté dans le SOA, atteignant un maximum d'environ 5 mW pour $I_{\text{SOA}} = 80\text{ mA}$ et $I_{\text{Laser}} = 100\text{ mA}$. Au-delà de cette valeur, la puissance sature puis décroît pour des courants plus élevés dans le SOA, ce qui reflète l'apparition d'effets non-linéaires et/ou thermiques (courbes rouge et orange de la Figure 6.19-a). Par ailleurs, une diminution du courant seuil du laser est constatée lorsque le SOA est plus fortement polarisé, effet attribué à une rétroaction optique parasite. Le transmetteur a été testé à $T = 20\text{ }^{\circ}\text{C}$.

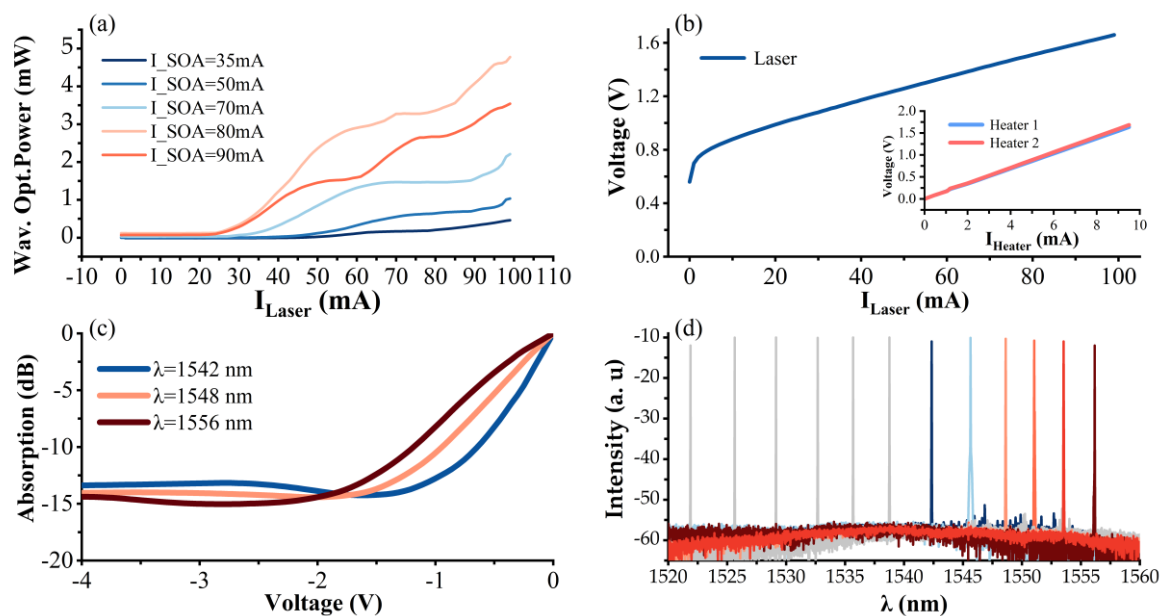


Figure 6.19 : (a) Caractéristique L-I du laser accordable mesurée pour différentes valeurs de courant injecté dans le SOA (I_{SOA}). (b) Caractéristique V-I mesurée du laser et des résistances chauffantes (encart). (c) Absorption optique mesurée en fonction de la tension de polarisation inverse appliquée à l'EAM pour différentes longueurs d'onde. (d) Spectres optiques superposés pour différents modes de fonctionnement du laser accordable, montrant la plage de longueurs d'onde dans laquelle l'EAM peut fonctionner (courbes colorées)

La caractéristique V-I du laser est présentée en Figure 6.19-b. La résistance différentielle extraite est de $7,5\text{ }\Omega$ pour $I_{\text{Laser}} = 100\text{ mA}$, ce qui est cohérent avec les valeurs rapportées pour des dispositifs similaires [30]. L'encart montre la réponse électrique des résistances chauffantes intégrées, avec une résistance série d'environ $175\text{ }\Omega$. Ces chauffages permettent d'accorder la longueur d'onde d'émission du laser avec un pas d'environ $2,7\text{ nm}$ par actionneur, ce qui conduit à une plage de réglage spectrale totale d'environ 40 nm .

La réponse en extinction statique de l'EAM a également été mesurée et est illustrée en Figure 6.19-c. L'absorption optique varie avec la tension de polarisation inverse appliquée et dépend

de la longueur d'onde d'essai. Les mesures ont été réalisées avec un courant injecté de 60 mA dans la section laser et 80 mA dans le SOA. L'absorption croît avec la tension jusqu'à saturation autour de -2 V, traduisant l'entrée dans un régime de forte déplétion. On constate par ailleurs que, pour des longueurs d'onde plus élevées, une tension plus importante est nécessaire pour atteindre la même absorption, ce qui correspond à l'éloignement progressif du bord de bande du modulateur. Le rapport d'extinction maximal atteint 16 dB à 1556 nm, et reste supérieur à 14,5 dB à 1542 nm.

Enfin, la Figure 6.19-d présente une superposition des spectres de sortie obtenus pour différentes conditions de pilotage des résistances chauffantes. Les spectres gris correspondent aux longueurs d'onde pour lesquelles l'EAM n'est pas transparent à l'état bloqué, tandis que les spectres colorés indiquent la plage dans laquelle le modulateur absorbe efficacement et fournit un rapport d'extinction élevé. La plage d'accord s'étend sur environ 15 nm (1542–1557 nm) tout en maintenant un rapport de suppression de modes latéraux (SMSR) ≥ 50 dB, ce qui est comparable aux résultats déjà rapportés dans la littérature pour des modulateurs III-V/Si [31].

Après la caractérisation statique, les performances dynamiques du transmetteur accordable ont été évaluées. La réponse électro-optique de l'EAM a été mesurée à l'aide d'un analyseur de réseau vectoriel (VNA) pour différentes longueurs de modulateurs (100, 150 et 200 μm). Les mesures ont été effectuées à $\lambda = 1551$ nm, avec $I_{\text{Laser}} = 70$ mA, $I_{\text{SOA}} = 80$ mA et une tension de polarisation de -3 V. Les résultats montrent une augmentation notable de la bande passante lorsque la longueur du modulateur diminue, atteignant 38 GHz pour $L = 100$ μm (contre 20 GHz pour $L = 200$ μm). Cette amélioration est cohérente avec la réduction de la constante RC pour des modulateurs plus compacts.

La transmission a été évaluée sur toute la bande modulable du EAM (15 nm, de 1542 à 1557 nm) avec un pas de 3 nm. Les mesures de BER (Figure 6.20) montrent des transmissions sans erreur ($\text{BER} < 10^{-9}$) en configuration back-to-back et après 2 km de SSMF pour la majorité des longueurs d'onde testées. Le rapport d'extinction dynamique varie entre 8.5 dB ($\lambda = 1554$ nm) et 4.5 dB ($\lambda = 1542$ nm). Au-delà, la transmission se dégrade progressivement, avec des pénalités de puissance de 0.5 dB à 2.8 dB pour des distances allant jusqu'à 8 km. L'utilisation conjointe de l'EAM et du SOA, aux chirps opposés, permet une compensation partielle du chirp, prolongeant les distances de transmission par rapport à des DMLs classiques. Enfin, des tests de modulation large signal ont été réalisés à plus haut débit. À 50 Gbps NRZ, un DER de 4.7 dB a été obtenu. Le dispositif a ensuite été poussé jusqu'à 56 Gbps, où un DER de 4 dB a été mesuré malgré un certain jitter et une déformation du signal due aux limites de la chaîne RF. Ces résultats, obtenus sans traitement numérique du signal (DSP) ni égalisation, confirment la capacité du transmetteur accordable III-V/Si à fonctionner à très haut débit.

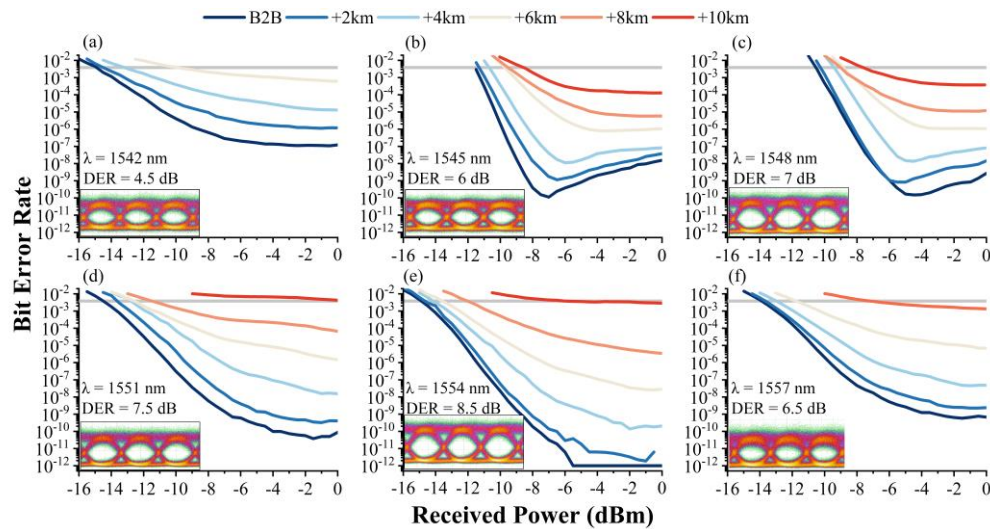


Figure 6.20 : Taux d'erreur binaire mesuré à 32 Gbps en fonction de la puissance reçue pour différentes longueurs d'onde, sur des distances allant du back-to-back jusqu'à 10 km avec un pas de 2 km. L'encart présente les diagrammes de l'œil pour chaque longueur d'onde, et les lignes horizontales grises indiquent la limite du HD-FEC.

Ces résultats valident la faisabilité de sources III-V/Si intégrées accordables et modulées en externe, capables de combiner accordabilité spectrale, puissance de sortie suffisante, fort rapport d'extinction et hautes vitesses de modulation. Ils mettent en évidence le potentiel de telles architectures pour les réseaux optiques courts et métropolitains, notamment dans le contexte des interconnexions optiques à haut débit et des systèmes DWDM de prochaine génération.

Conclusion et perspectives

Cette thèse a porté sur la conception et la réalisation de transmetteurs III-V/Si à haut débit en bande C, destinés aux réseaux optiques métropolitains et d'accès. L'intégration hétérogène de matériaux III-V sur circuits photoniques silicium a été explorée comme solution compacte et économe en énergie pour des systèmes IM-DD à très haute capacité.

Dans un premier temps, deux architectures de lasers DFB ont été étudiées : un SBG-DFB à faible κ ($\Delta\nu = 118$ kHz, SMSR > 45 dB, BW = 14 GHz, transmission NRZ jusqu'à 25 Gbps) et un $\lambda/4$ -DFB ($\kappa = 54$ cm⁻¹, BW = 21 GHz, transmission NRZ jusqu'à 32 Gbps et PAM-4 à 50 Gbps). Les tests de tolérance au rétro-couplage optique ont montré une robustesse remarquable, avec des BER maintenus sous le seuil HD-FEC jusqu'à -17 dB de feedback.

Ensuite, la modulation externe a été étudiée via l'intégration de DFB + EAM + SOA sur une même pile épitaxiale. L'EML co-intégré a délivré 16 mW de puissance sur puce et permis une transmission NRZ à 32 Gbps, avec un DER de 6 dB et des pénalités inférieures à 1 dB sur 5 km. L'impact du SOA sur les rétro-réflexions a également été caractérisé. Enfin, une architecture plus avancée a été démontrée : un laser accordable largement intégrable avec EAM et SOA. Le dispositif a offert une gamme d'accord de 40 nm, une bande passante électro-optique de

38 GHz, et des transmissions fiables jusqu'à 56 Gbps NRZ avec des pénalités modérées sur 10 km de fibre SSMF.

Ces résultats confirment le potentiel des émetteurs intégrés III-V/Si pour des liens optiques compacts, performants et compatibles avec des architectures WDM à haut débit. Des extensions naturelles de ce travail consistent à :

- Explorer l'O-band, particulièrement adaptée aux communications courtes distances (centres de données), où la dispersion est minimale.
- Optimiser les effets de rétroaction optique/électrique, non seulement pour la stabilité mais aussi comme levier actif de réduction de bruit et d'élargissement de bande passante. En particulier, le concept de double feedback optique-électrique, déjà étudié théoriquement dans l'équipe, apparaît prometteur pour affiner le contrôle spectral et dynamique des lasers.
- Développer des architectures reconfigurables, tirant parti de la large accordabilité démontrée, pour des applications DWDM flexibles.

Ces avancées ouvrent la voie à la prochaine génération de transmetteurs hybrides III-V/Si, capables d'allier compacité, efficacité énergétique et très haut débit pour les futurs réseaux optiques.

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7. Appendices

A – CONFINEMENT FACTORS AND MODAL GAIN

Confinement factor

We adopt the time-harmonic convention $e^{+i\omega t}$. The rigorous definition of the optical confinement factor uses the time-averaged Poynting flux,

$$S_z = \frac{1}{2} \mathbb{R}\{\mathbf{E} \times \mathbf{H}^*\} \cdot \hat{z} \quad (A - 1)$$

The electric field, $\mathbf{E}$, and the magnetic field, $\mathbf{H}$, so that :

$$\Gamma = \frac{\iint_{\Omega_{act}} S_z d_x d_y}{\iint_{\Omega} S_z d_x d_y} \quad (A - 2)$$

We use a right-handed (x,y,z) frame with z along propagation. For any fixed z, Ω denotes the full transverse cross-section of the waveguide. $\Omega_{act} \subset \Omega$, denotes the active material region, i.e., the area that contributes to stimulated emission (typically the QWs). In the rectangular case used in the main text, $\Omega_{act} = \{(x, y) | x \in [-\omega/2, \omega/2], y \in [-d/2, d/2]\}$, where ω is the lateral width and d the active thickness.

In non-magnetic, weakly guiding III–V/Si waveguides this reduces to the familiar scalar overlap :

$$\Gamma \approx \frac{n_{act}}{\bar{n}} \frac{\iint_{\Omega_{act}} |U(x, y)|^2 d_x d_y}{\iint_{\Omega} |U(x, y)|^2 d_x d_y} \quad (A - 3)$$

We denote by n_{act} the refractive index in the active region and $\bar{n}$ the cross-section average index is defined as :

$$\bar{n} = \frac{\iint_{\Omega} n(x, y) |U(x, y)|^2 dx dy}{\iint_{\Omega} |U(x, y)|^2 dx dy} \quad (A - 4)$$

Any overall normalization of U cancels in the ratio. If U is normalized so that $\iint_{\Omega} |U(x, y)|^2 dx dy = 1$, then Eq. (A - 3) further simplifies to :

$$\Gamma \approx \frac{n_{act}}{\bar{n}} \iint_{\Omega_{act}} |U(x, y)|^2 dx dy \quad (A - 5)$$

Modal gain

If the gain is distributed over sub-regions m , define :

$$\Gamma_m = \frac{\iint_{\Omega_m} S_z dx dy}{\iint_{\Omega} S_z dx dy}, \quad \Gamma = \sum_m \Gamma_m \quad (A - 6)$$

Let g_{mat} be the material gain of the active medium and α_i the internal loss. The modal gain is :

$$g_{mod} = \Gamma g_{mat} - \alpha_i \quad (A - 7)$$

And, for quantities per layer, $g_{mod} = \sum_m \Gamma_m g_{mat,m} - \alpha_i$.

Standing-wave description along z and axial averaging

In cavities with counter-propagating waves, the longitudinal field can be written as a superposition of forward and backward waves. A convenient representation is :

$$\mathbf{E}(x, y, z) = U(x, y) \cos(\beta_z z) \quad (A - 8)$$

Where $\mathbf{U}_i(x, y)$ denotes the transverse electric-field profile of the guided mode and $\beta_z (= 2\pi/\lambda_z \equiv 2\pi\bar{n}/\lambda)$ is the z -component of the propagation constant k . In writing (A – 1), we have neglected the imaginary part of β , and hence are ignoring the growth of the fields in the cavity.

Consider an active slab of thickness d , width w , and active length L_A embedded in a cavity of total length L . If the group velocity and β are approximately the same in the active and passive sections, the small-signal modal gain per unit length obtained by volume-averaging a (possibly space-dependent) local gain $\bar{g}(x, y, z)$ is

$$g = \frac{\iiint \bar{g}(x, y, z) |U(x, y)|^2 2\cos^2(\beta z) dx dy dz / L}{\iint |U(x, y)|^2 dx dy} \quad (A - 9)$$

The factor $2\cos^2(\beta z) = 1 + \cos(2\beta z)$ arises from the standing-wave pattern, if the axial integration spans many optical periods, the $\cos(2\beta z)$ term averages to zero.

Assuming $\bar{g}$ is constant within the active region (gain g inside, zero outside) and setting the origin at the center of the active section, (A – 9) separates into a transverse confinement and an axial confinement :

$$g = \Gamma_z \Gamma_{xy} g \quad (A - 10)$$

Where :

$$\Gamma_{xy} = \frac{n_{act}}{\bar{n}} \frac{\int_{-\frac{\omega}{2}}^{\frac{\omega}{2}} \int_{-\frac{d}{2}}^{\frac{d}{2}} |U(x, y)|^2 dx dy}{\iint_{\Omega} |U(x, y)|^2 dx dy} \quad (A - 11)$$

It is often convenient to write $\Gamma_z = \Gamma_{fill} \Gamma_{enh}$ with a fill factor :

$$\Gamma_z = L_A / L \quad (A - 12)$$

And a standing-wave enhancement :

$$\Gamma_z = 1 + \frac{\sin(\beta L_A)}{\beta L_A} \quad (A - 13)$$

B – List of Publications and Communications

This appendix lists the scientific works carried out during the PhD, published in peer-reviewed journals or presented at international conferences.

Peer-reviewed Journals

- [1] **A. Souleiman** et al., "56 Gbps externally modulated widely tunable lasers with SOA boosters heterogeneously integrated on silicon," *Opt. Express* 32, 37036-37045 (2024)
- [2] J. M. Ramirez, **A. Souleiman** et al., "Low- κ , narrow linewidth III-V-on-SOI distributed feedback lasers with backside sampled Bragg gratings," *Opt. Express* 30, 36717-36726 (2022).
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- [5] **A. Souleiman** et al., "High Optical Feedback-Tolerance of Distributed Feedback III-V-on-SOI Laser for High-Speed Isolator-Free Operation", *J. Opt. Soc. Am. B, Feature Issue on Recent Advances in Integrated Photonics*, under review, 2025.

International Conference Presentations

- [1] **A. Souleiman** et al., "Electro-absorption Modulation of a III-V-on-Silicon DFB Laser co-integrated with SOA boosters at 32 Gbps NRZ," *ECOC 2024; 50th European Conference on Optical Communication*, Frankfurt, Germany, 2024, pp. 786-789.
- [2] **A. Souleiman** et al., "High-Speed Direct Modulation on III-V-on-SOI Distributed Feedback Lasers with Intrinsic Electro-Optical Bandwidth Over 20 GHz," *2023 Conference on Lasers and Electro-Optics Europe & European Quantum Electronics Conference (CLEO/Europe-EQEC)*, Munich, Germany, 2023, pp. 1-1, doi: 10.1109/CLEO/Europe-EQEC57999.2023.10231622
- [3] **Amin Souleiman** et al., "Recent progress on III-V/Si heterogeneous photonics circuits: scaling up maturity, versatility and wafer yield (Conference Presentation)", *Proc. SPIE PC13530, Integrated Optics: Design, Devices, Systems, and Applications VIII*, PC1353002 (5 June 2025); <https://doi.org/10.1117/12.3060700>
- [4] **Amin Souleiman** et al., "InP-based high-speed transceivers for silicon photonics", *Proc. SPIE PC13370, Smart Photonic and Optoelectronic Integrated Circuits 2025*, PC133700A (19 March 2025); <https://doi.org/10.1117/12.3041848>
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- [6] J. M. Ramirez, **A. Souleiman** et al., "Heterogeneously integrated III-V-on-SOI lasers for optical communications," *Proc. SPIE PC12424, Integrated Optics: Devices, Materials, and Technologies XXVII*, PC124240B (2023);
- [7] J. M. Ramirez, **A. Souleiman** et al., "Heterogeneously integrated lasers and electro-absorption modulators for silicon photonics," *Proc. SPIE PC12889, Integrated Optics: Devices, Materials, and Technologies XXVIII*, PC1288908 (13 March 2024); <https://doi.org/10.1117/12.3002761>

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National Conference Presentations

- [1] **A. Souleiman** et al., "Lasers à rétroaction distribuée III-V sur SOI à large bande passante électro-optique pour les télécommunications optiques" Journées Nationales d'Optique Guidée (JNOG), Lyon 2023.
- [2] **A. Souleiman** et al., "Caracterisation d'un laser DFB III-V/Si basé sur un couplage de type Bragg-exchange" Journées Nationales d'Optique Guidée (JNOG), Optique Nice 2022.

Titre : Émetteurs III-V-on-SOI à haut débit pour la communication de données à haute densité

Mots clés : Laser sur puce (DFB, Laser accordable), Lasers à modulation directe, Lasers à modulation externe (électro-absorption), Intégration hétérogène III-V / Si, Rétroaction optique

Résumé : Face aux exigences croissantes en matière de bande passante des systèmes de communication optique, qui doivent soutenir un trafic de données en constante augmentation, ainsi qu'à une prise de conscience accrue des enjeux liés au développement durable, il devient crucial de concevoir des dispositifs consommant moins d'énergie. Cela a conduit à l'exploration de nouvelles technologies favorisant une densité plus élevée et une efficacité énergétique optimisée. Parmi ces technologies, les circuits intégrés photoniques (PIC, pour Photonic Integrated Circuits) sur une plateforme hétérogène III-V sur silicium (SOI, pour Silicon-On-Insulator) se distinguent par leur potentiel d'intégration de divers composants, tout en maintenant des coûts de production compétitifs et en réduisant la consommation énergétique des futurs émetteurs.

L'intégration de matériaux III-V par collage moléculaire sur plaque de silicium permet de bénéficier d'un gain optique sur puce grâce à l'utilisation de ces matériaux, tout en tirant parti de la densité élevée et de la maturité des fonderies CMOS. Dans cette thèse, j'ai mené une étude théorique et expérimentale sur les lasers à rétroaction distribuée (DFB) modulés directement. En exploitant les propriétés de cette plateforme, nous avons intégré un modulateur par électroabsorption (MEA) avec ces DFB pour étudier la modulation externe. Pour atténuer les pertes de puissance optique induites par le modulateur, nous avons intégré un amplificateur optique à semiconducteurs (SOA). Une étude similaire a été menée pour un laser accordable, également modulé de manière externe avec un MEA, co-intégré avec un SOA.

Title : High-Speed III-V-on-SOI Transmitter for High-Density Data Communication

Keywords : On-chip Laser (DFB, Tunable Laser), Directly-Modulated Lasers, Externally-Modulated Lasers (Electro-absorption), Heterogeneous integration III-V / Si, Optical Feedback

Abstract : Given the increasing bandwidth demands of optical communication systems, which must support constantly rising data traffic, along with a heightened awareness of sustainability issues, it becomes crucial to design devices that consume less energy. This has led to the exploration of new technologies that promote higher density and optimized energy efficiency. Among these technologies, photonic integrated circuits (PICs) on a heterogeneous III-V-on-SOI (Silicon-On-Insulator) platform stand out for their potential to integrate various components while maintaining competitive production costs and reducing the energy consumption of future emitters.

The integration of III-V materials through molecular bonding onto silicon wafers allows for an optical gain on-chip due to the use of these materials, while leveraging the high density and maturity of CMOS foundries. In this thesis, I conducted a theoretical and experimental study on directly modulated distributed feedback (DFB) lasers. By exploiting the properties of this platform, we integrated an electro-absorption modulator (EAM) with these DFB lasers to investigate external modulation. To mitigate the optical power losses induced by the modulator, we integrated a semiconductor optical amplifier (SOA). A similar study was conducted for a tunable laser, also externally modulated with an EAM, co-integrated with an SOA.